

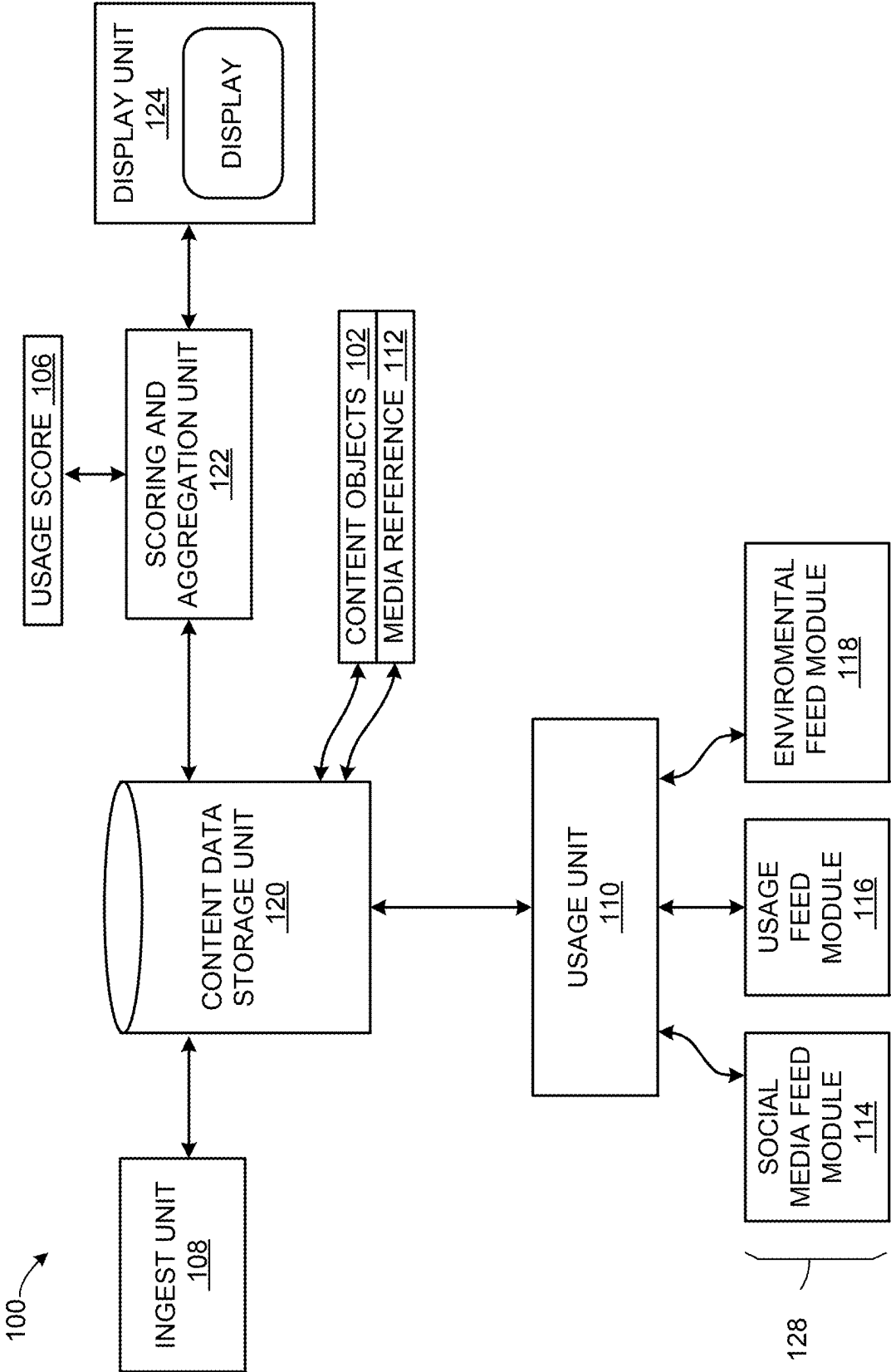


FIG. 1

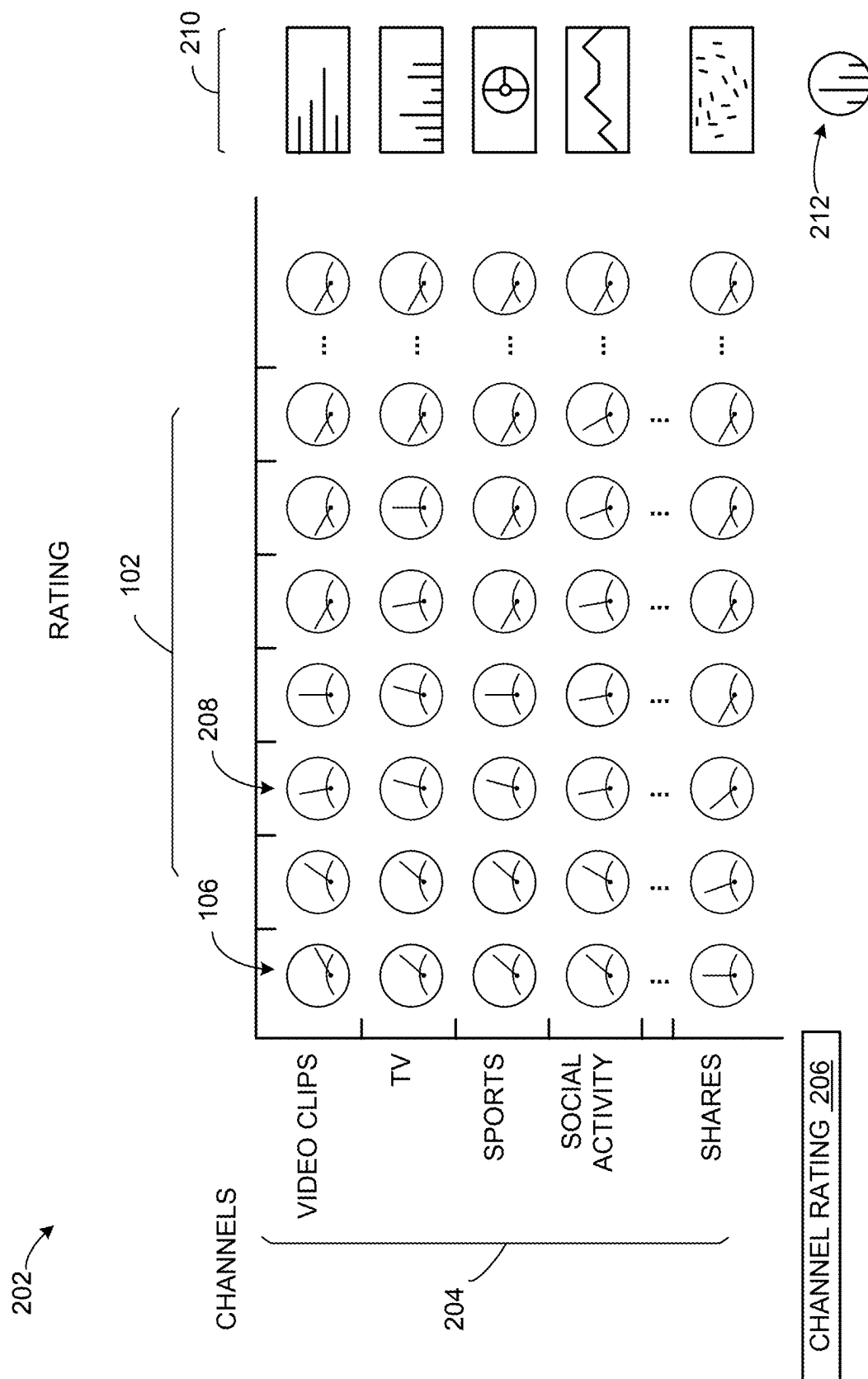


FIG. 2

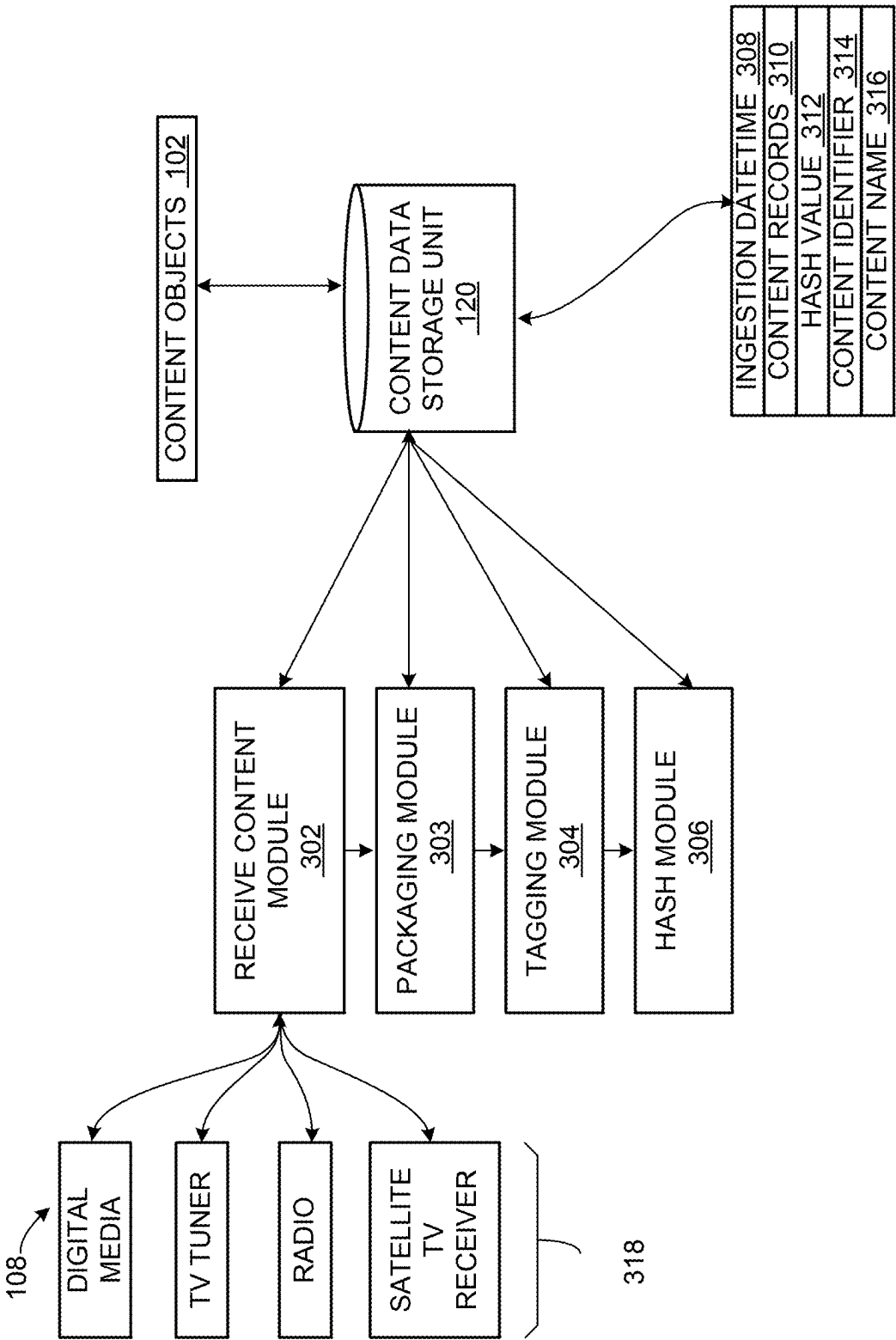


FIG. 3

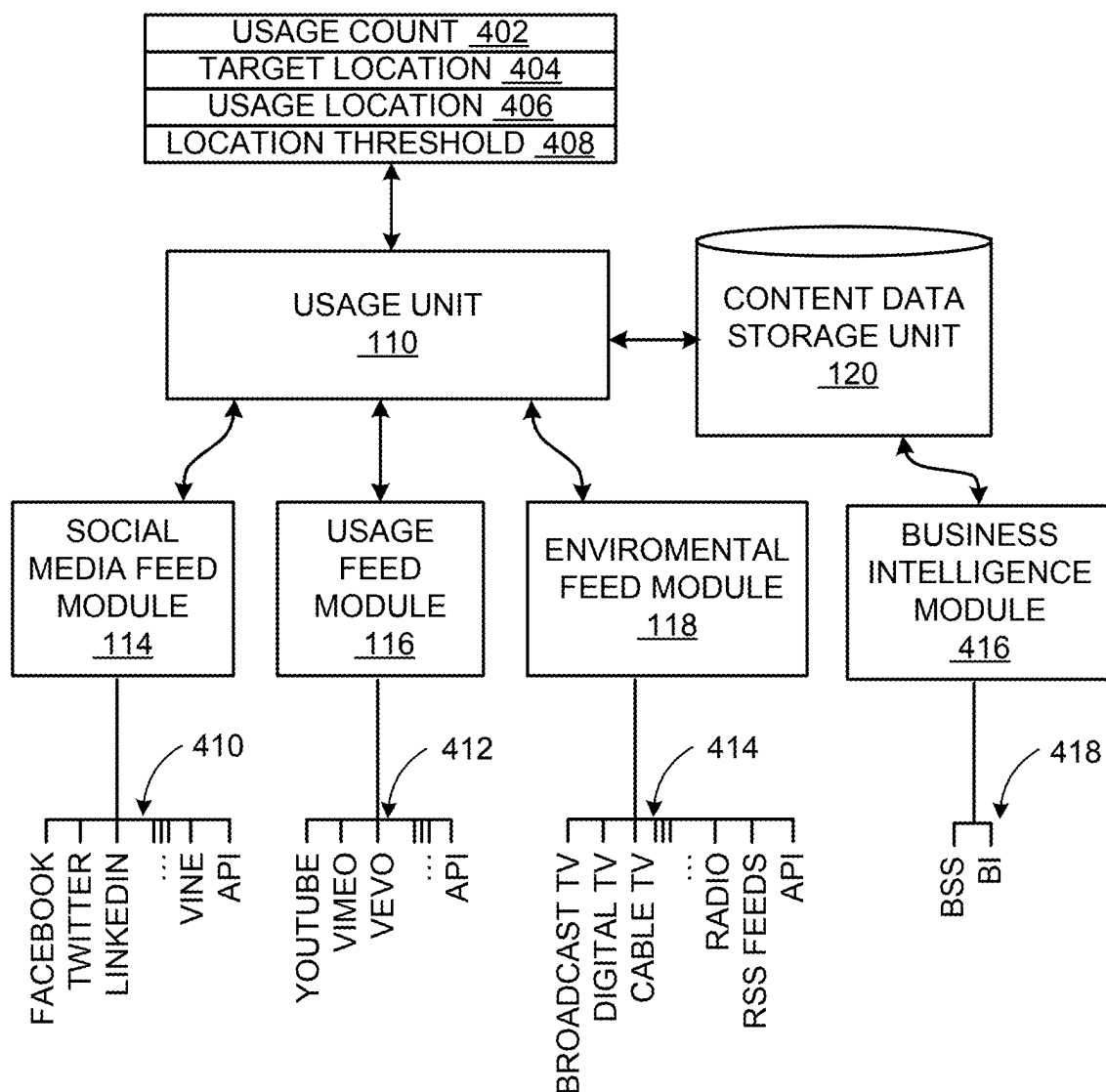


FIG. 4

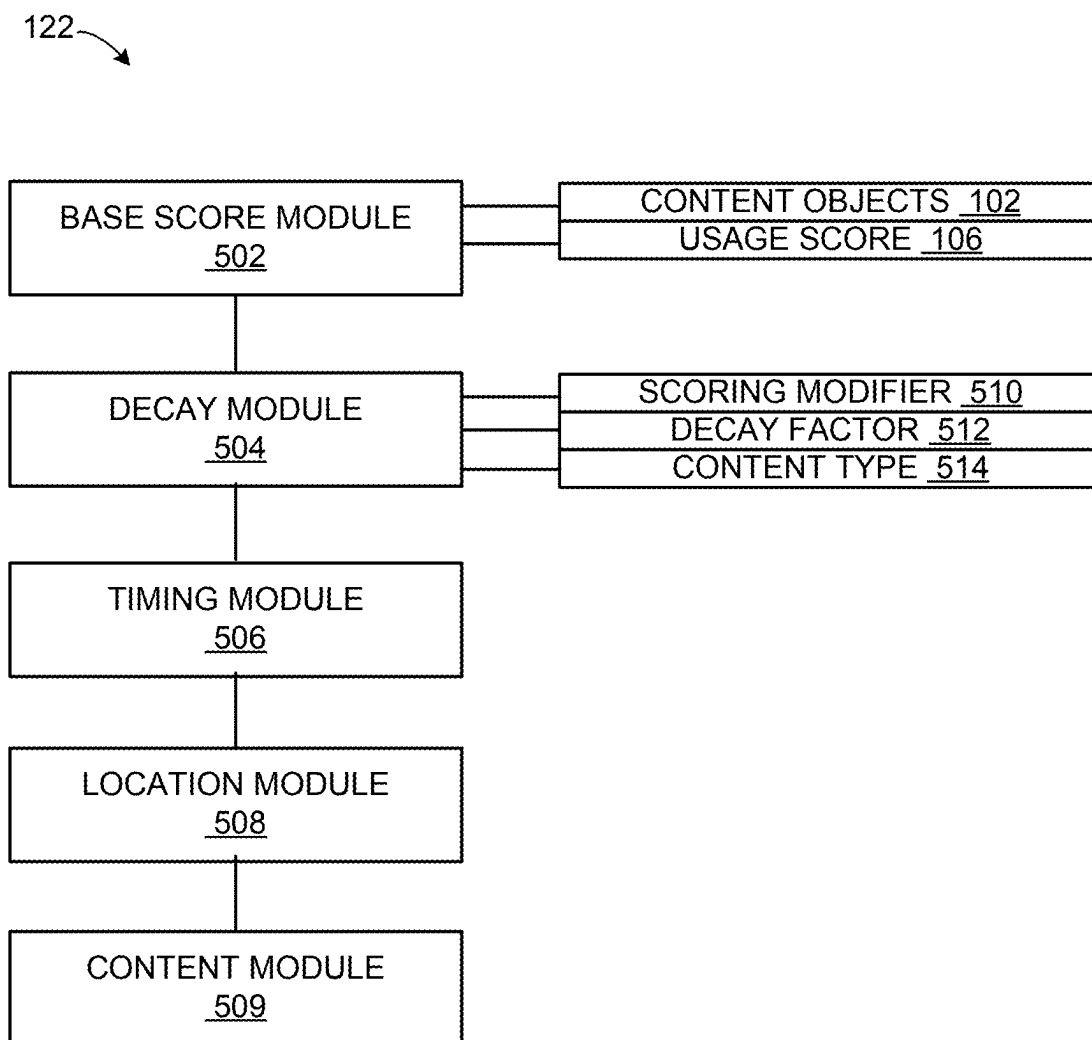


FIG. 5

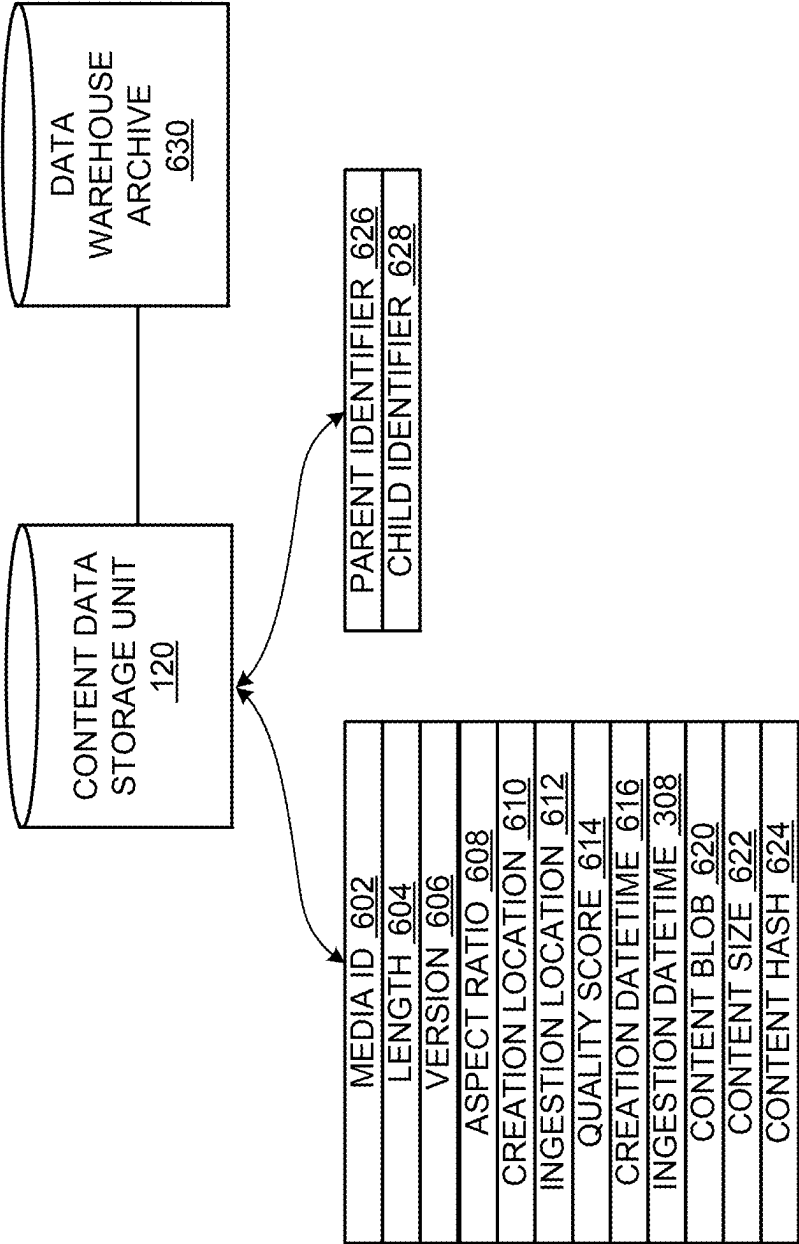


FIG. 6

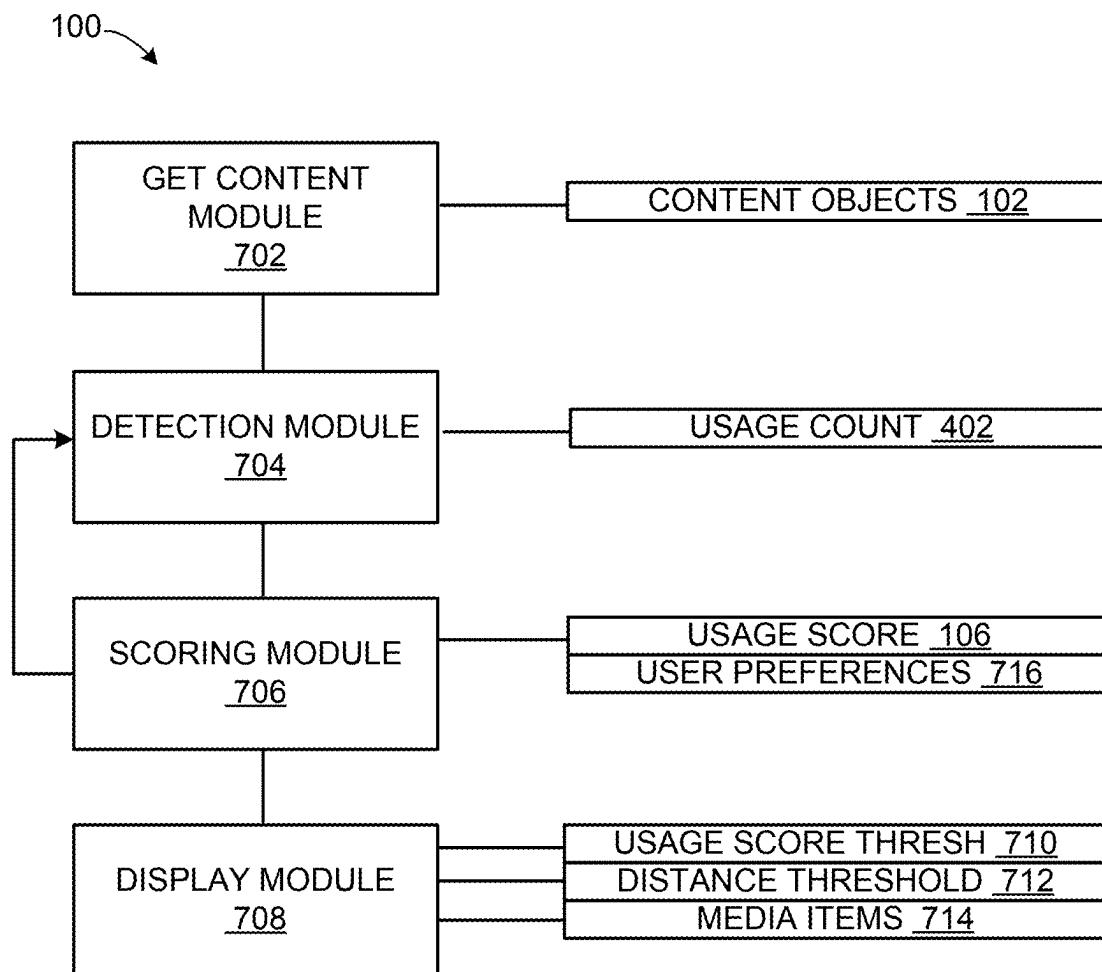


FIG. 7

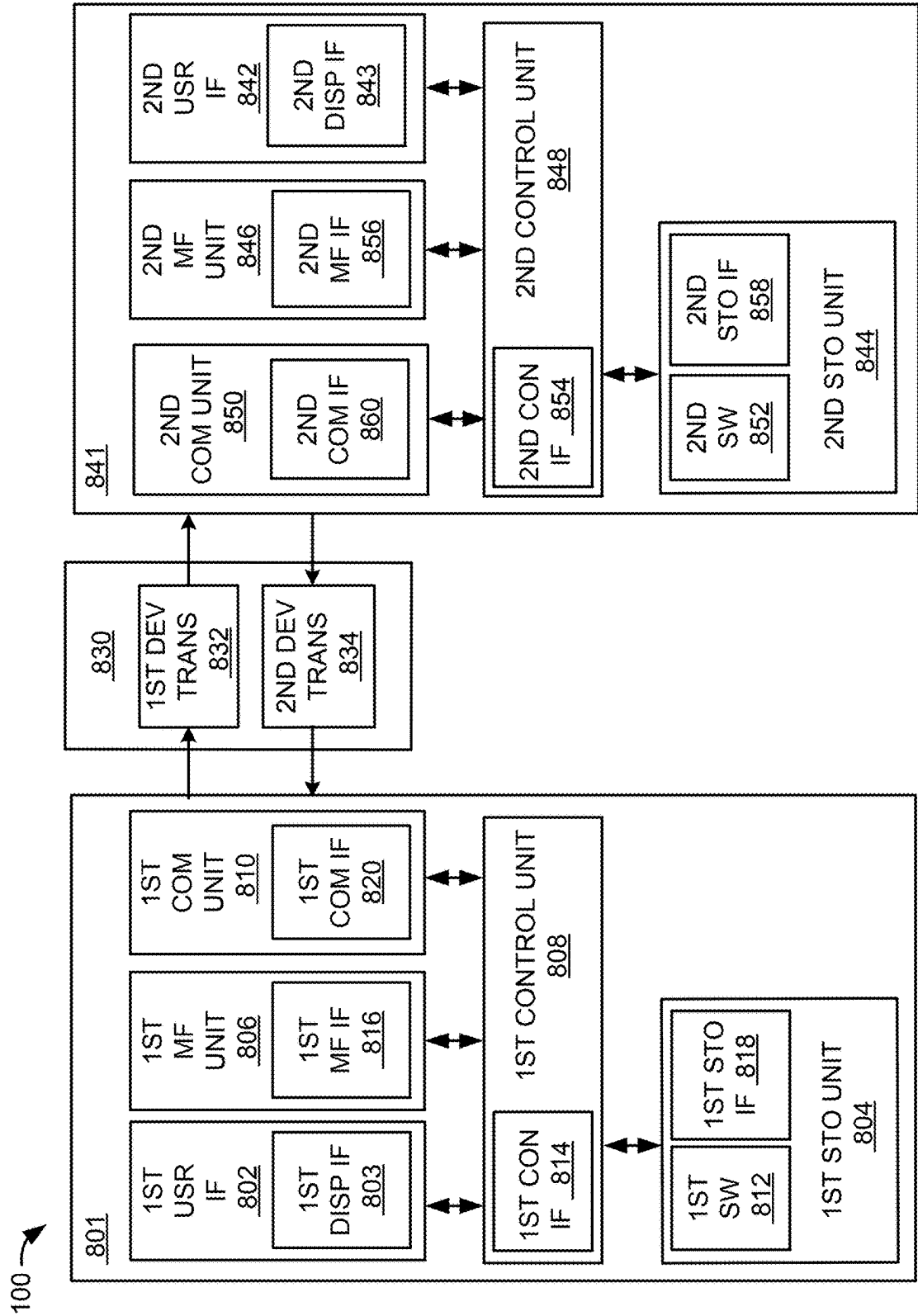


FIG. 8

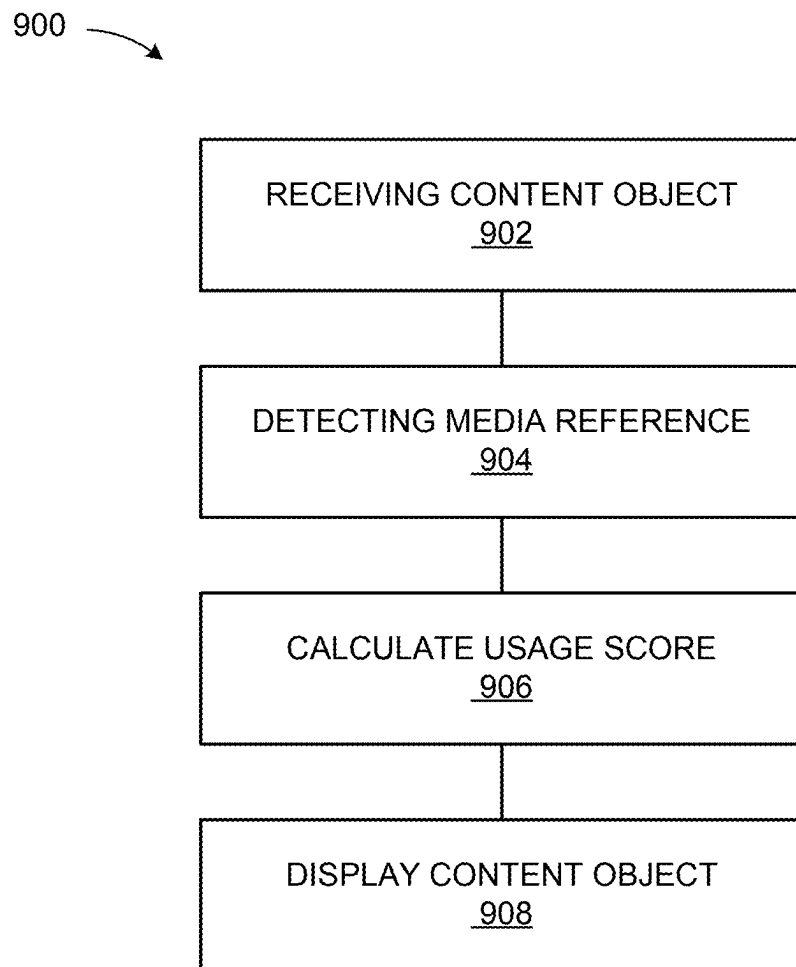


FIG. 9

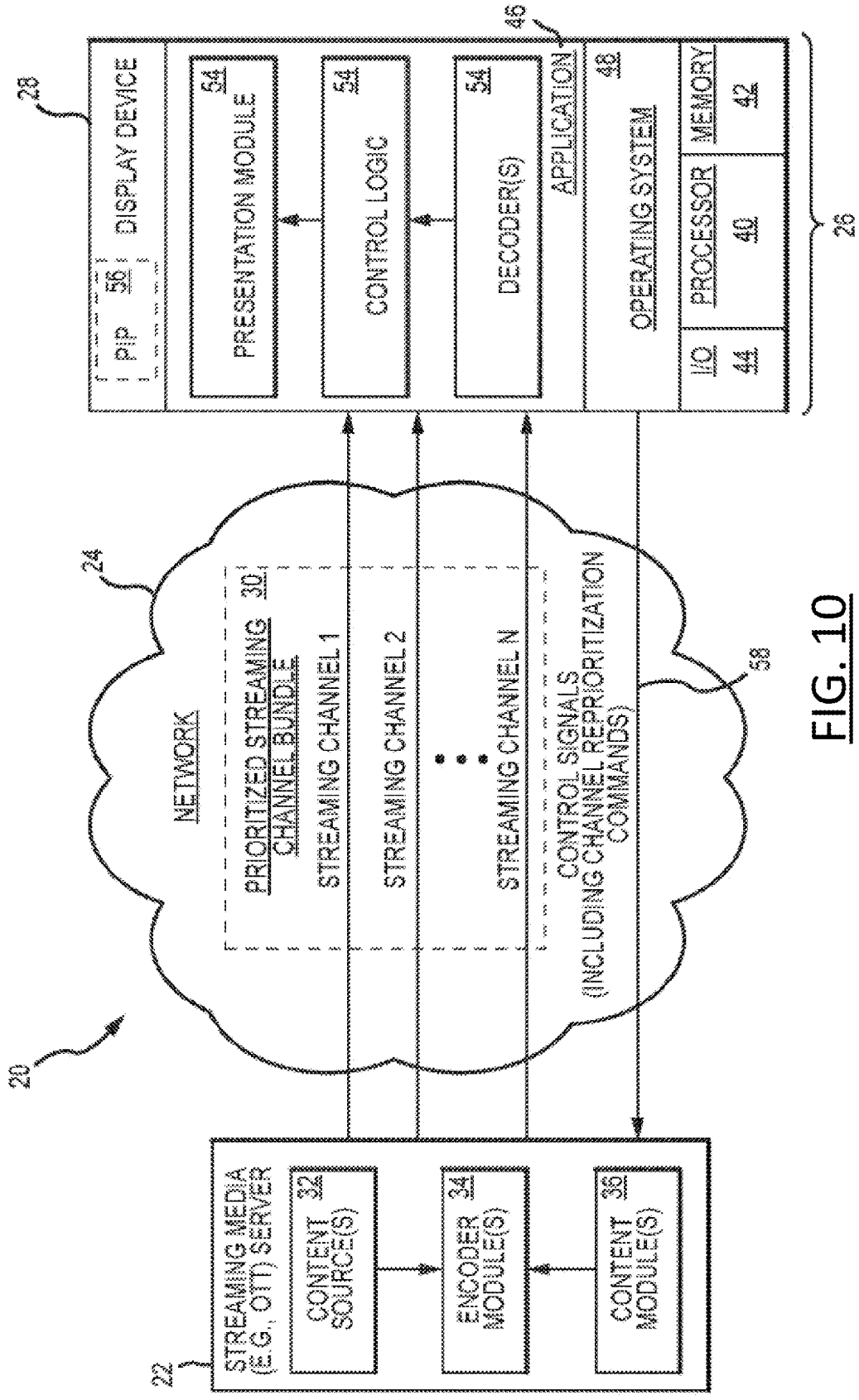


FIG. 10
PRIOR ART

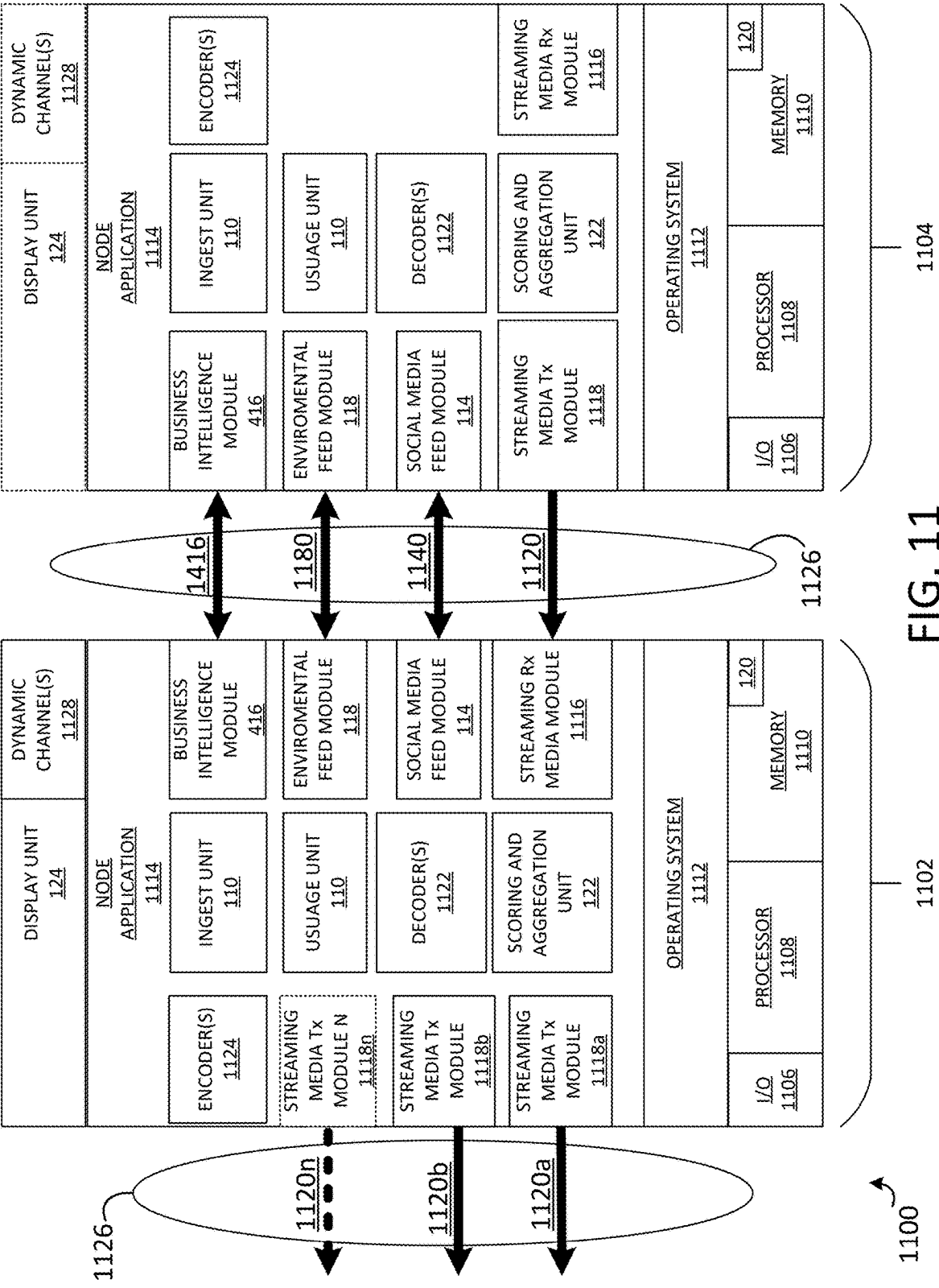
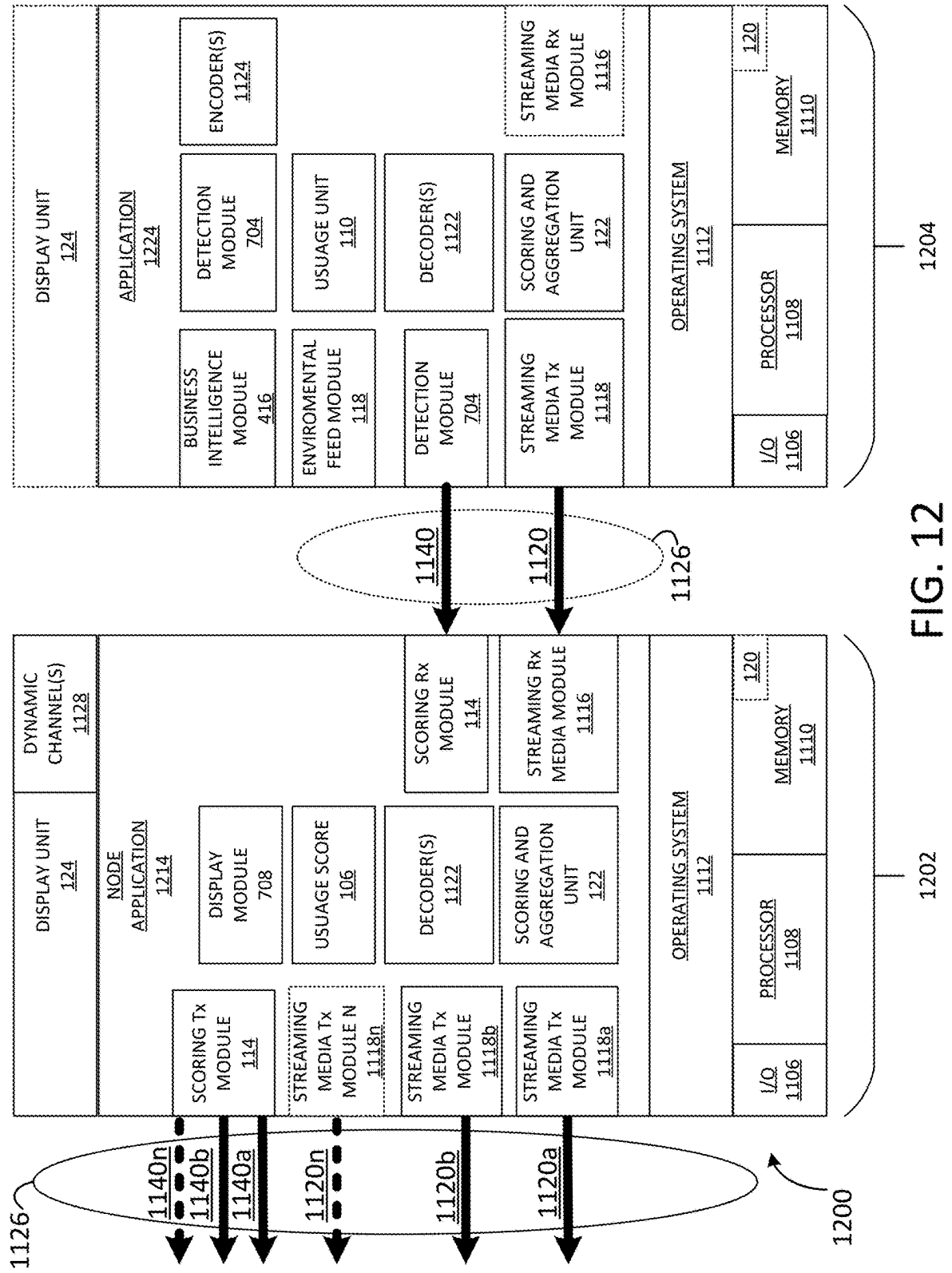
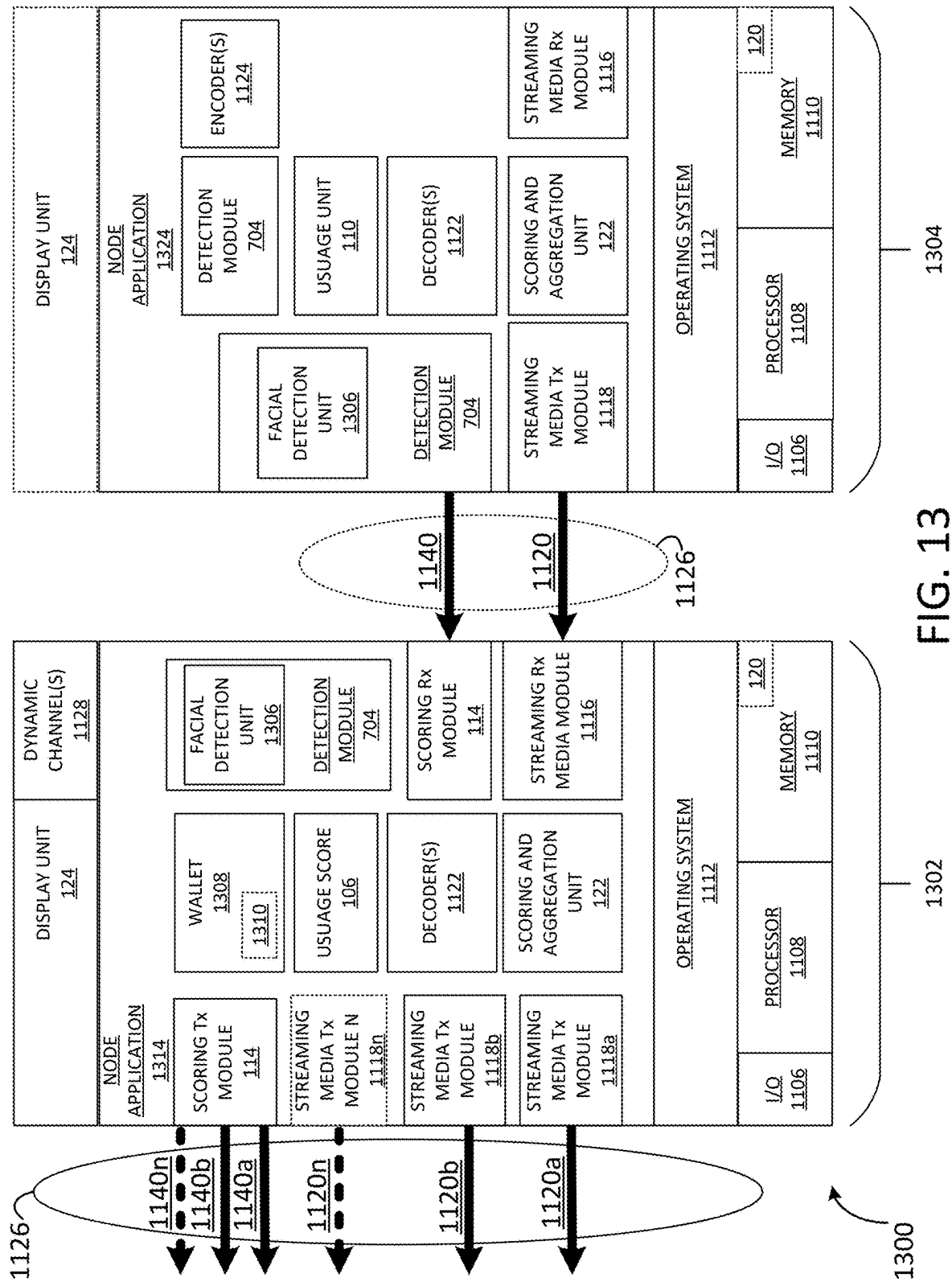


FIG. 11





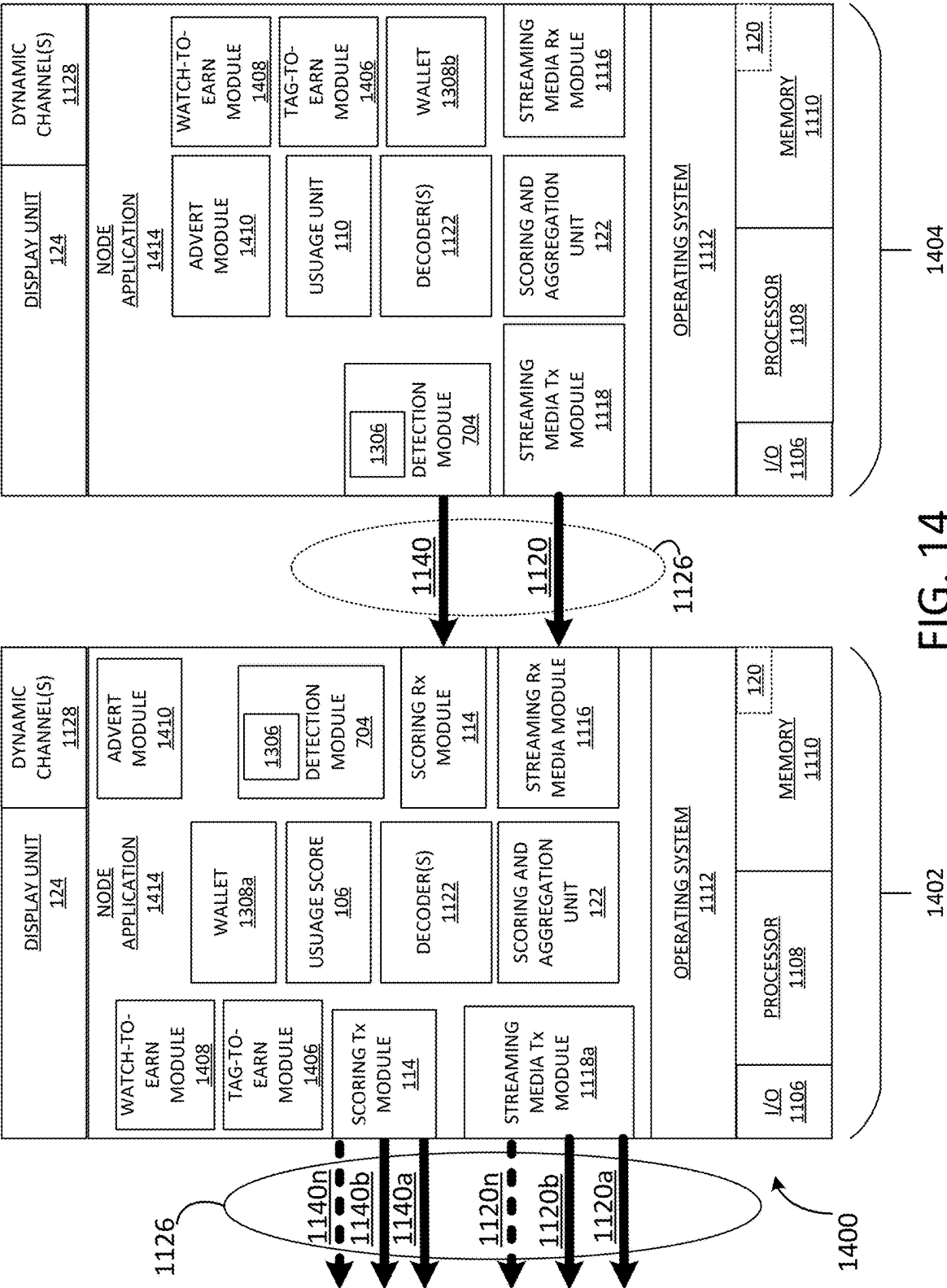
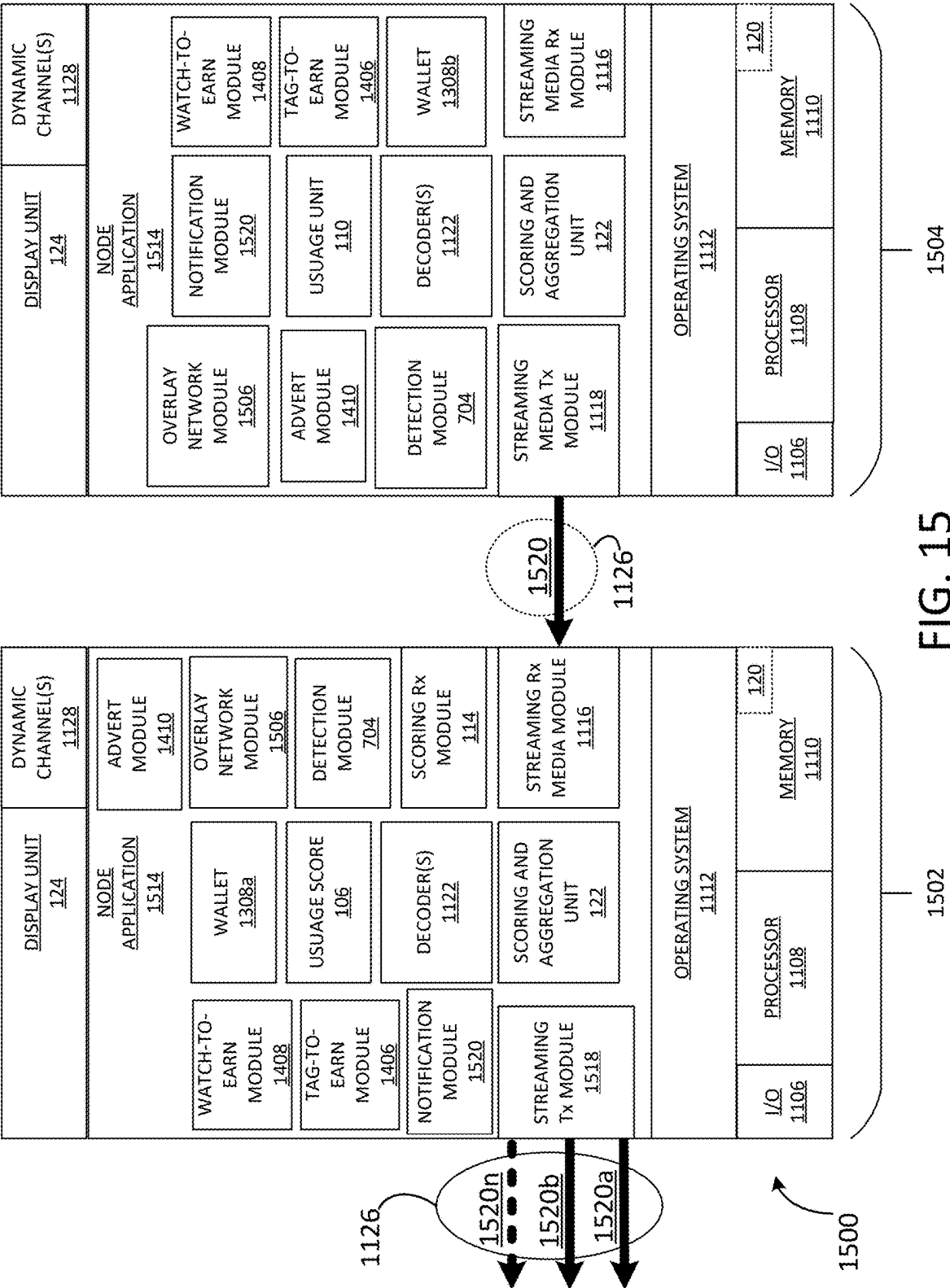


FIG. 14



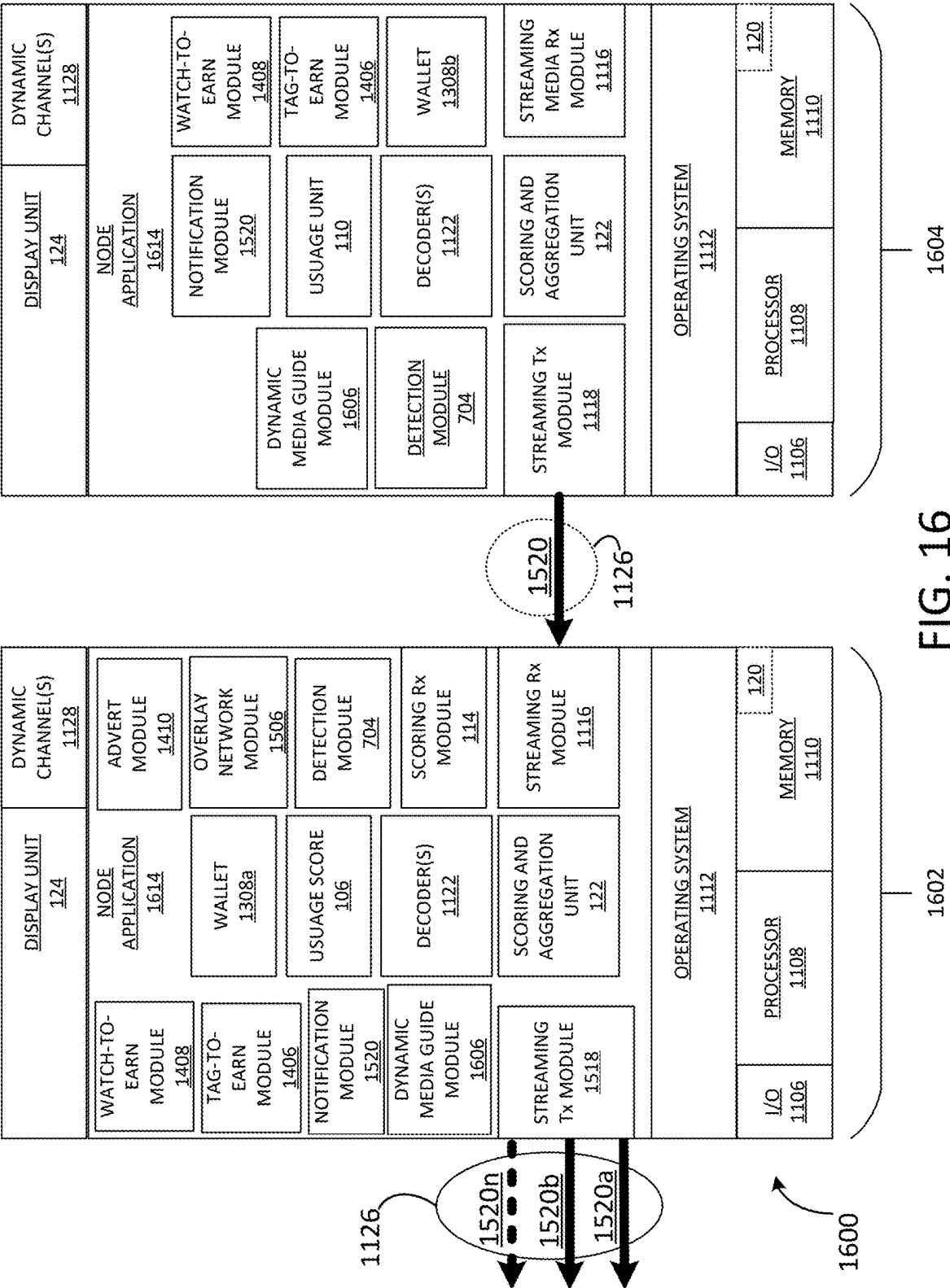


FIG. 16

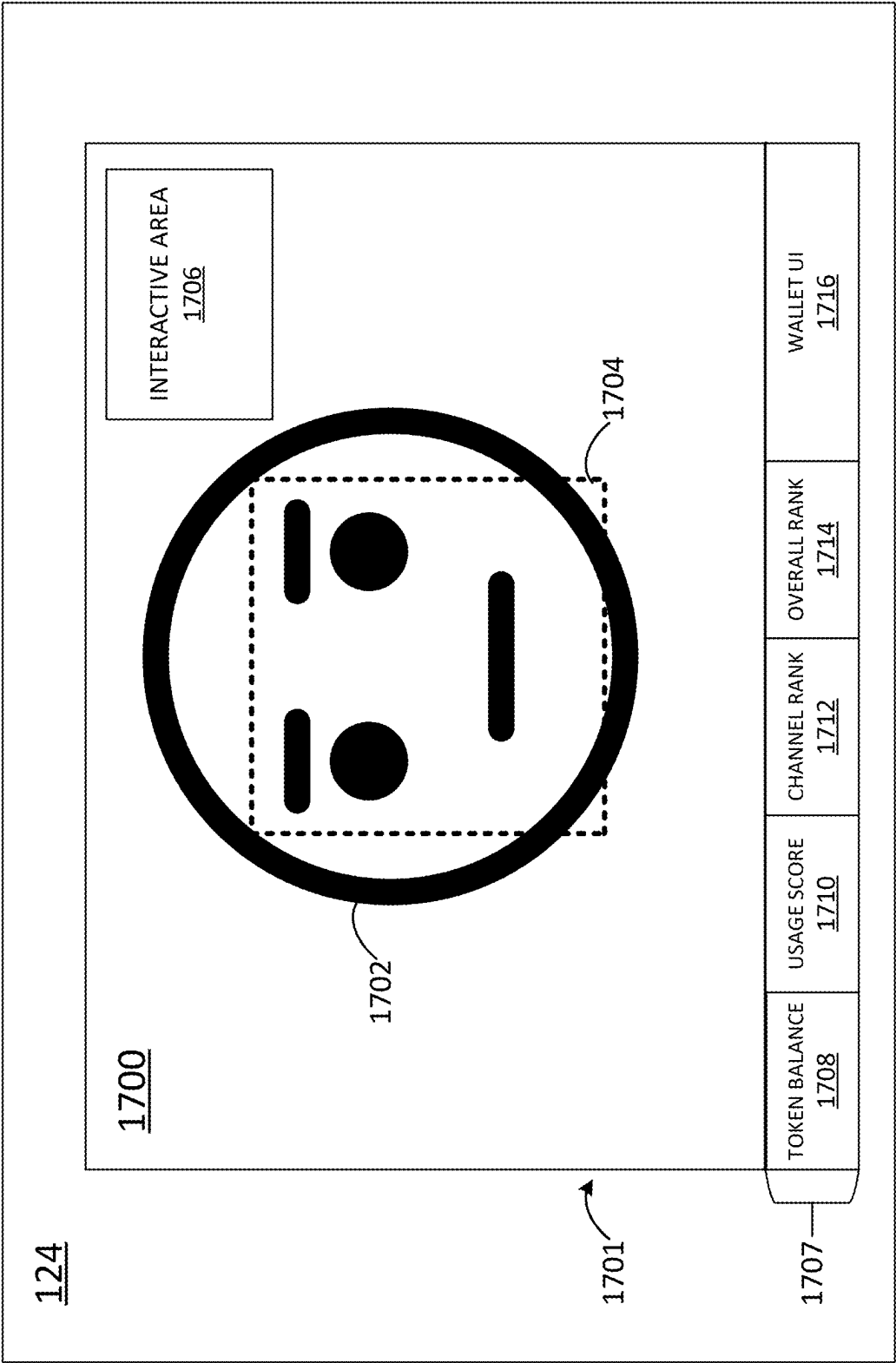


FIG. 17

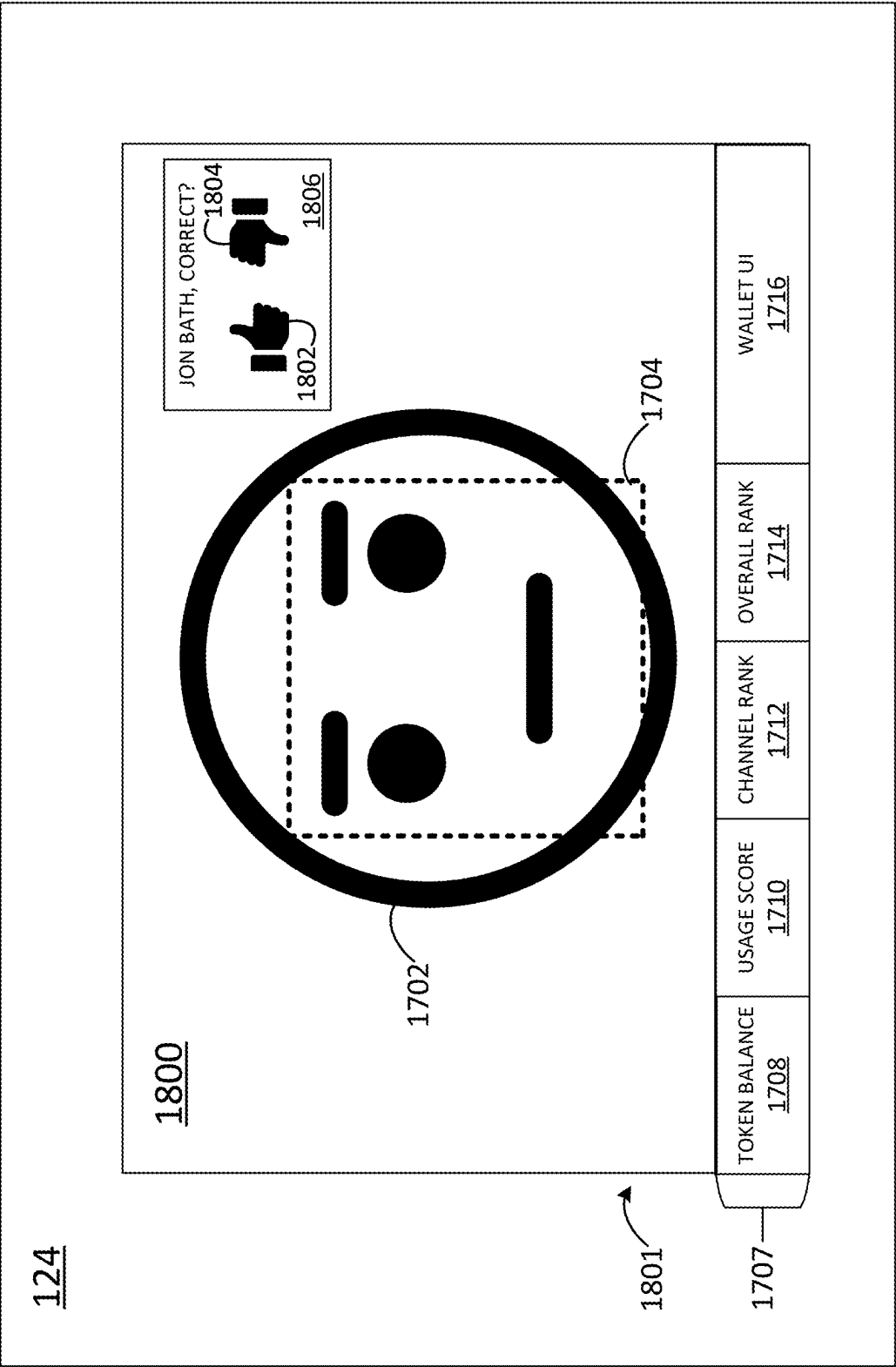


FIG. 18

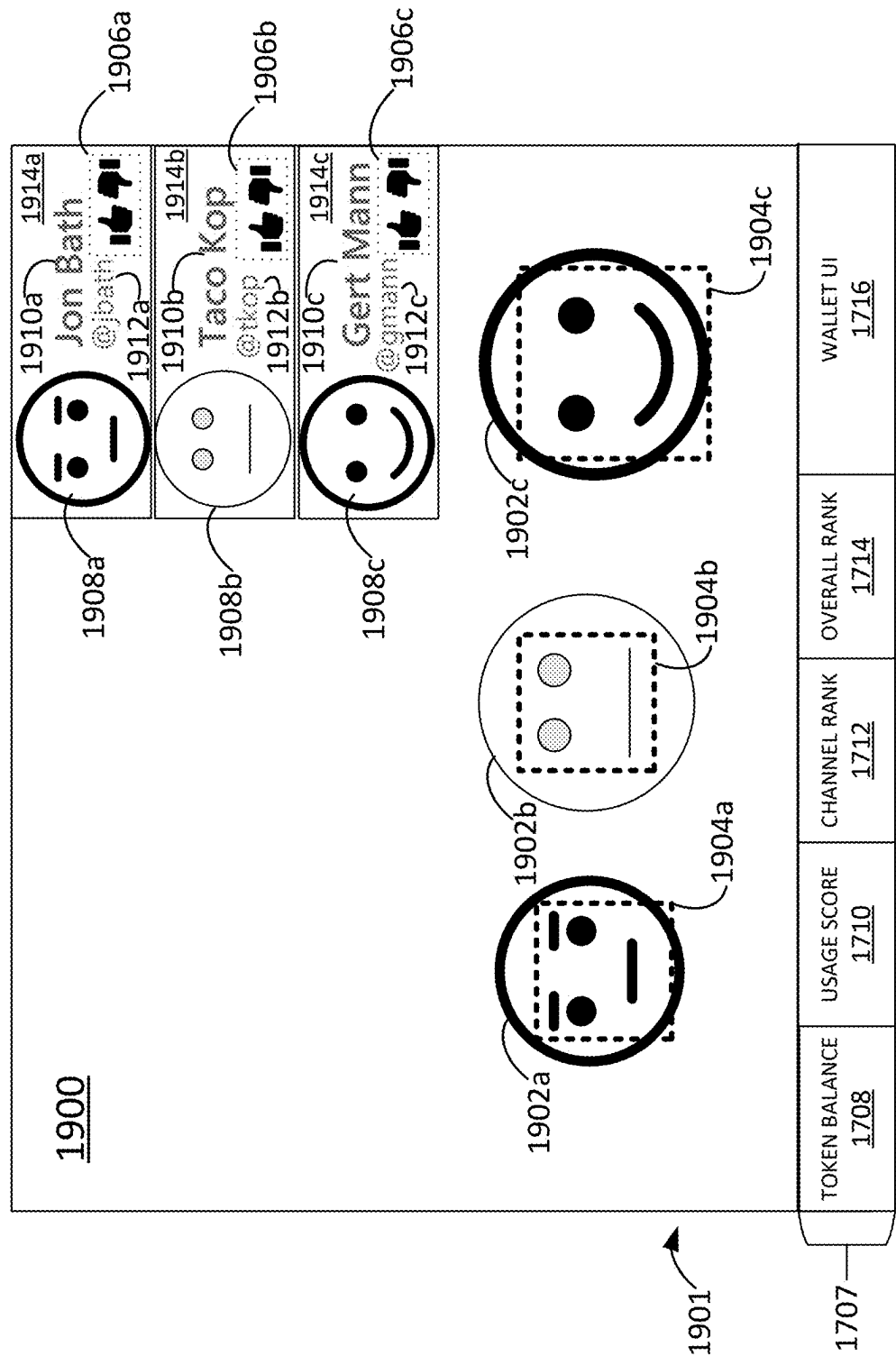


FIG. 19

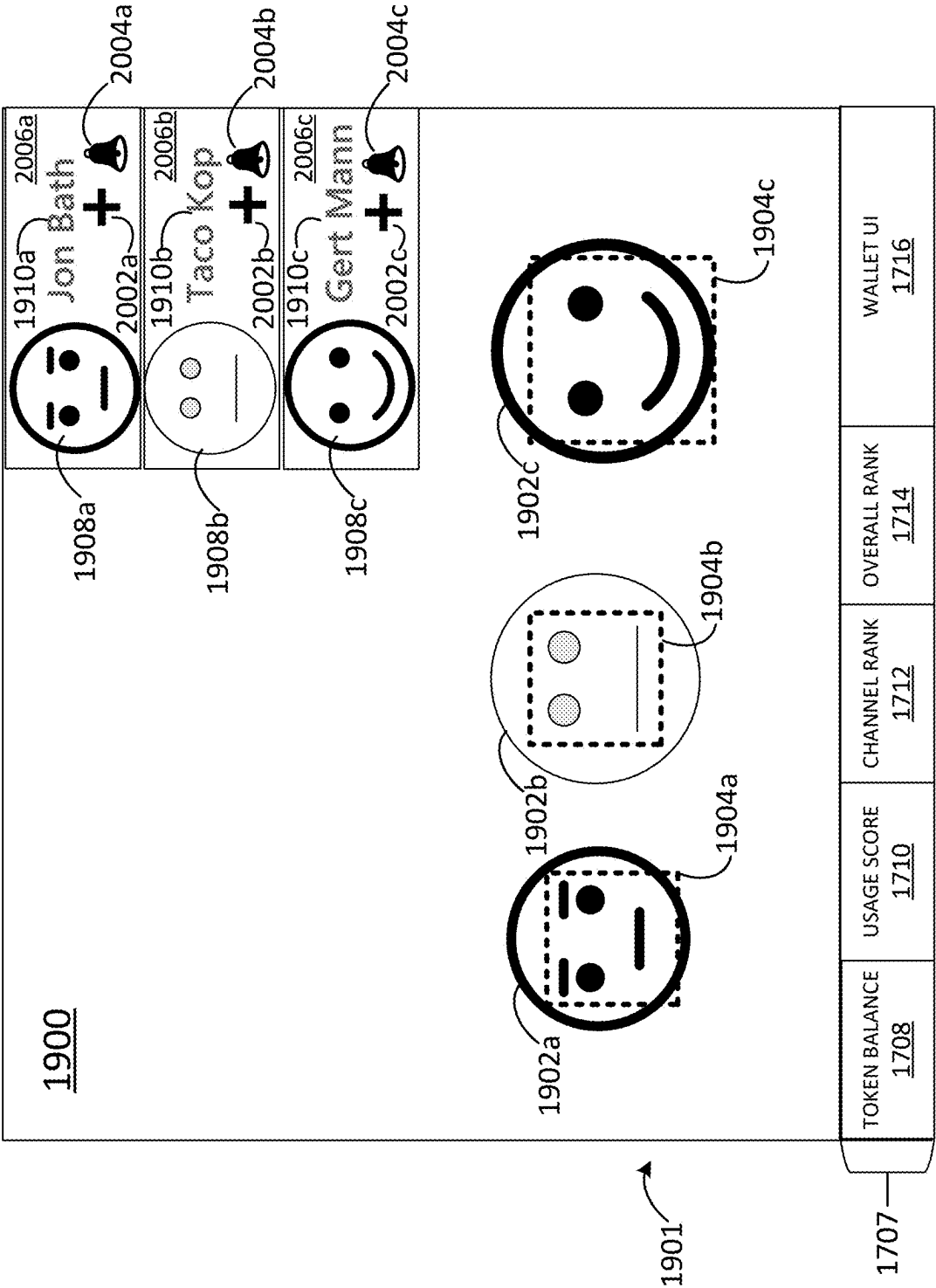


FIG. 20

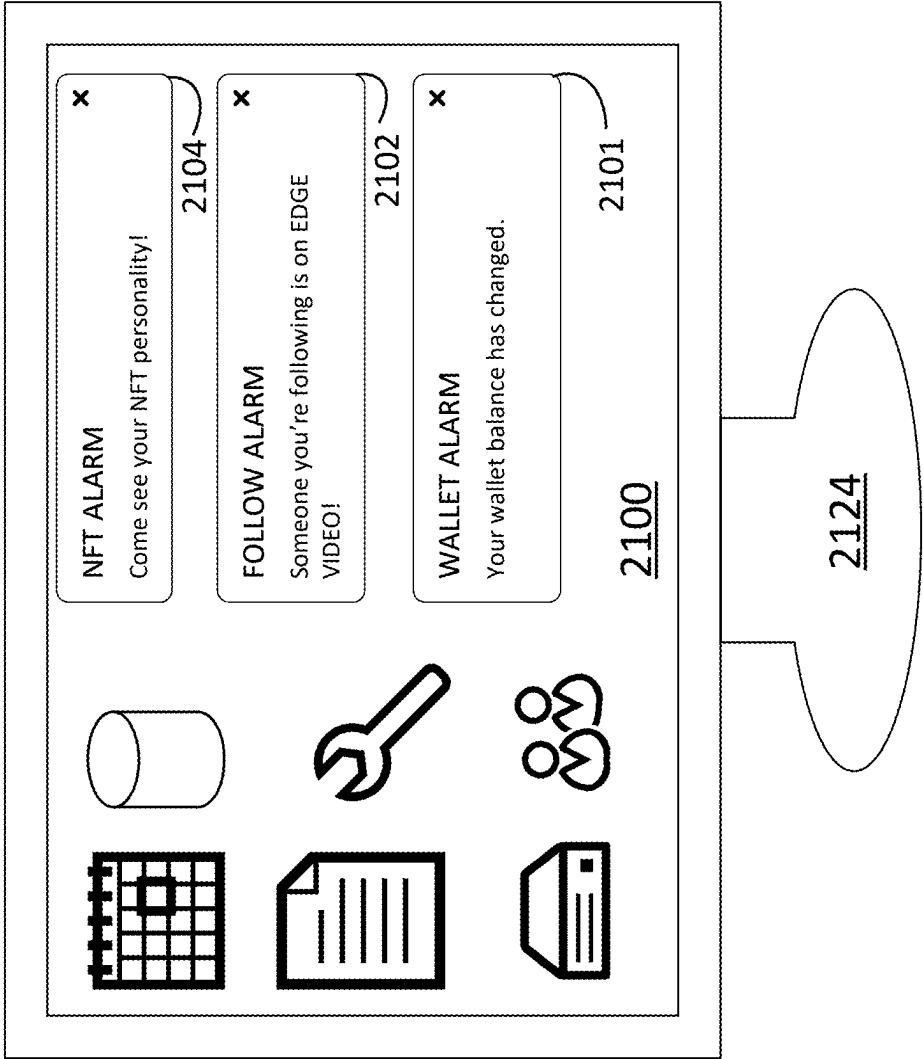


FIG. 21

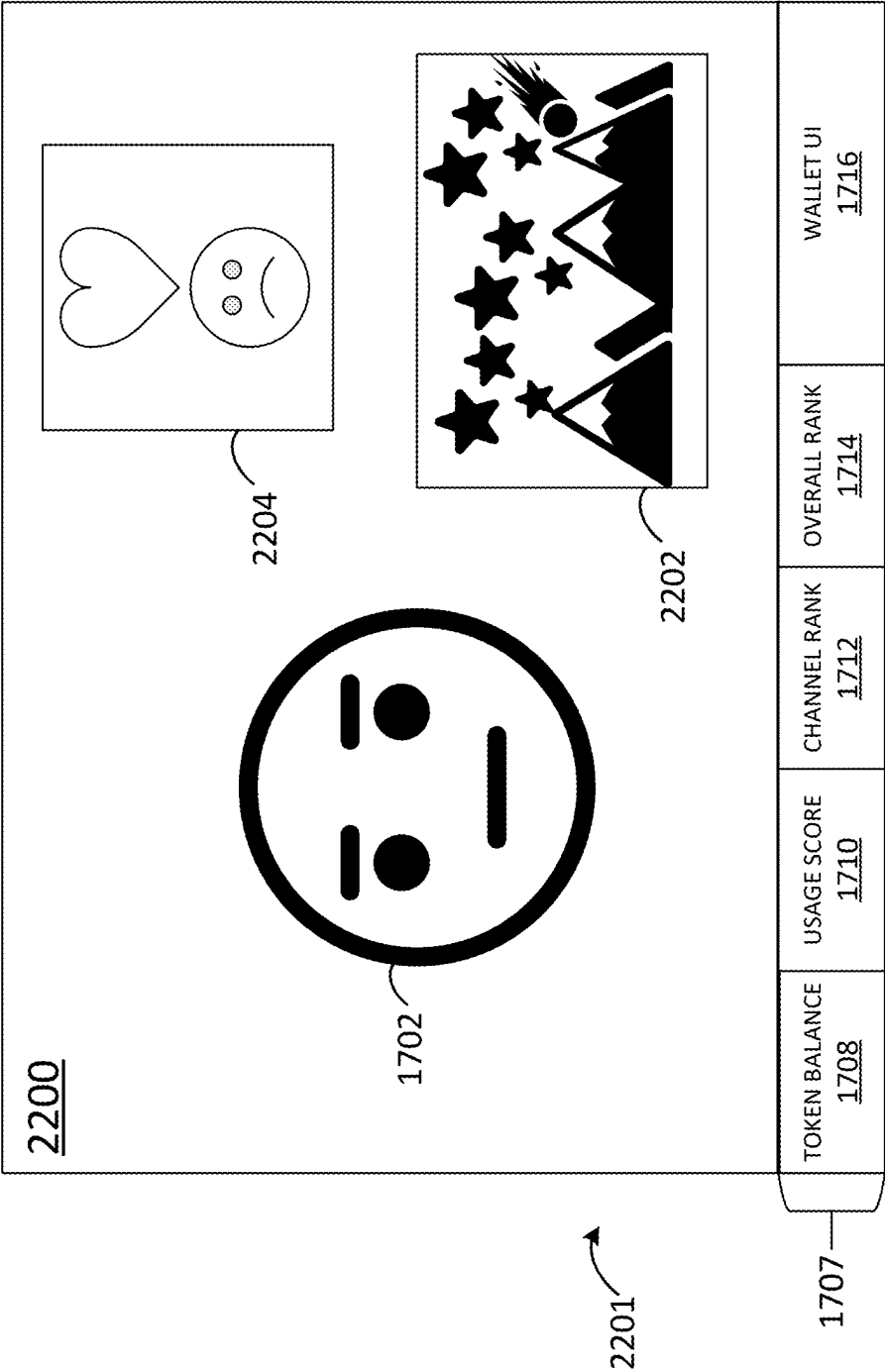


FIG. 22

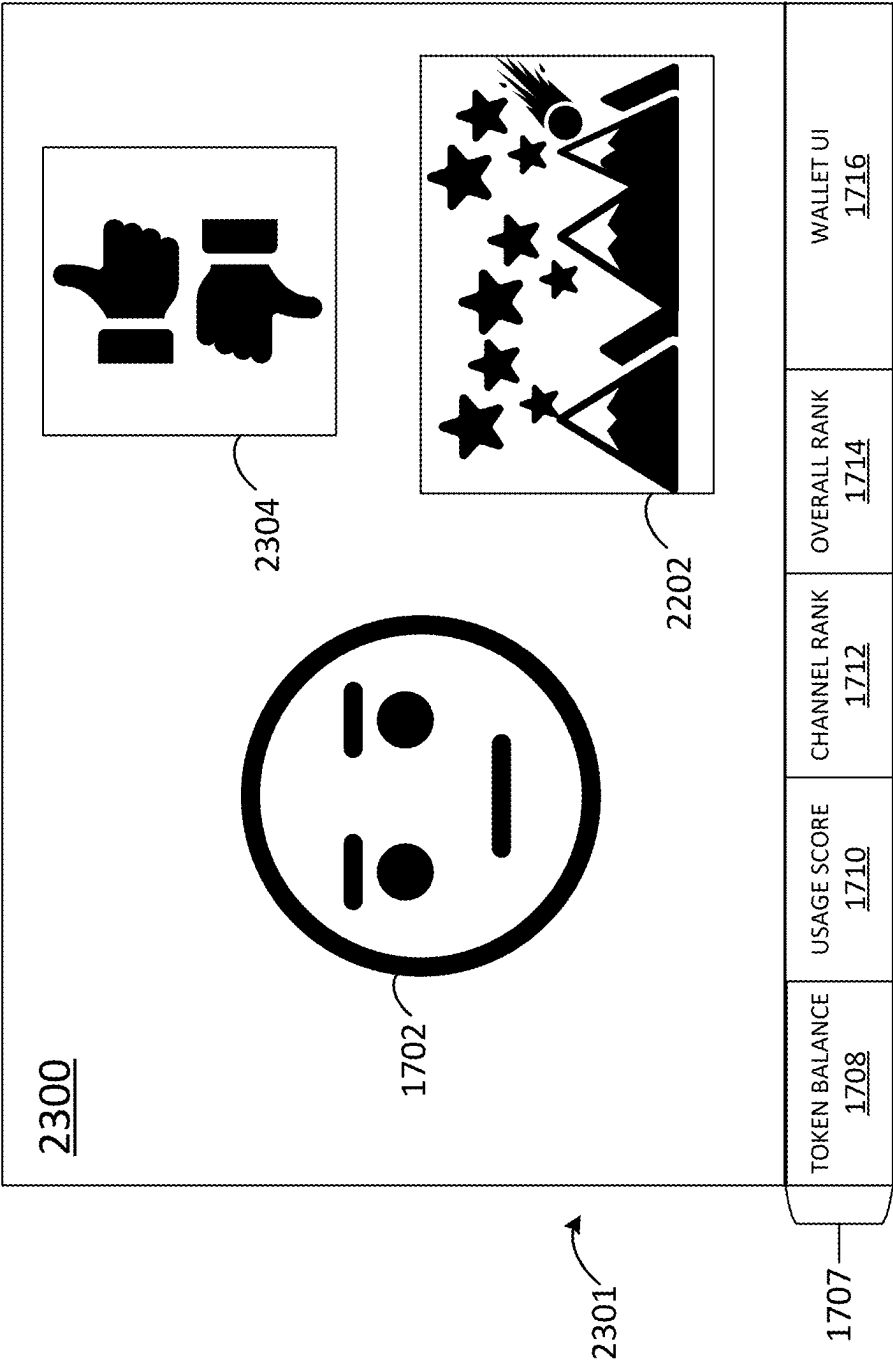


FIG. 23

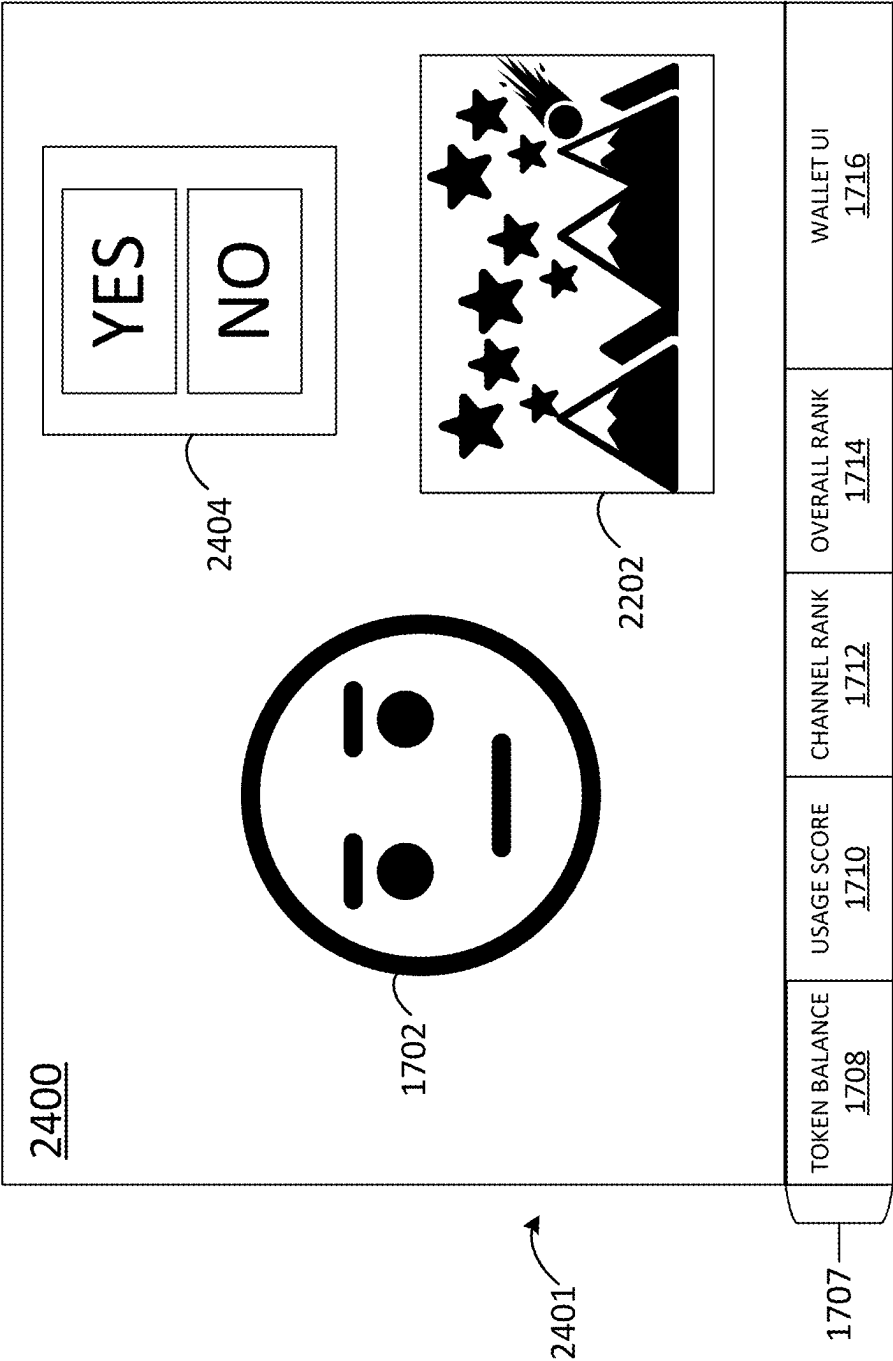


FIG. 24

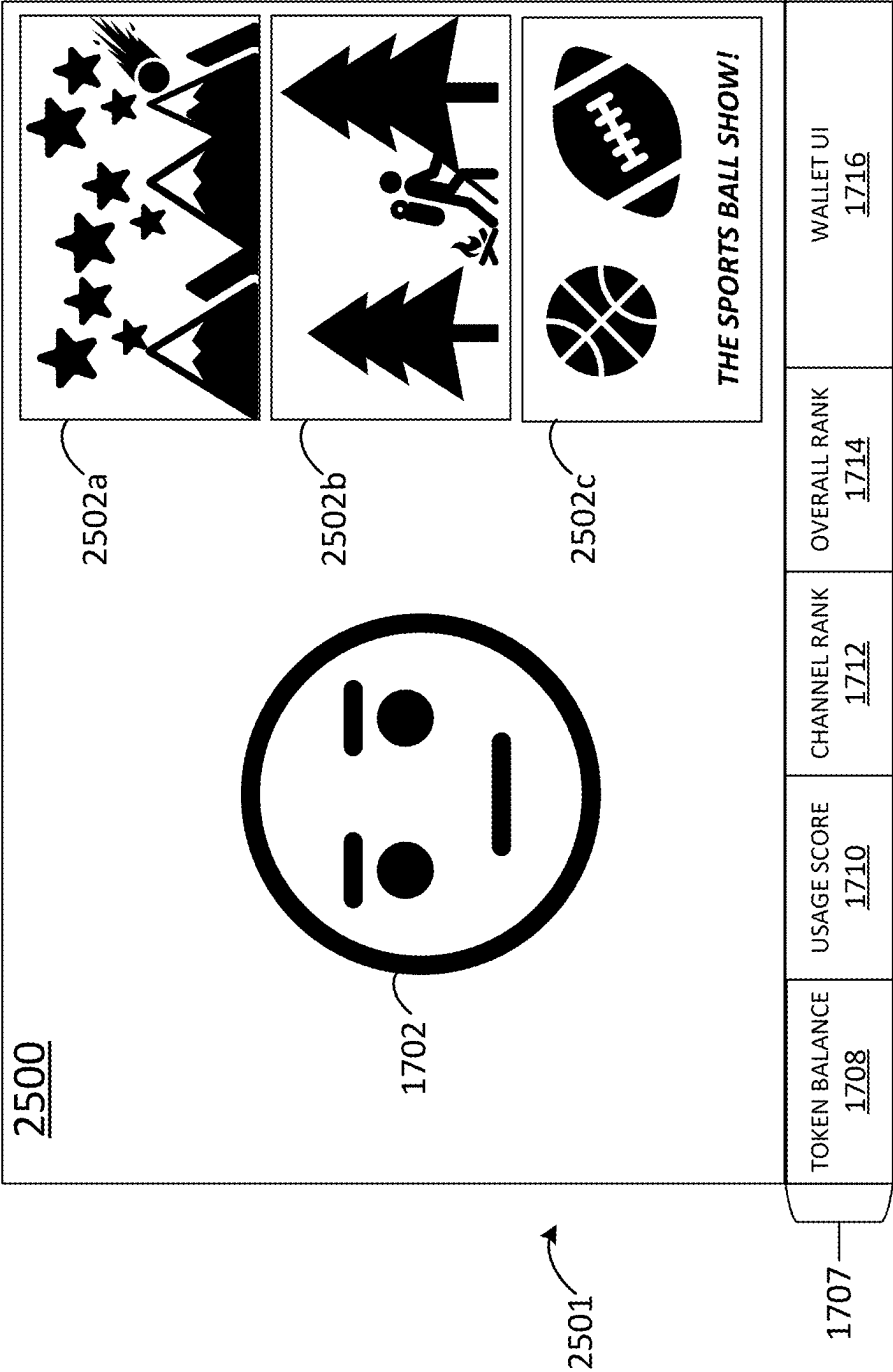


FIG. 25

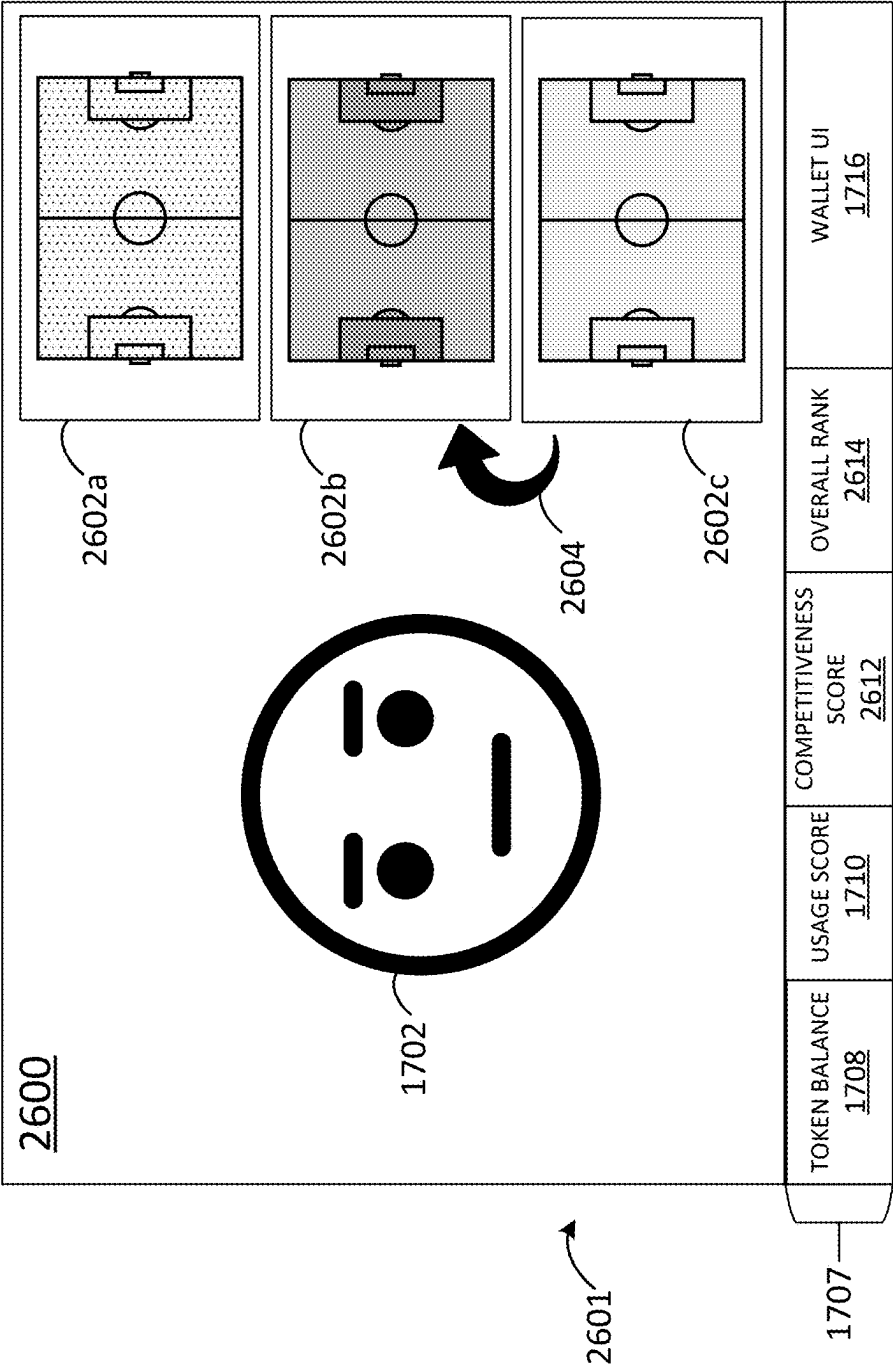


FIG. 26

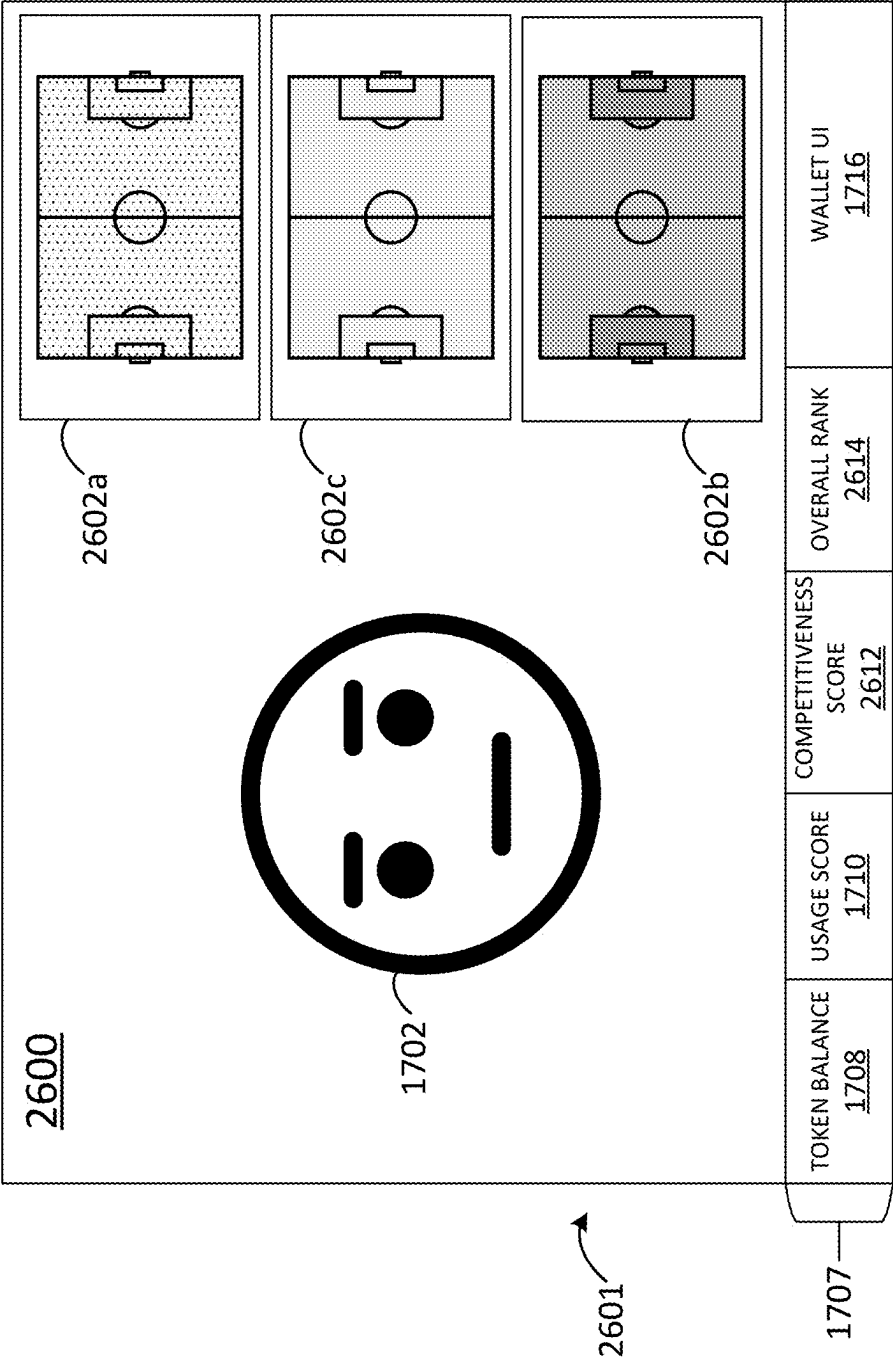


FIG. 27

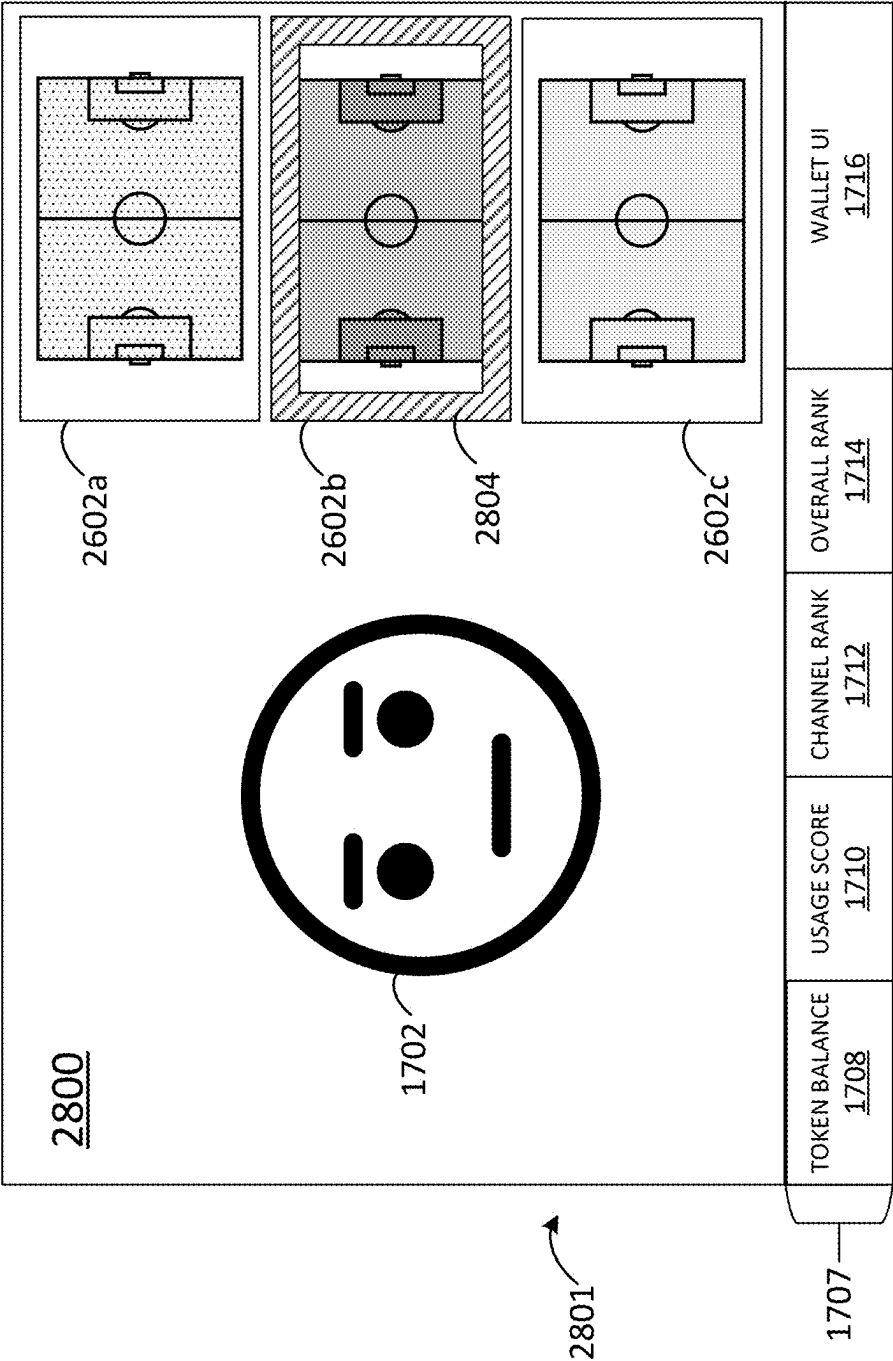
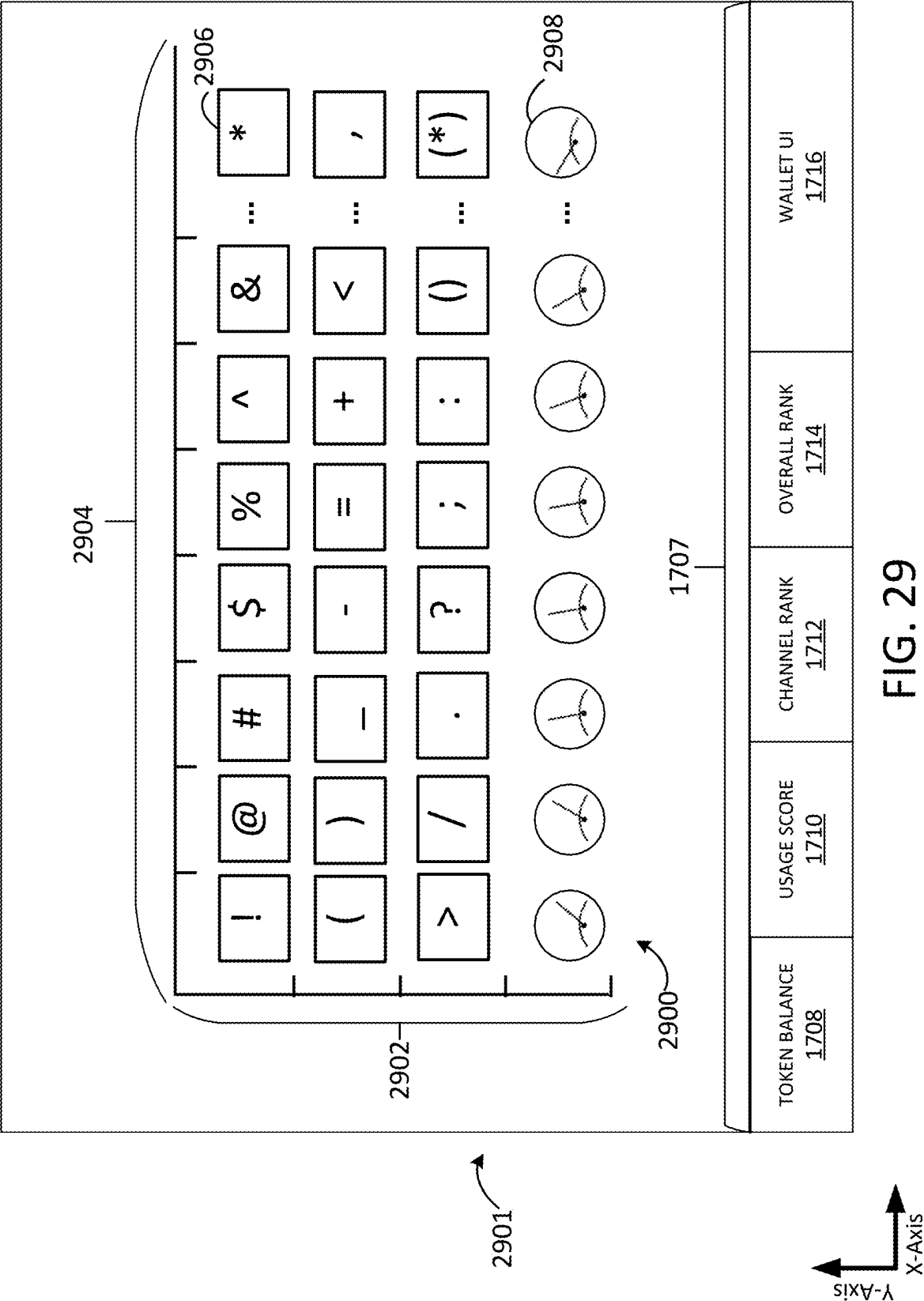
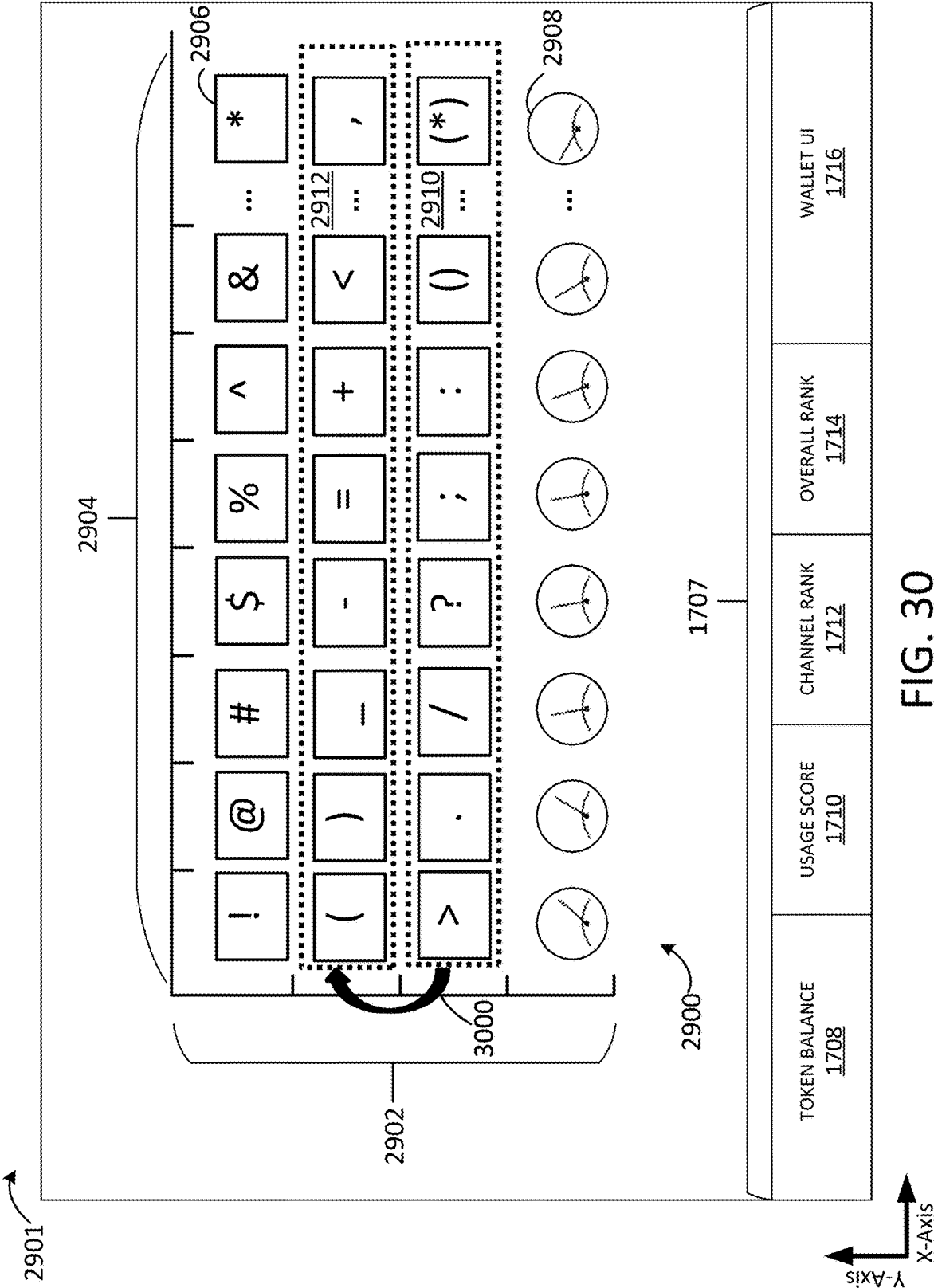


FIG. 28





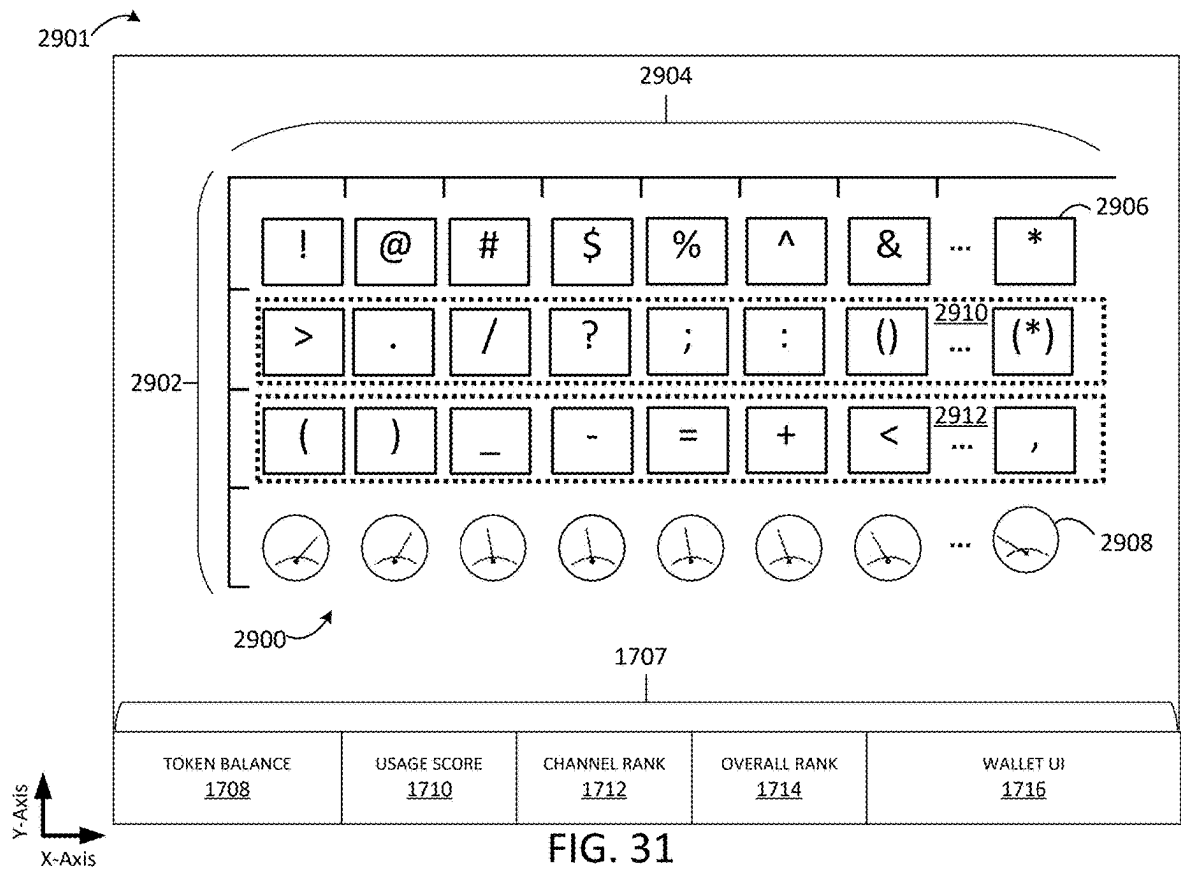
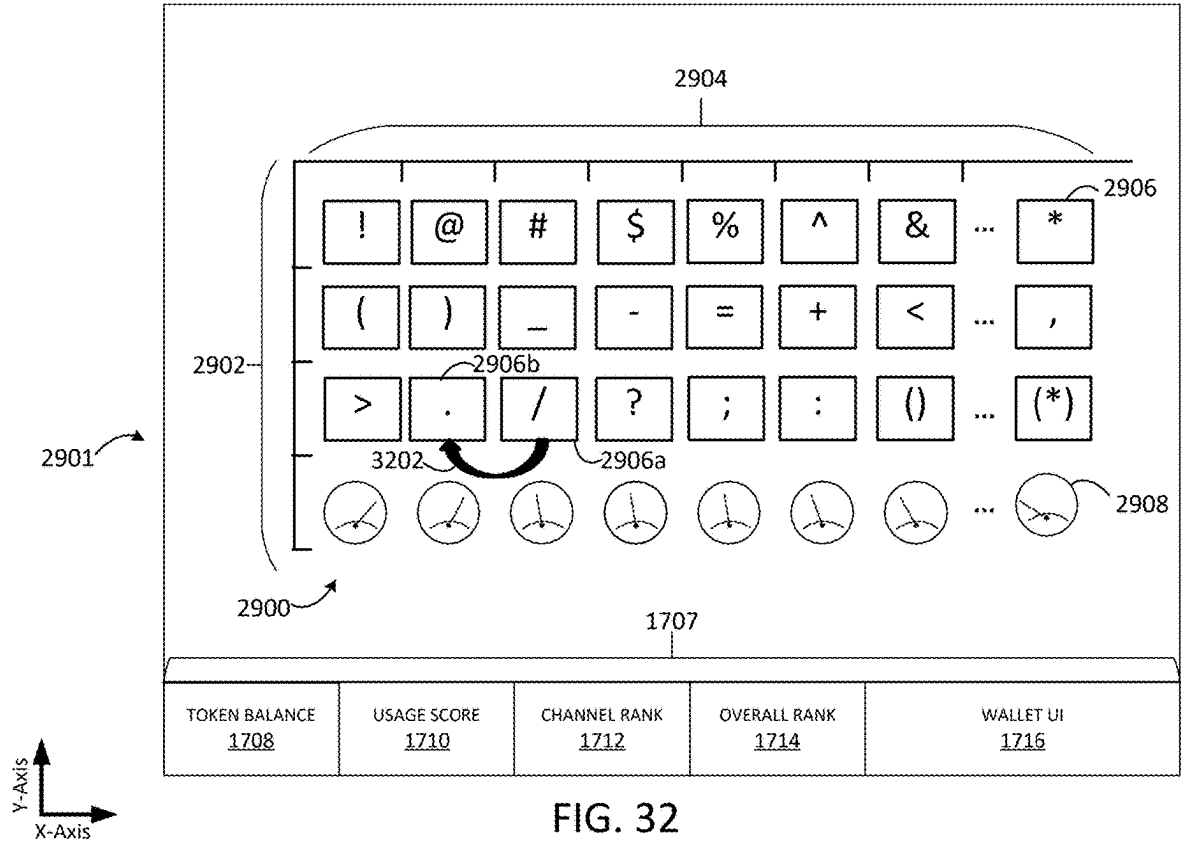


FIG. 31



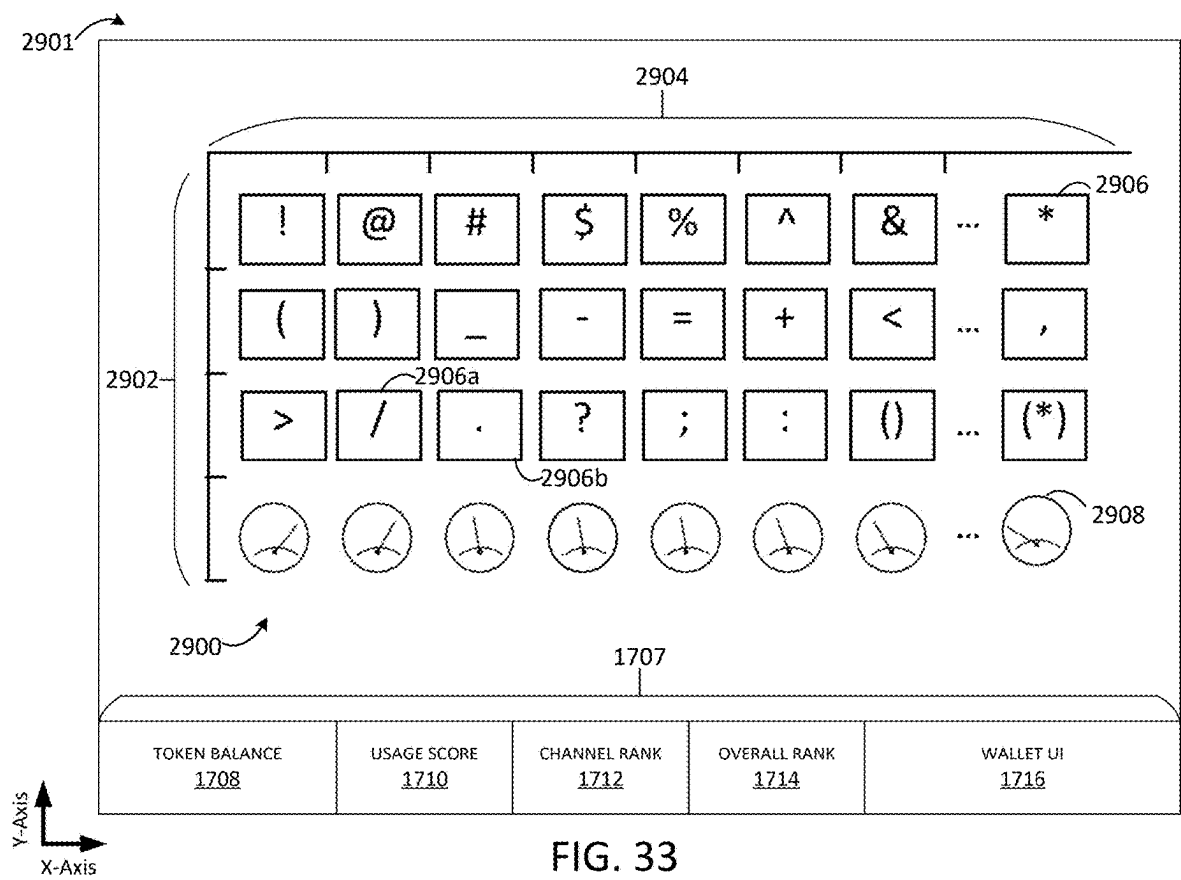
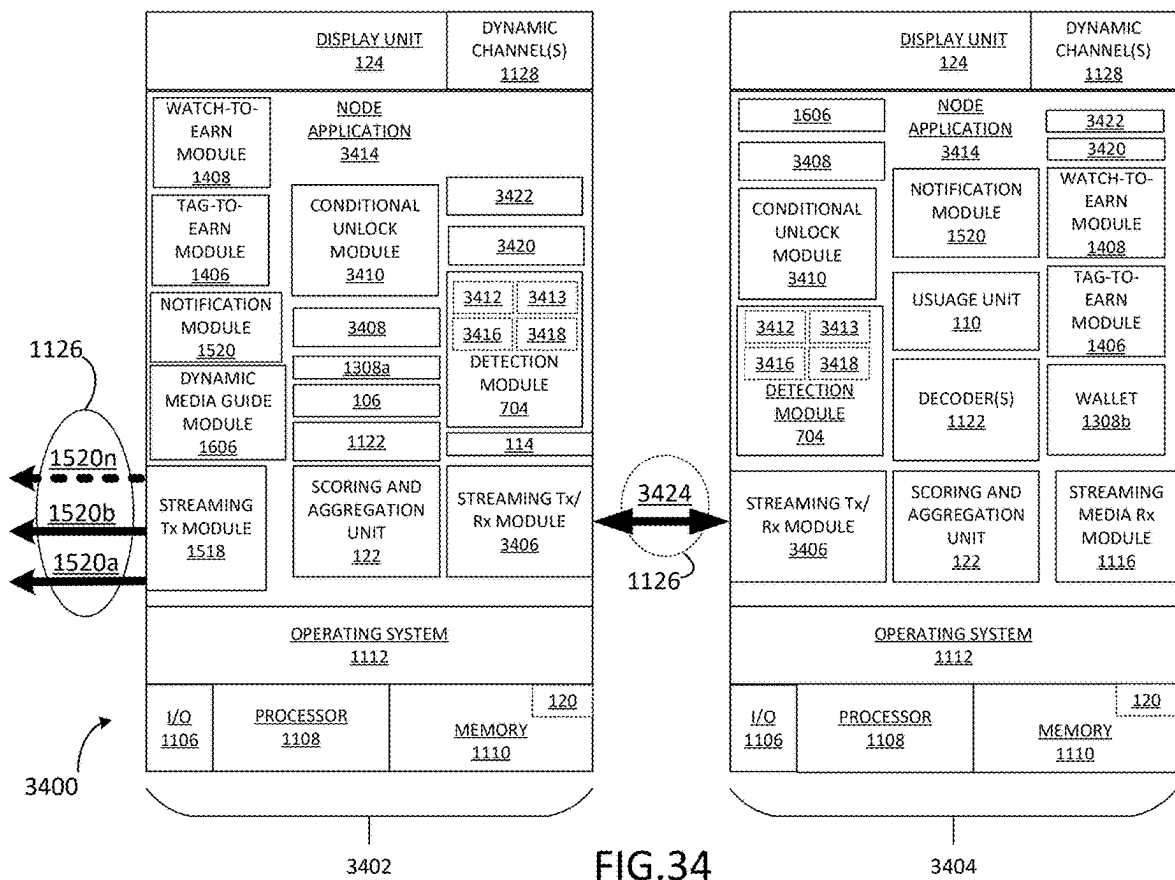


FIG. 33



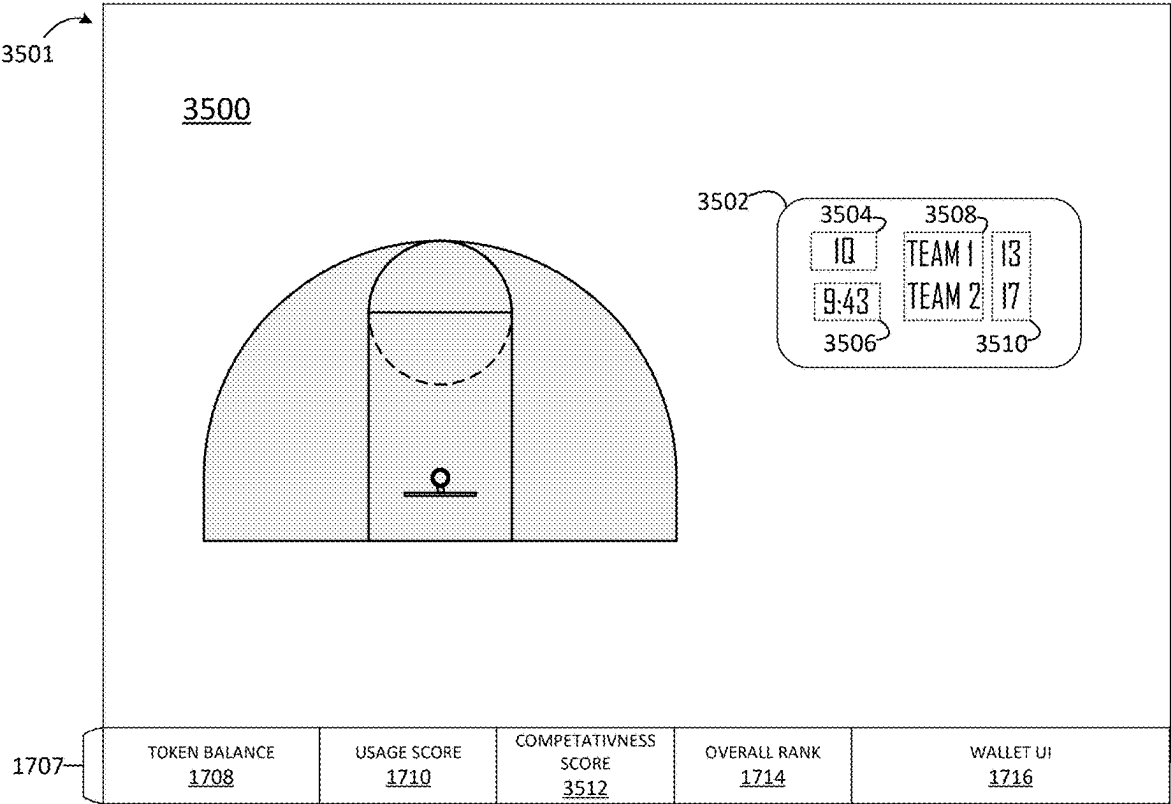


FIG. 35

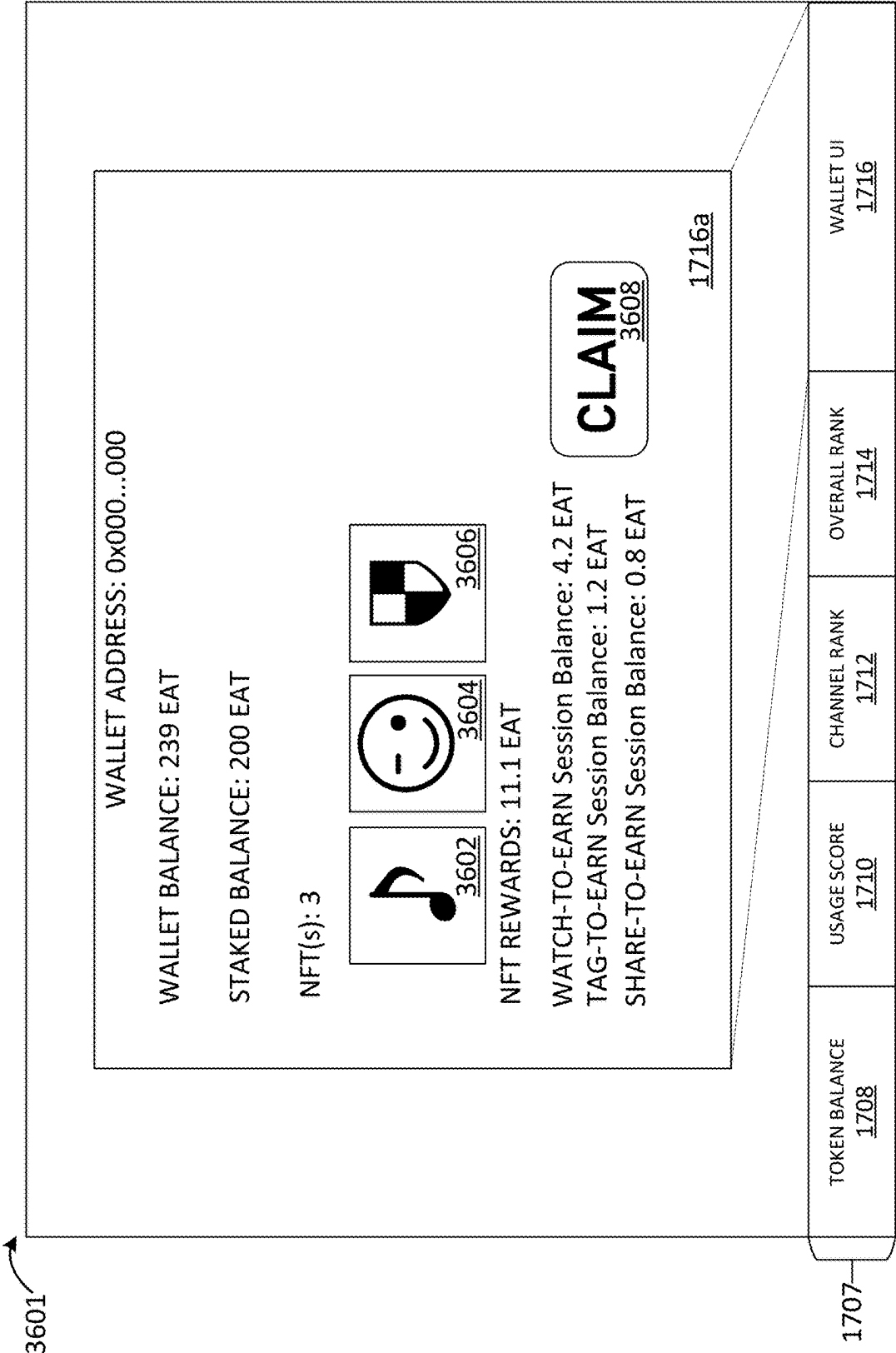
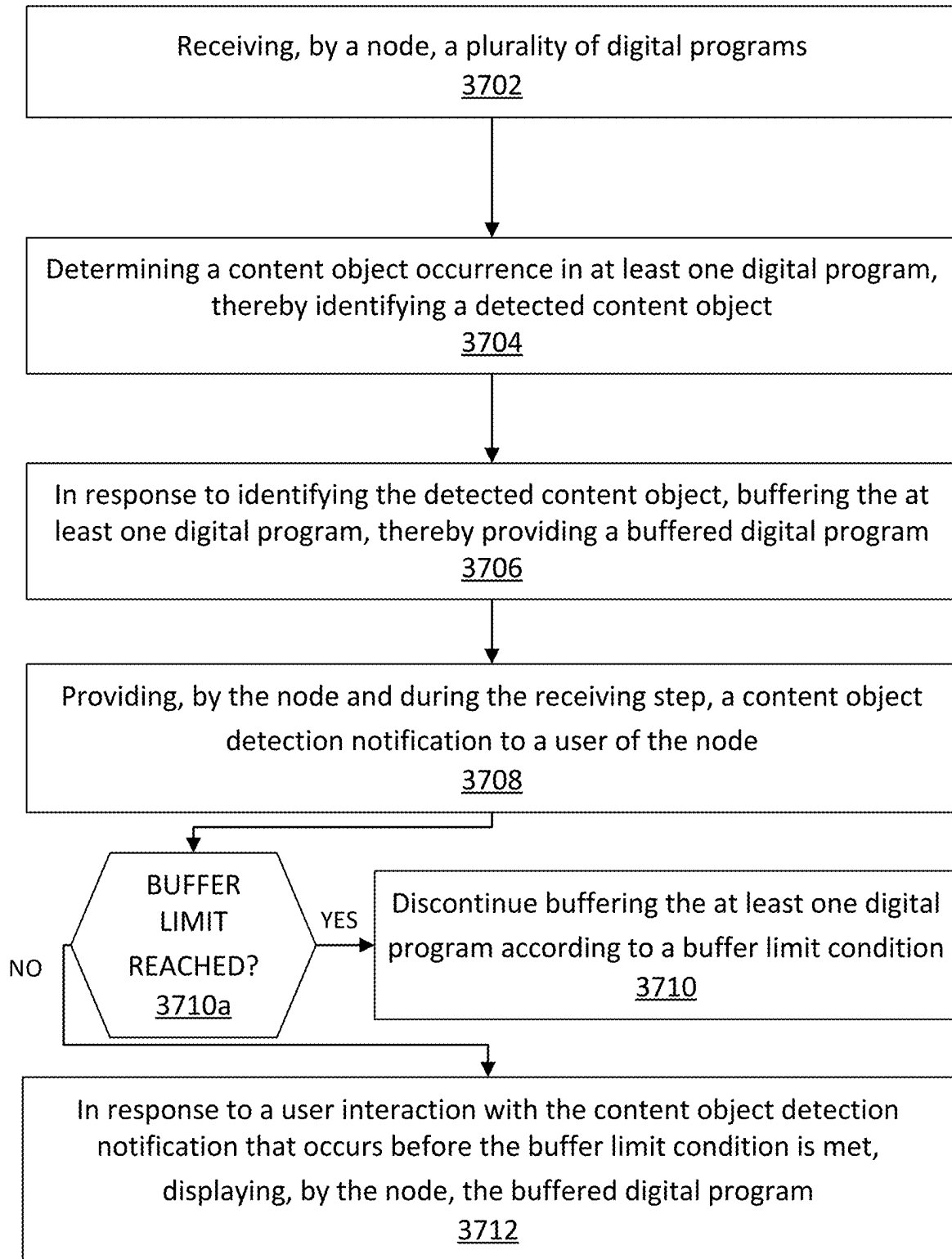
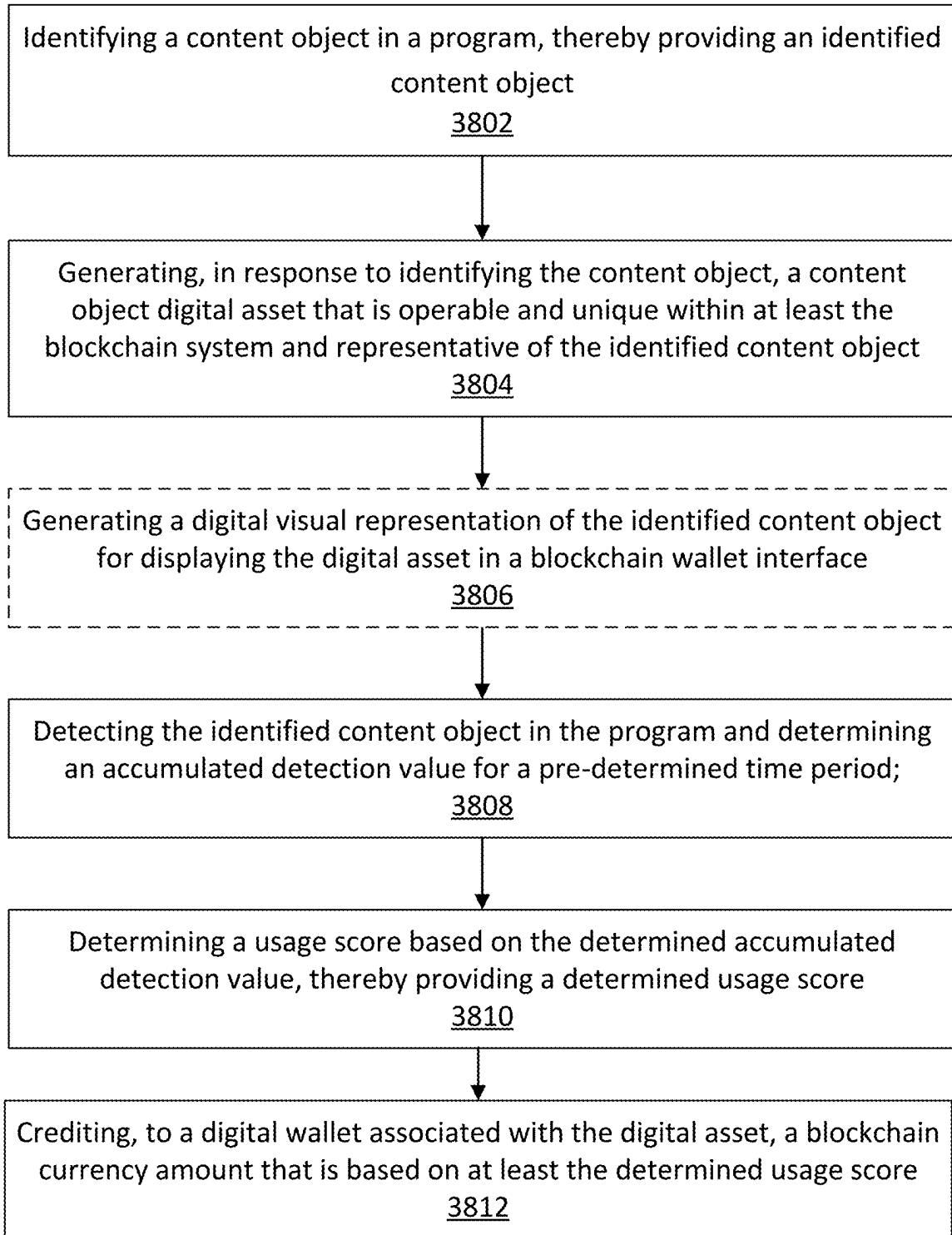


FIG. 36



3700

FIG. 37



3800 ↗

FIG. 38

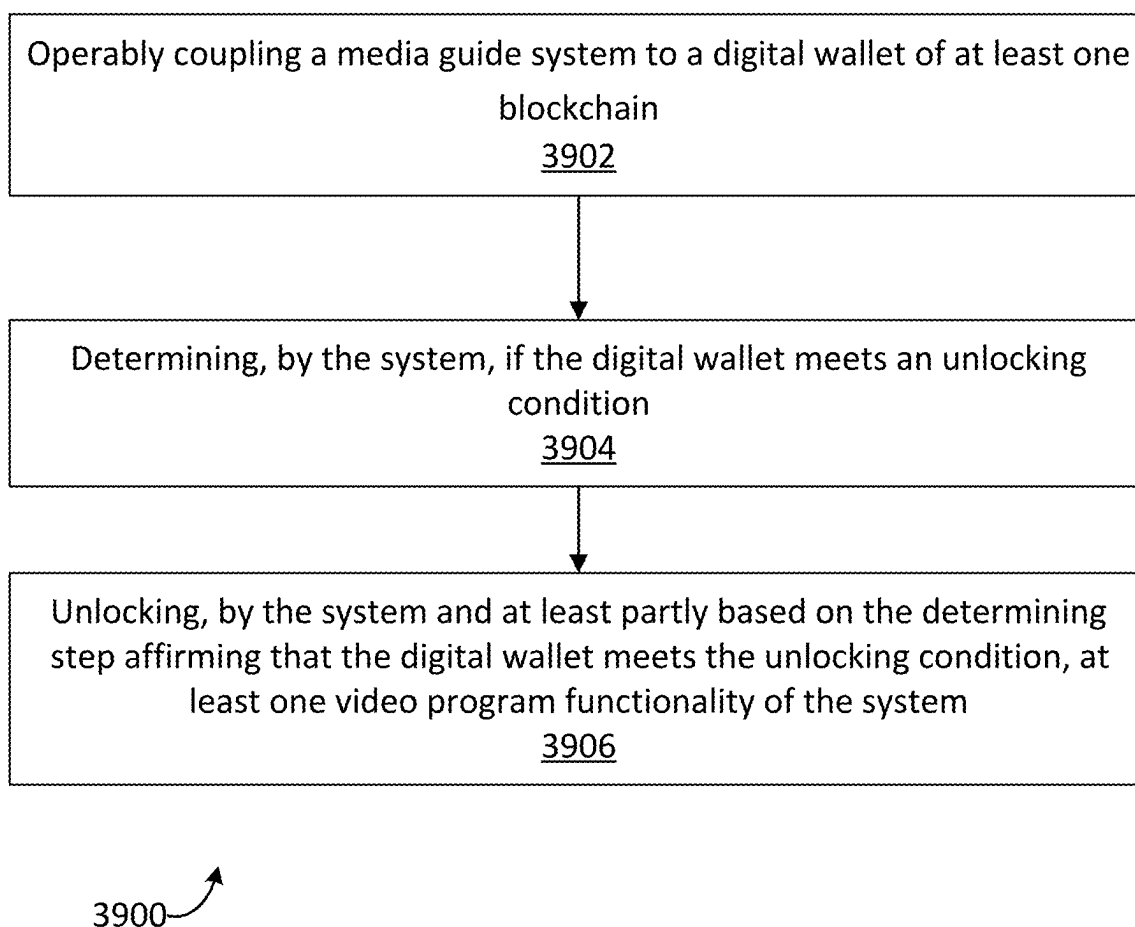


FIG. 39

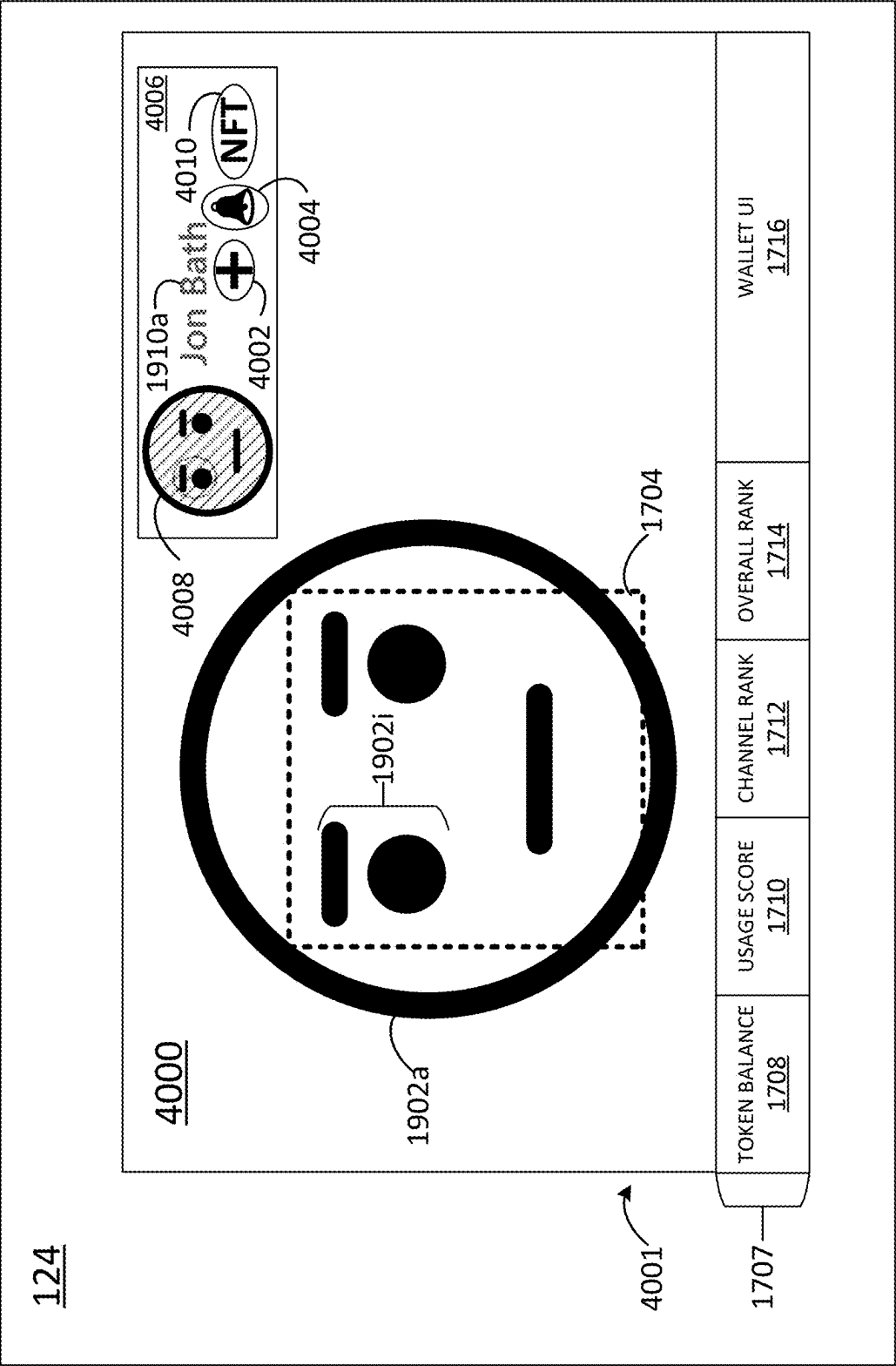


FIG. 40

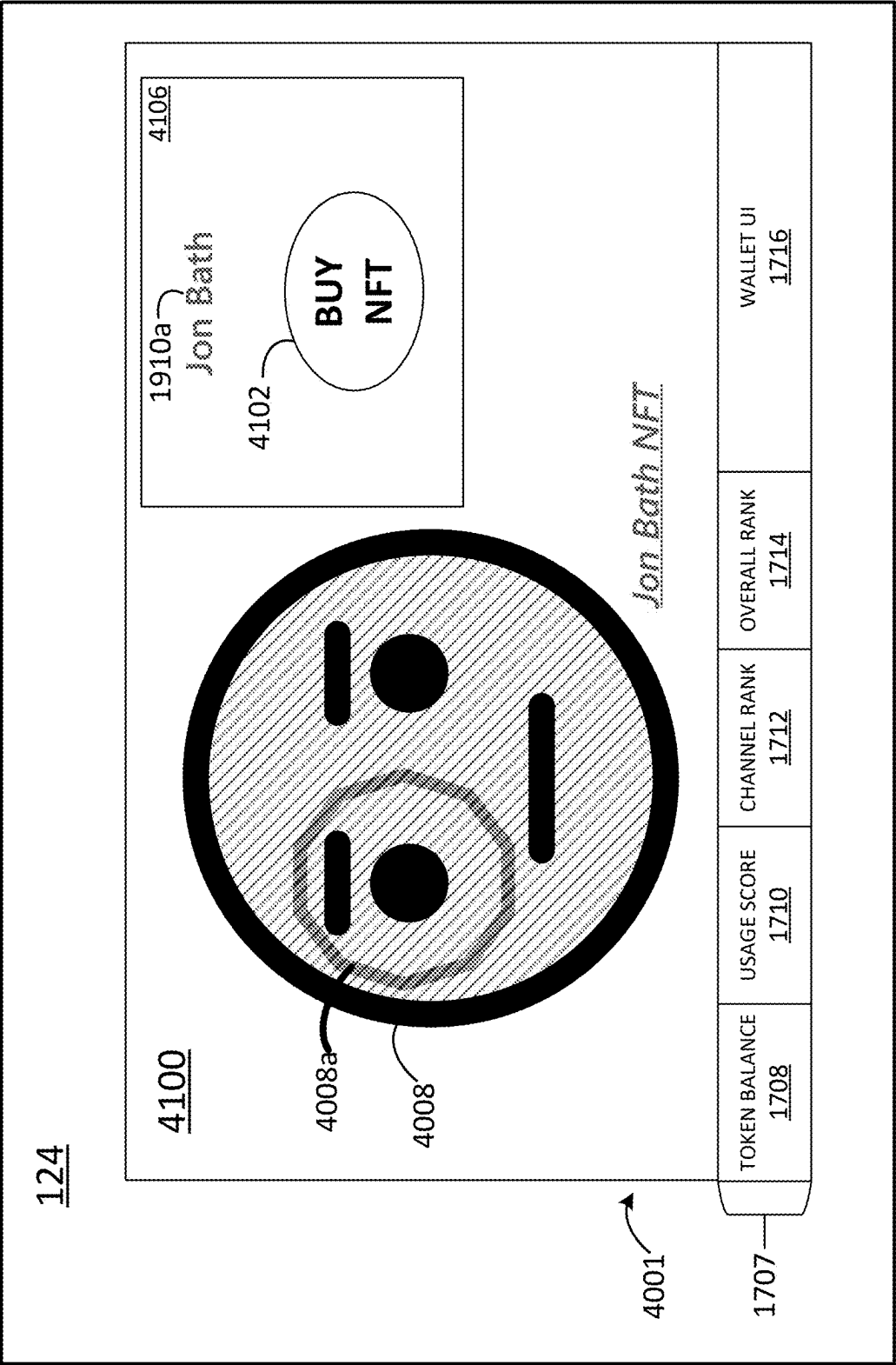
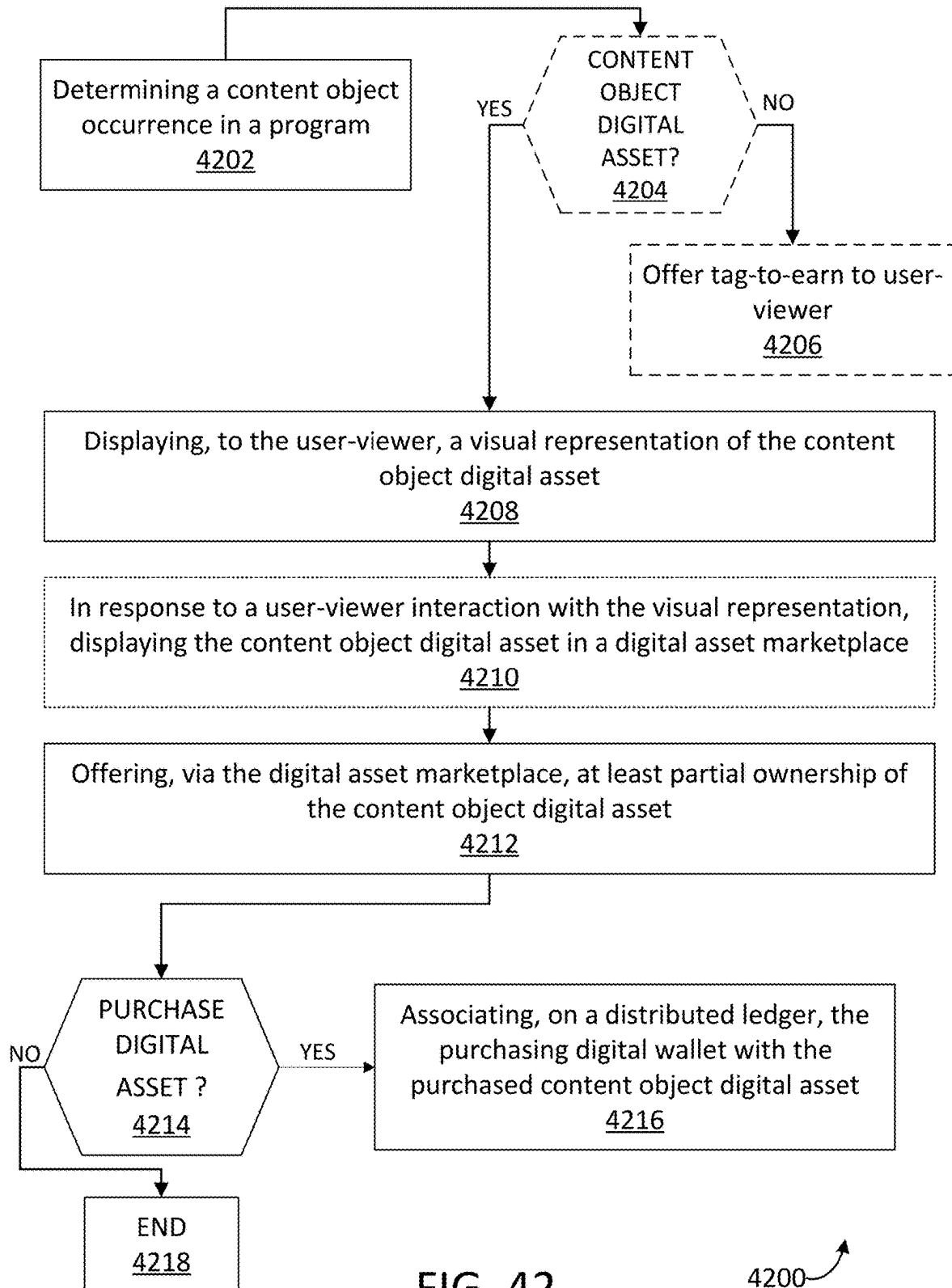


FIG. 41



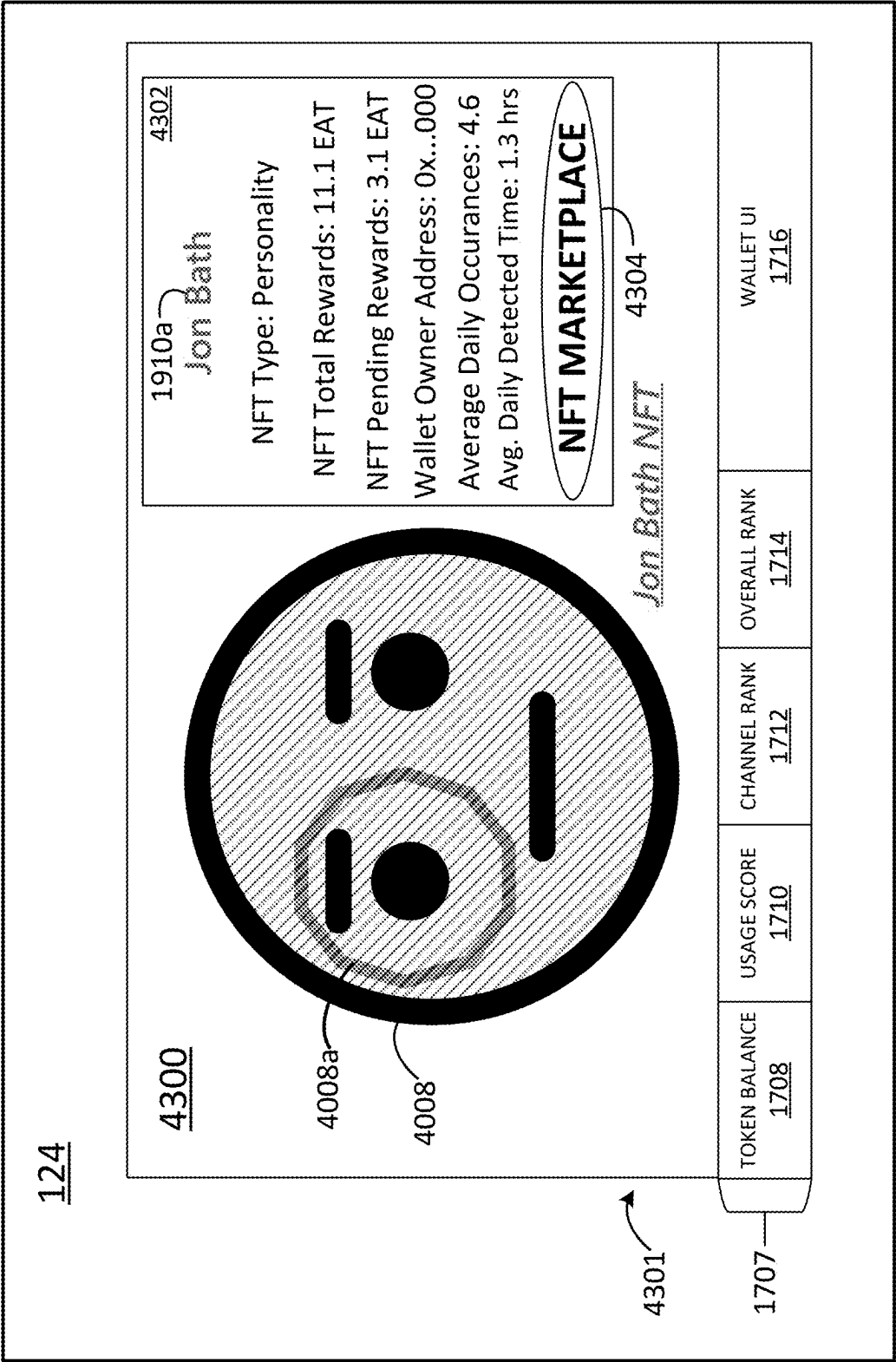
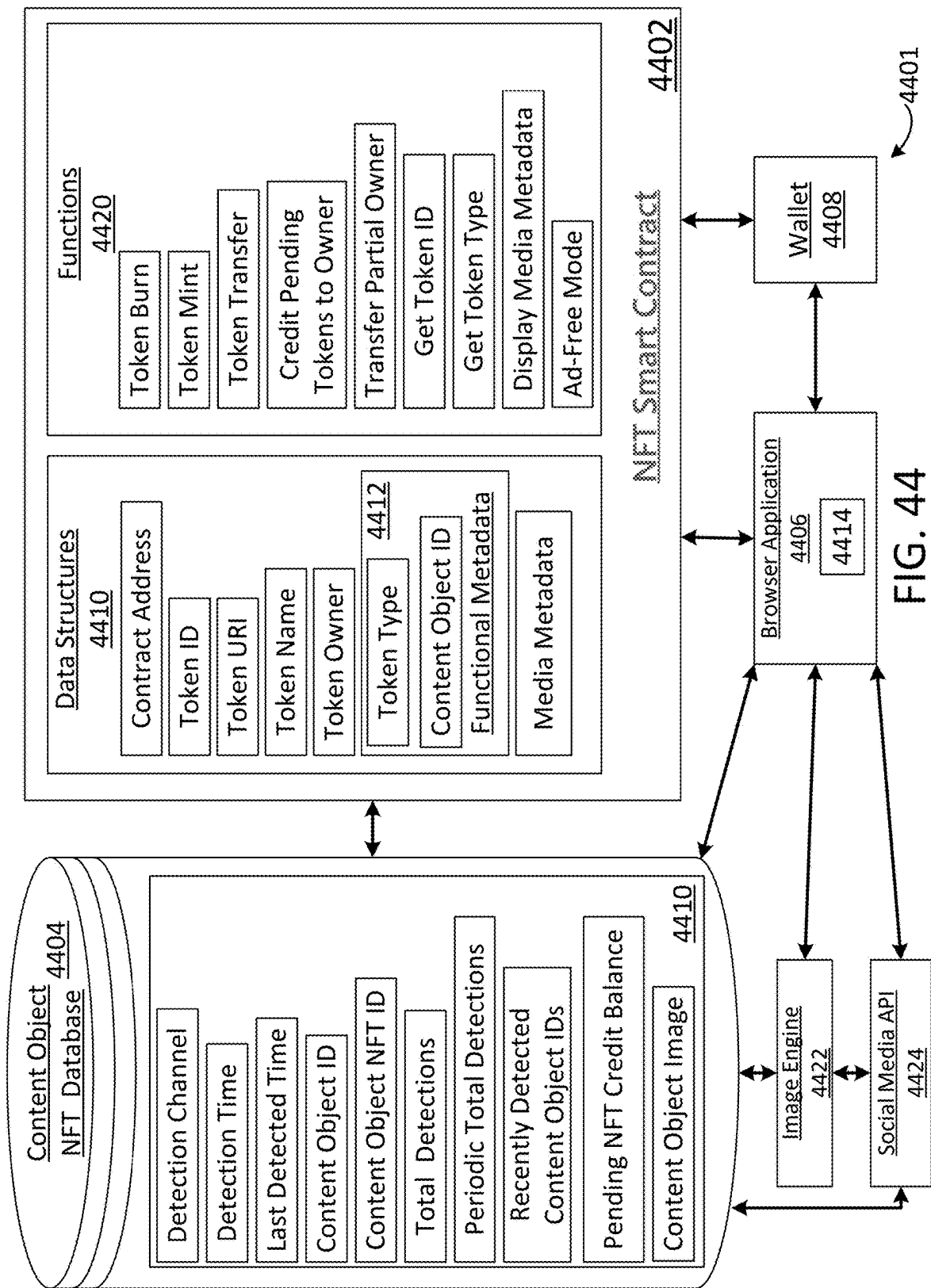


FIG. 43



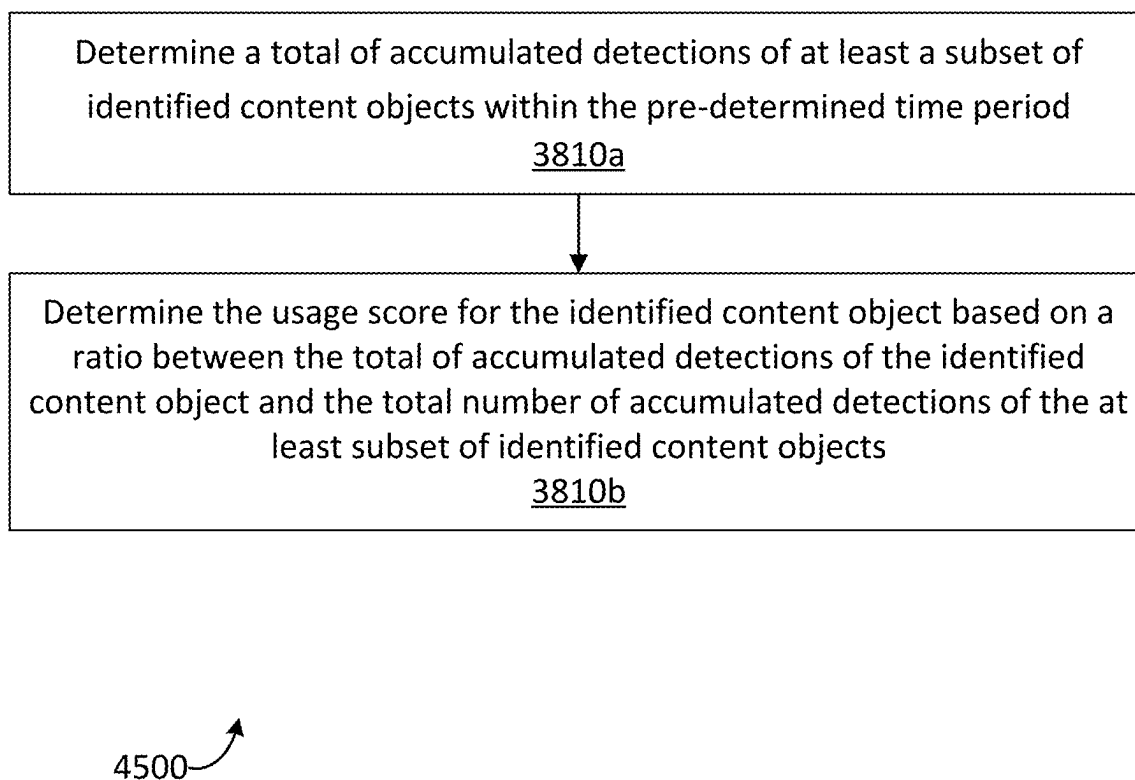


FIG. 45

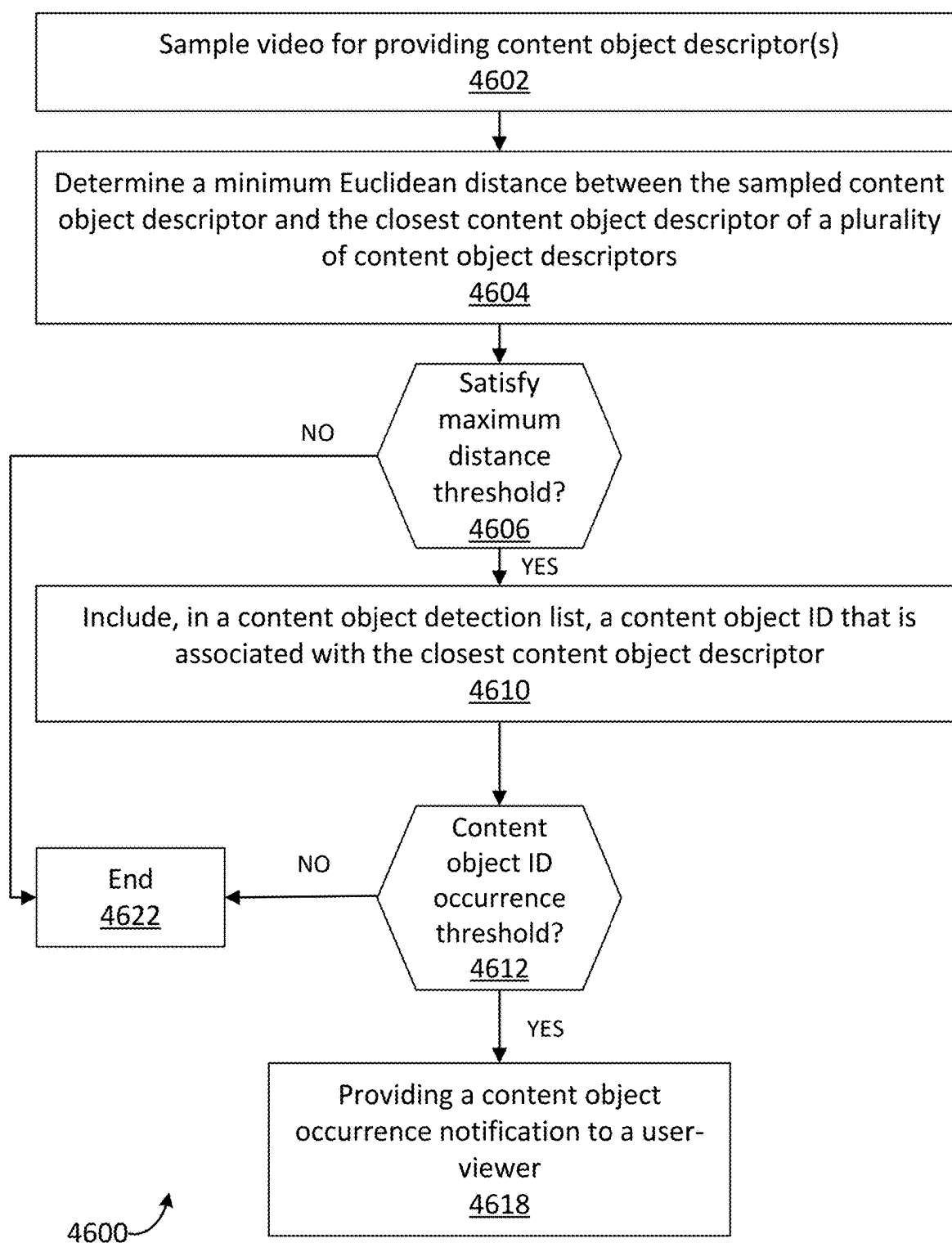
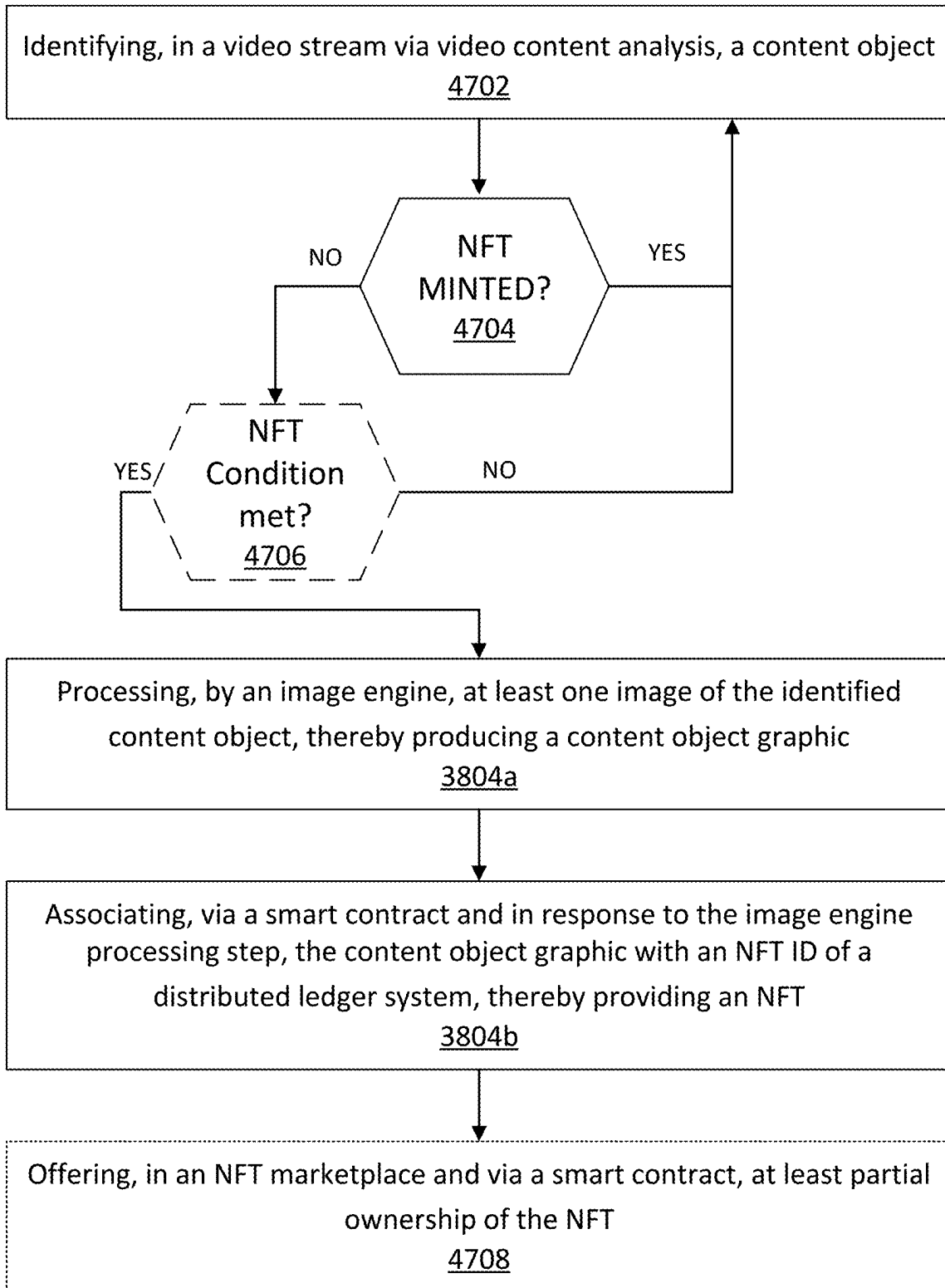
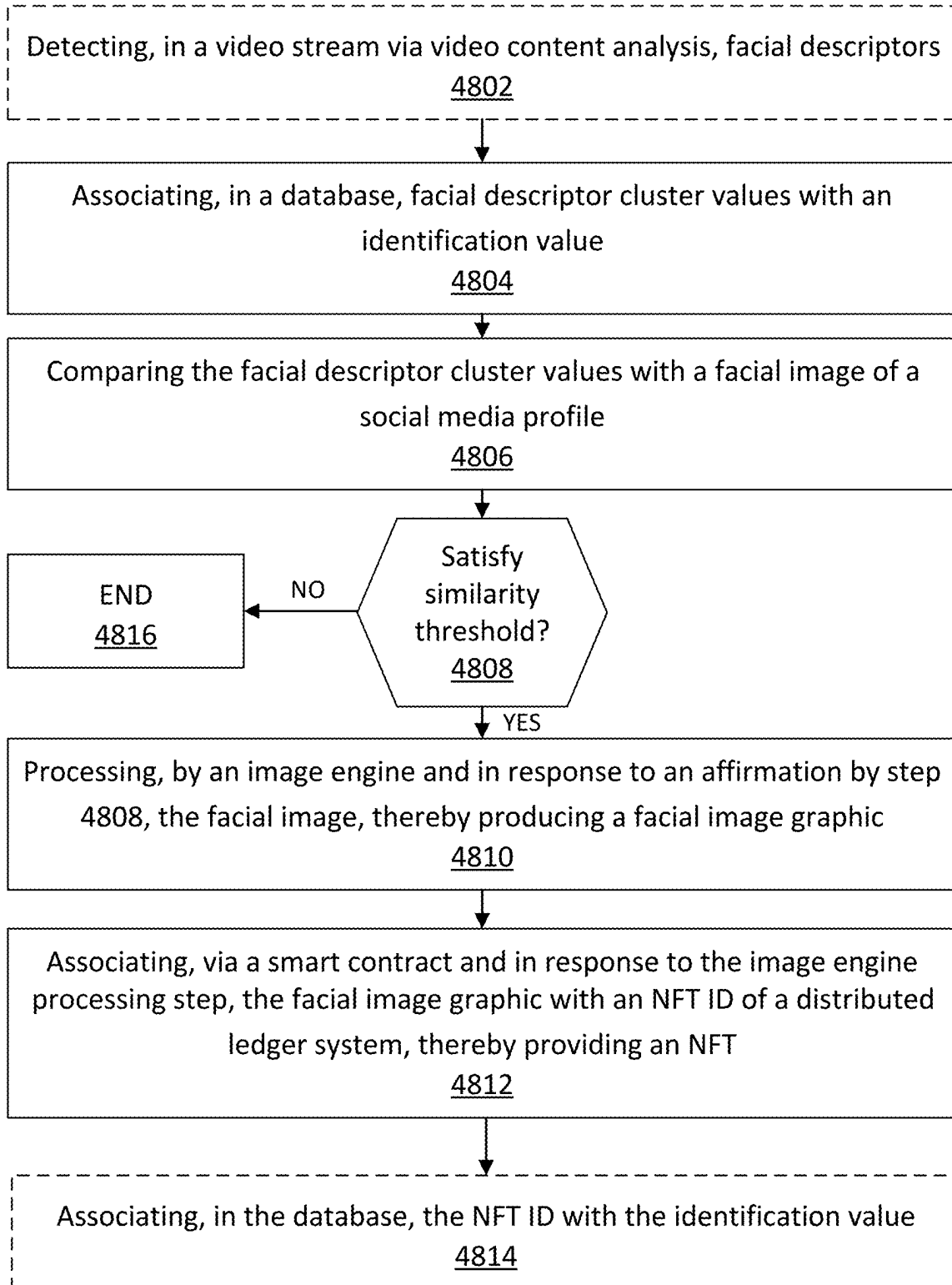


FIG. 46



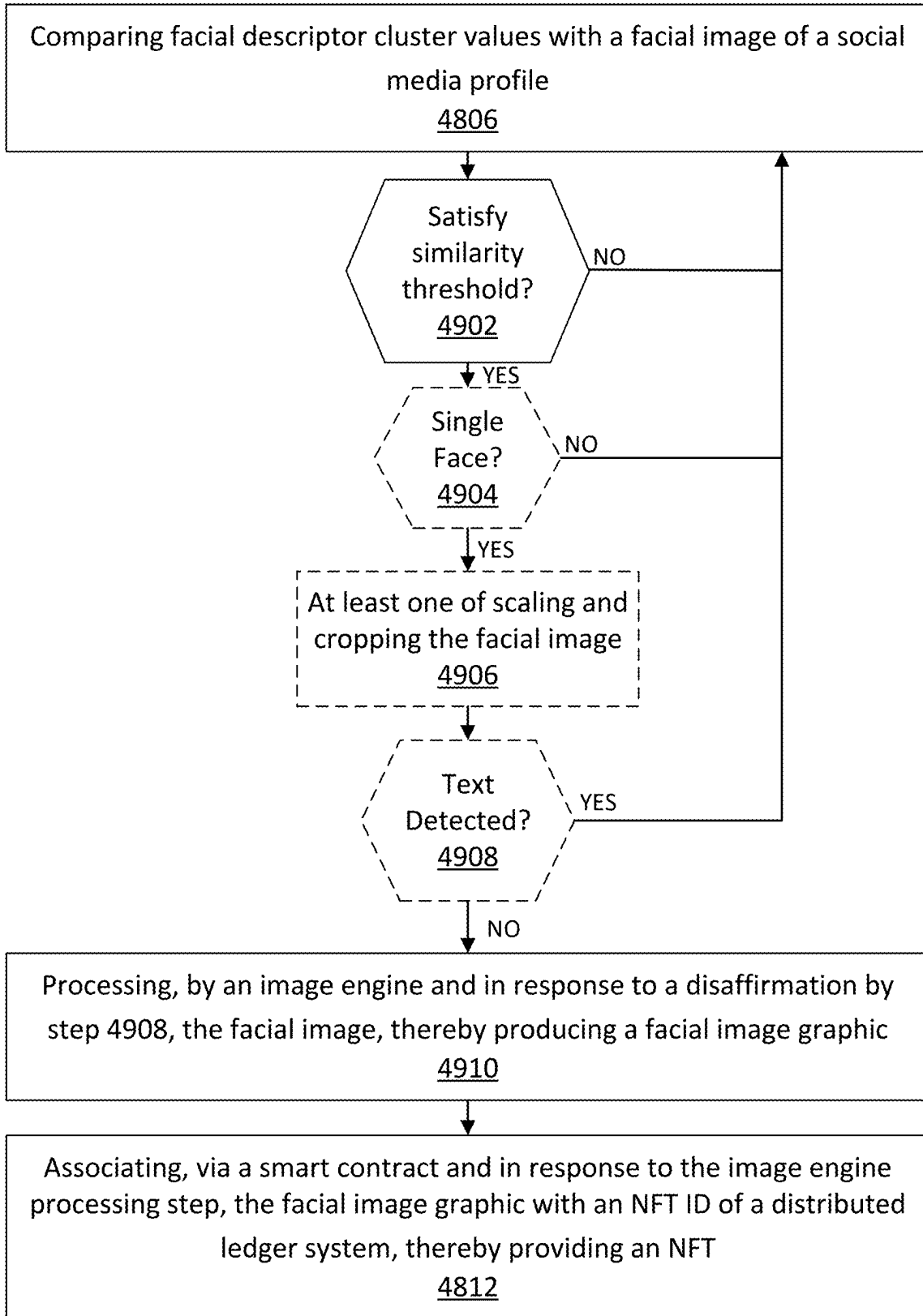
4700

FIG. 47



4800 ↗

FIG. 48



4900

FIG. 49

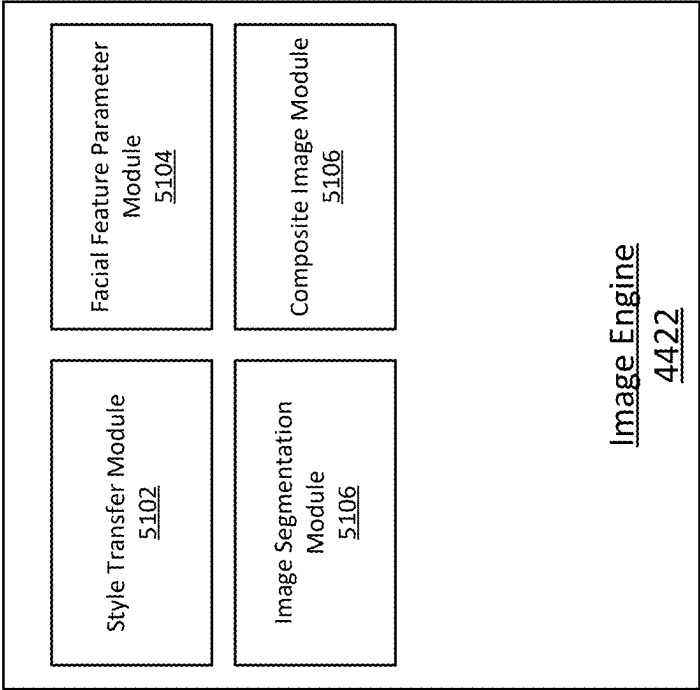


FIG. 51

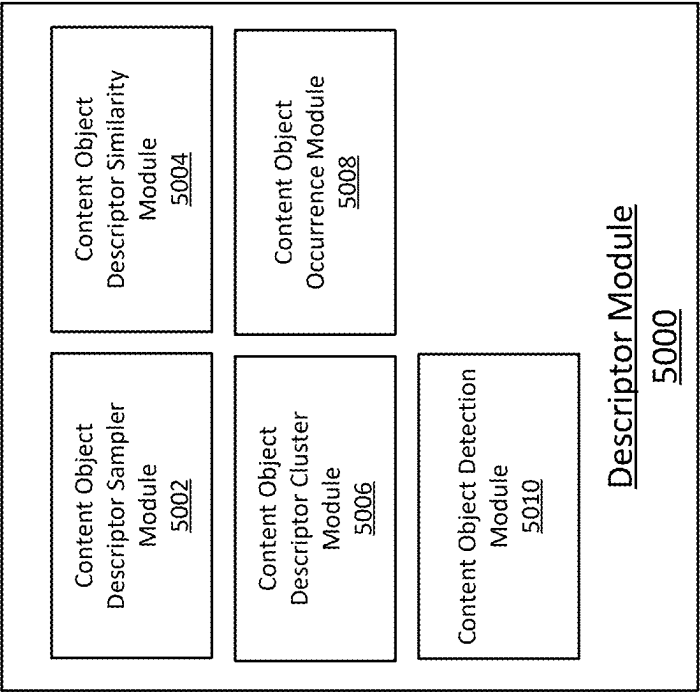
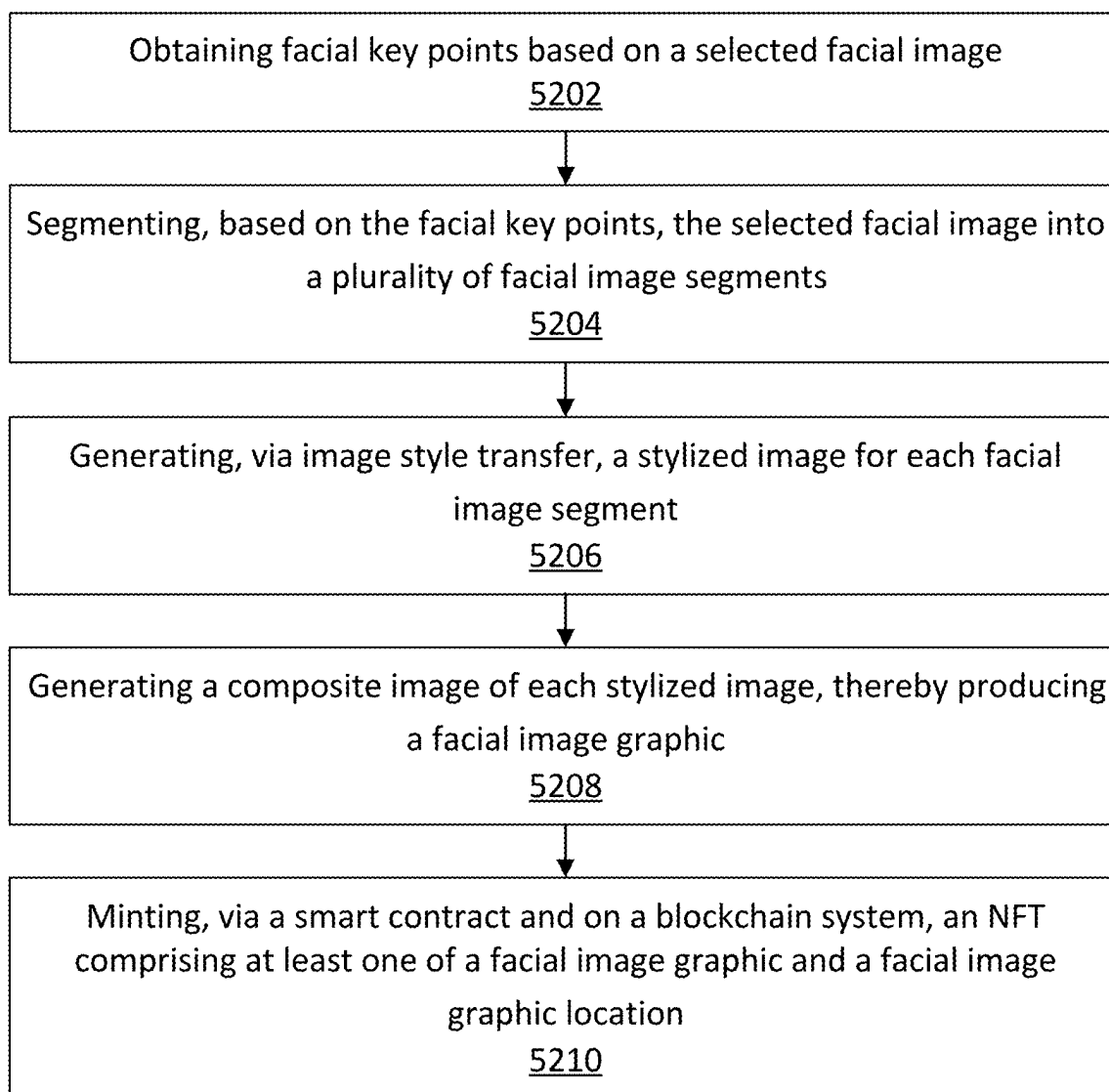
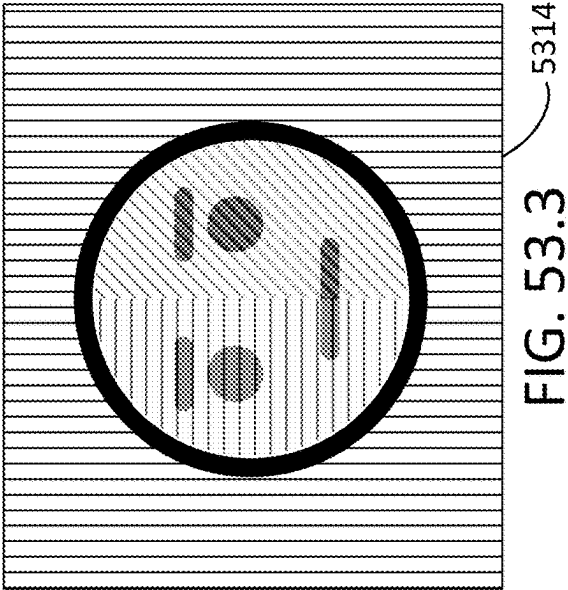
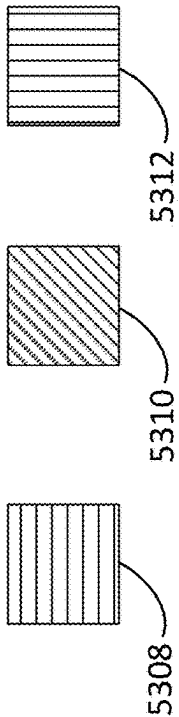
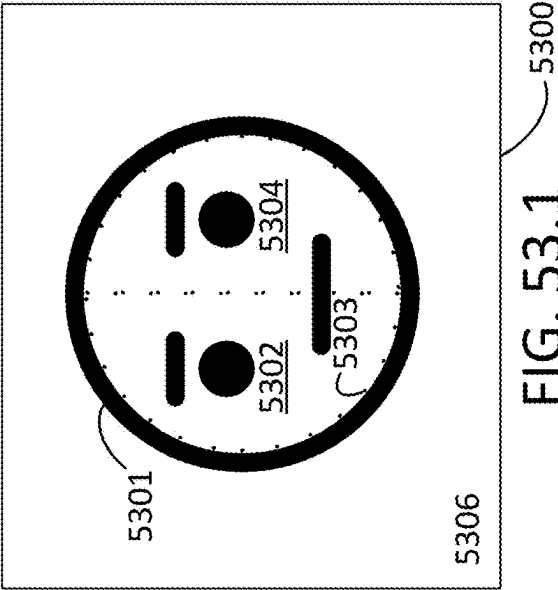


FIG. 50



5200 ↗

FIG. 52



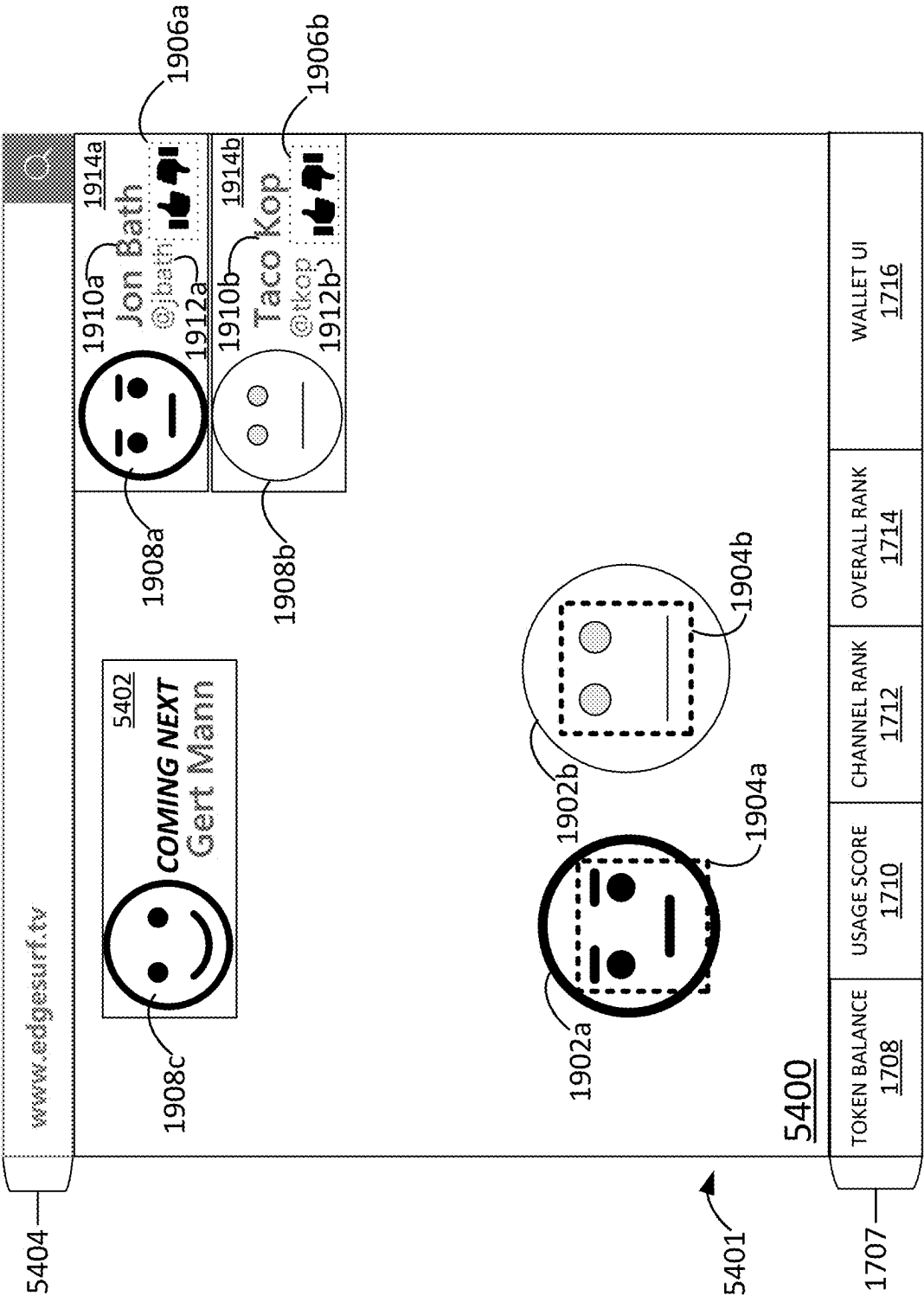


FIG. 54

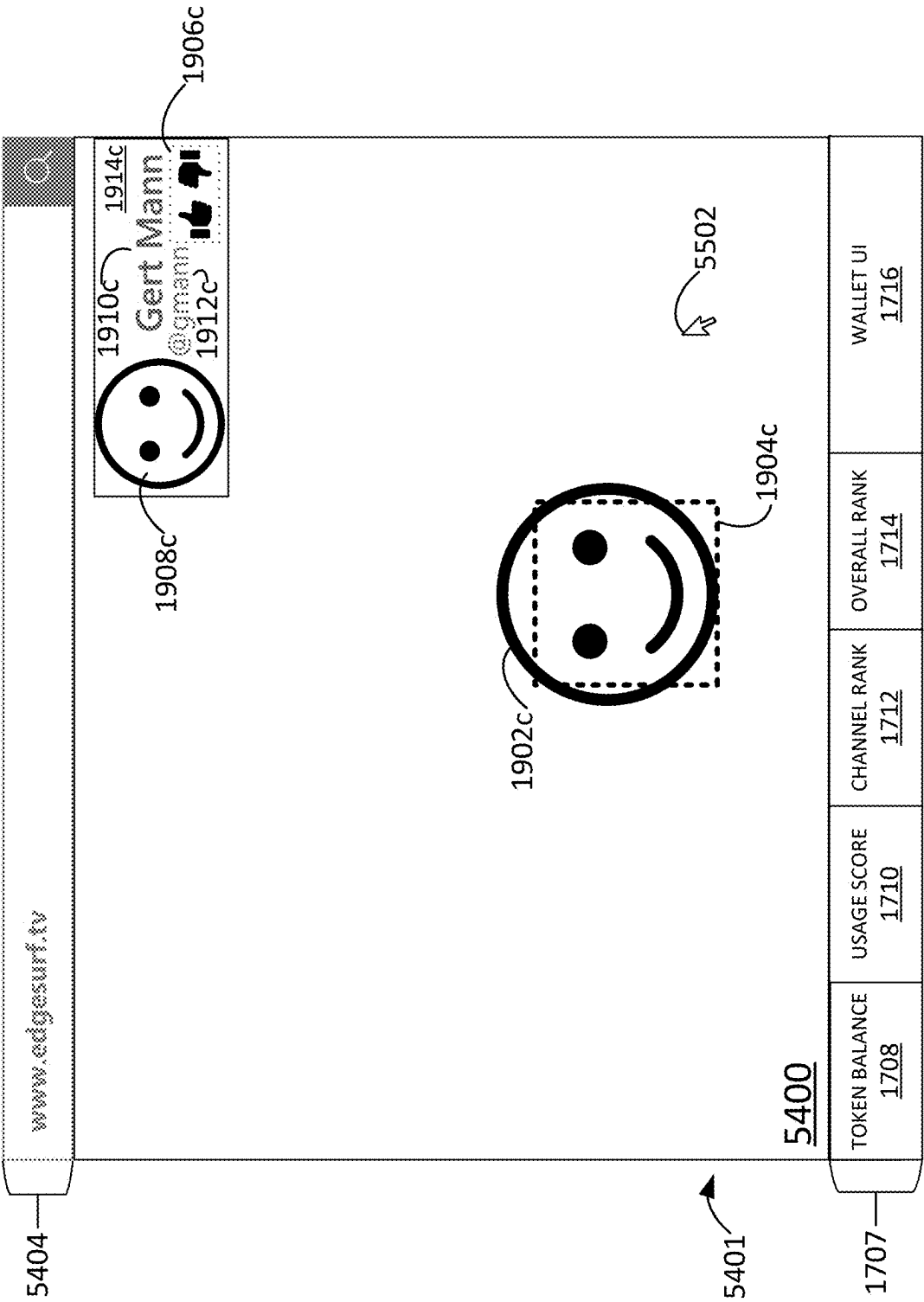


FIG. 55

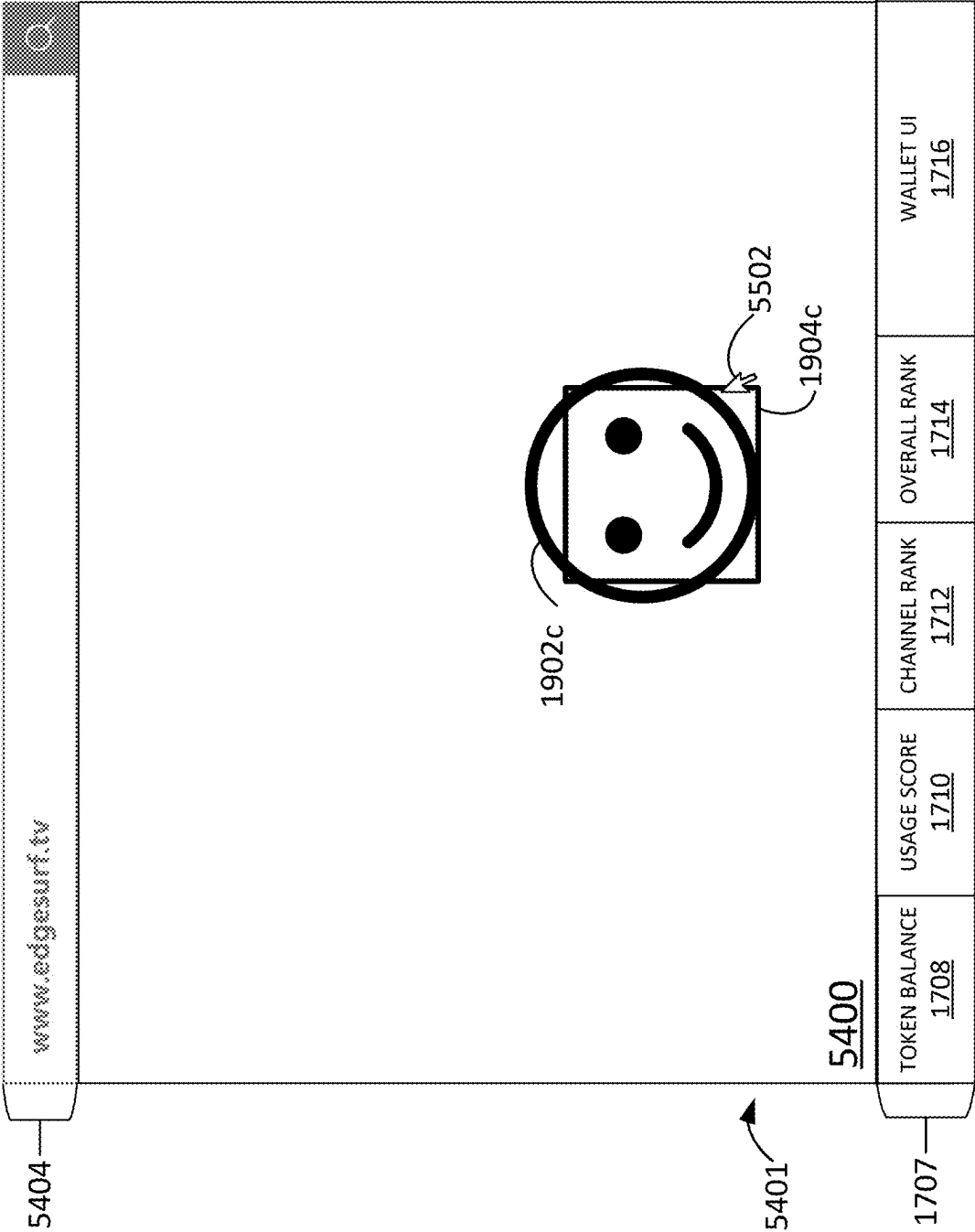


FIG. 56

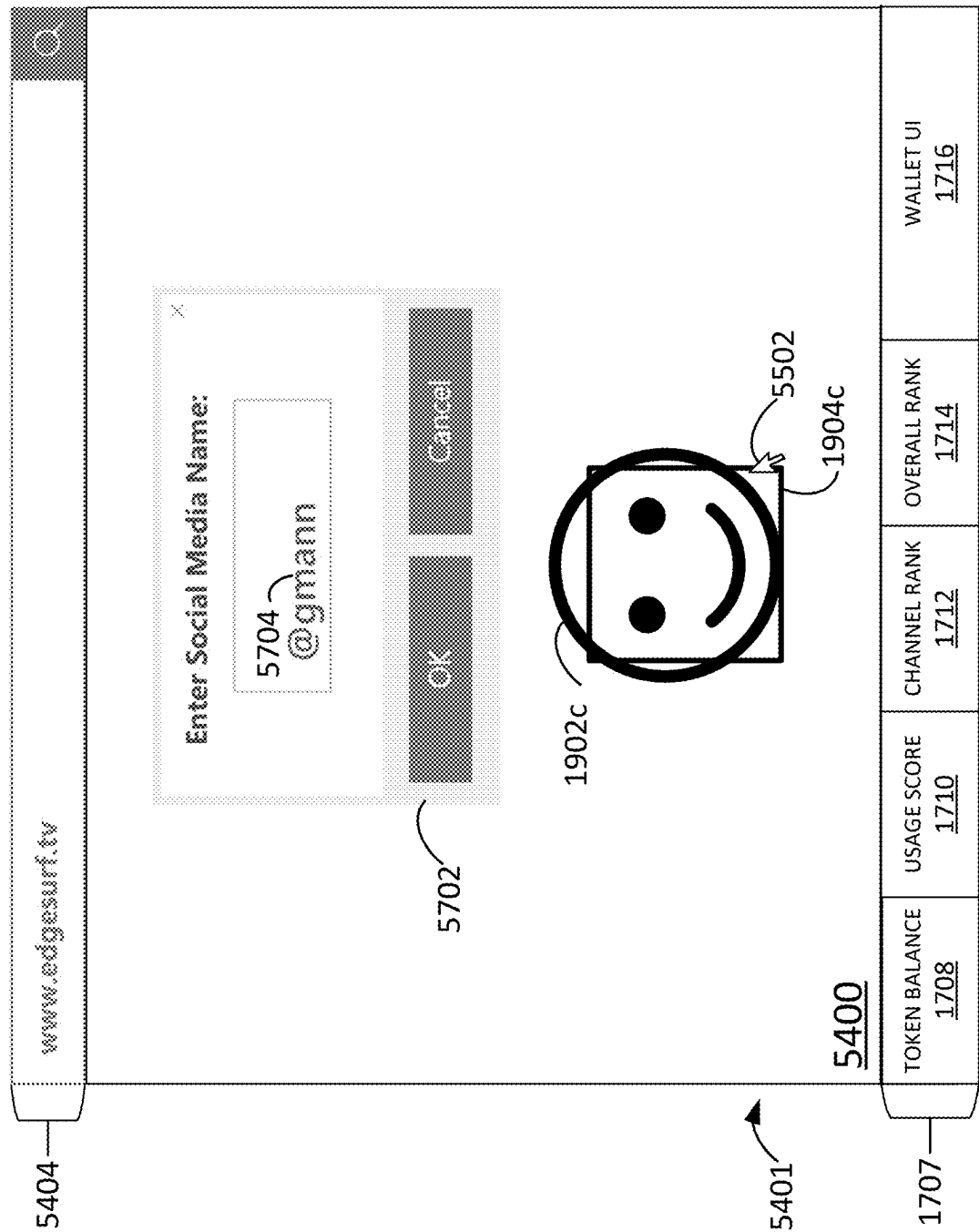
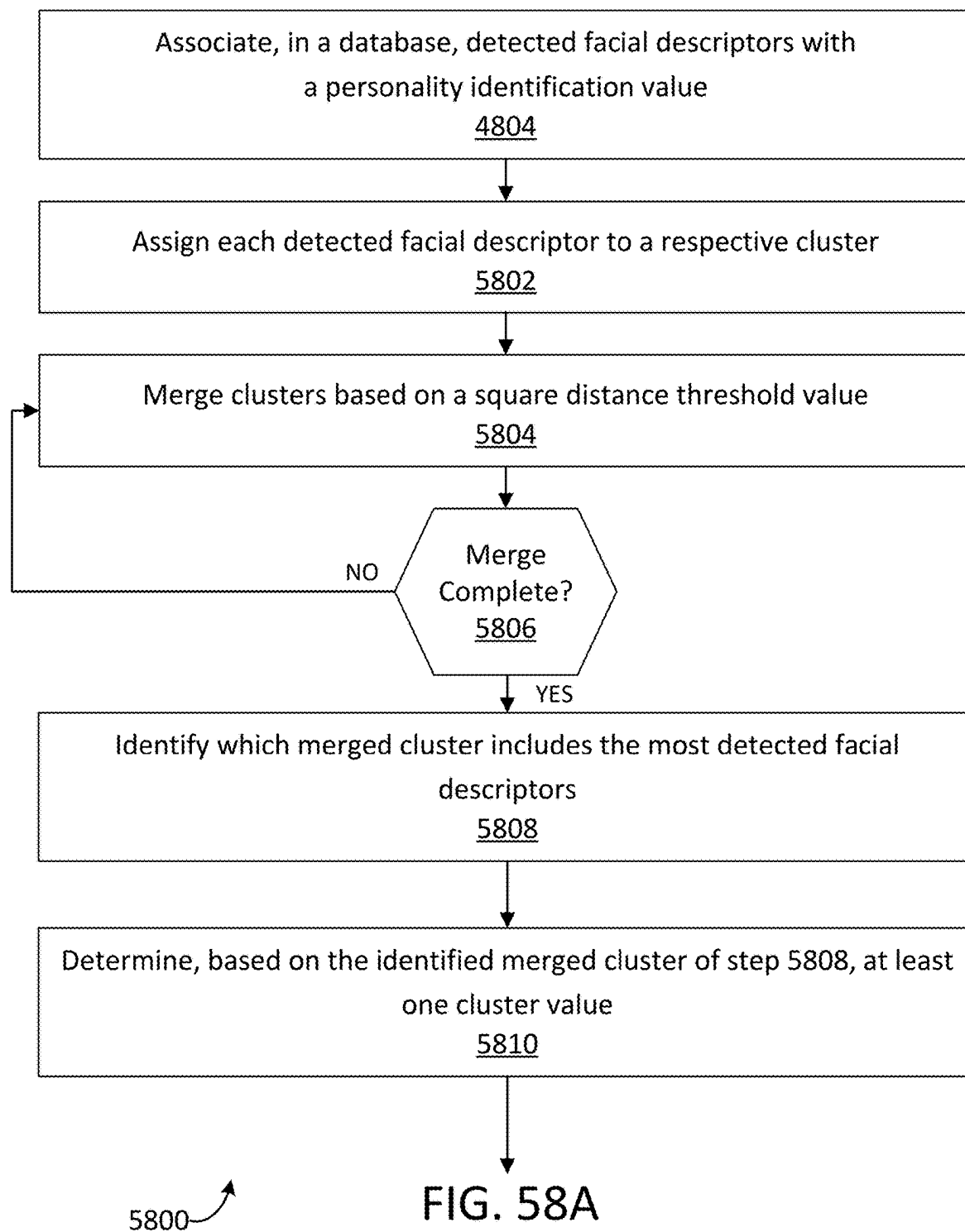
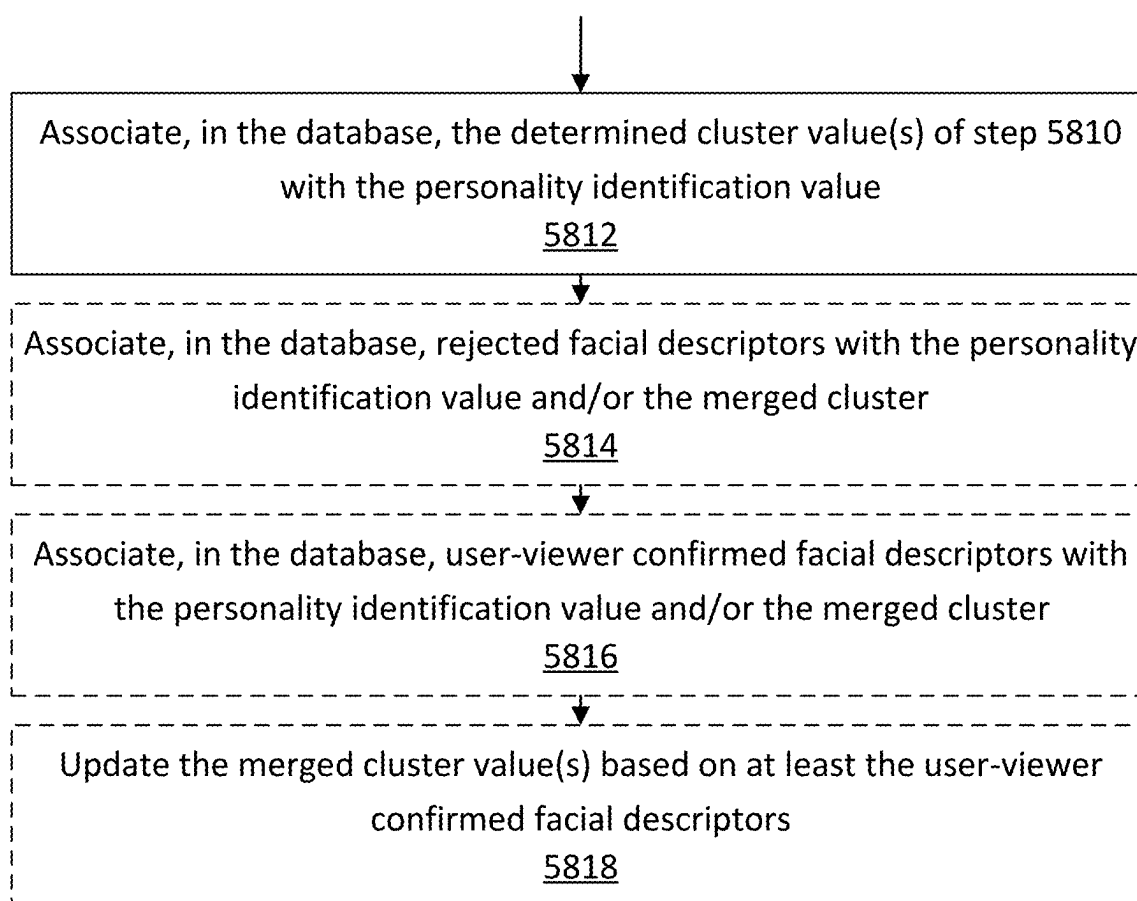


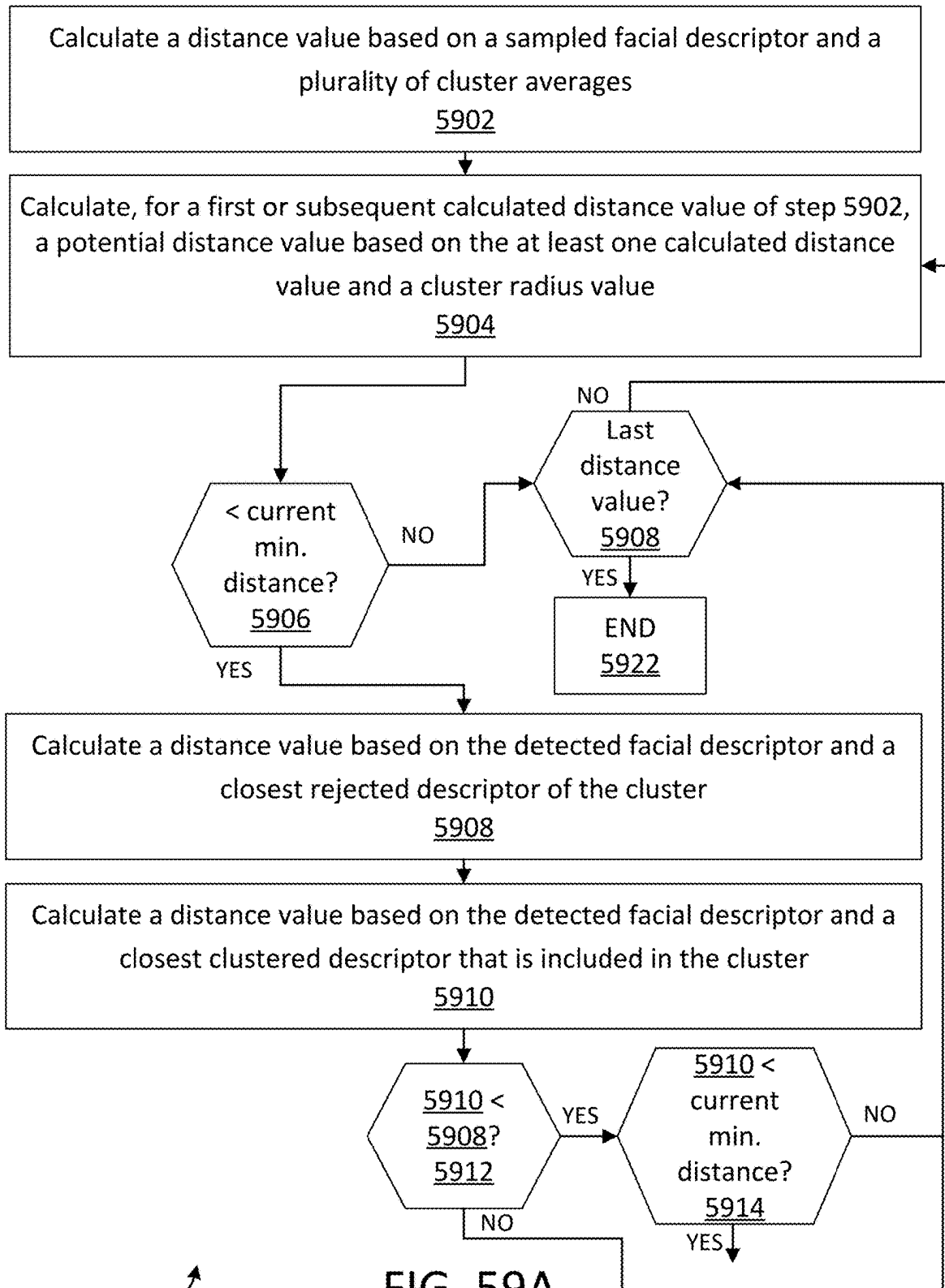
FIG. 57

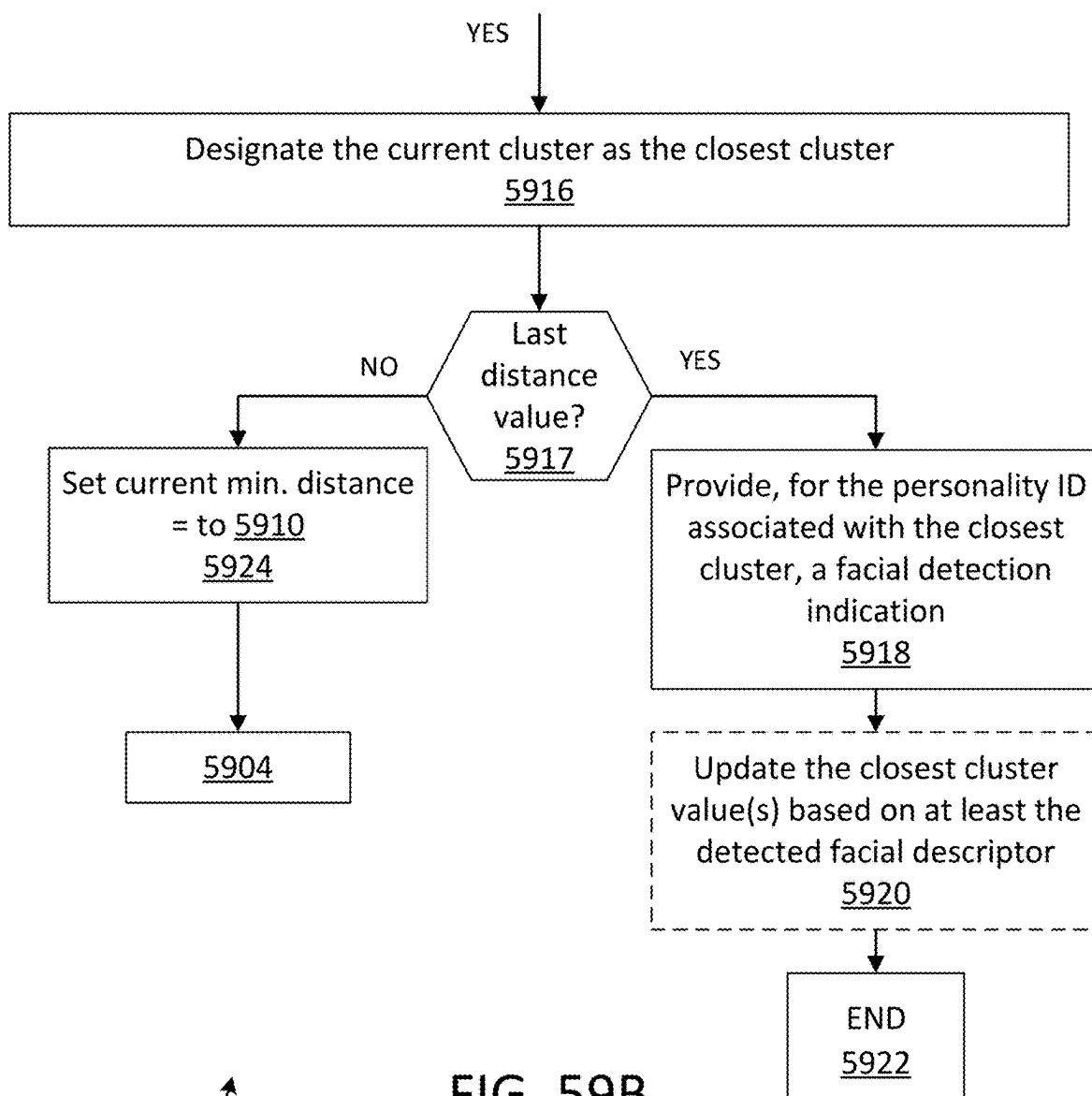




5800 ↗

FIG. 58B





5900 ↗

FIG. 59B

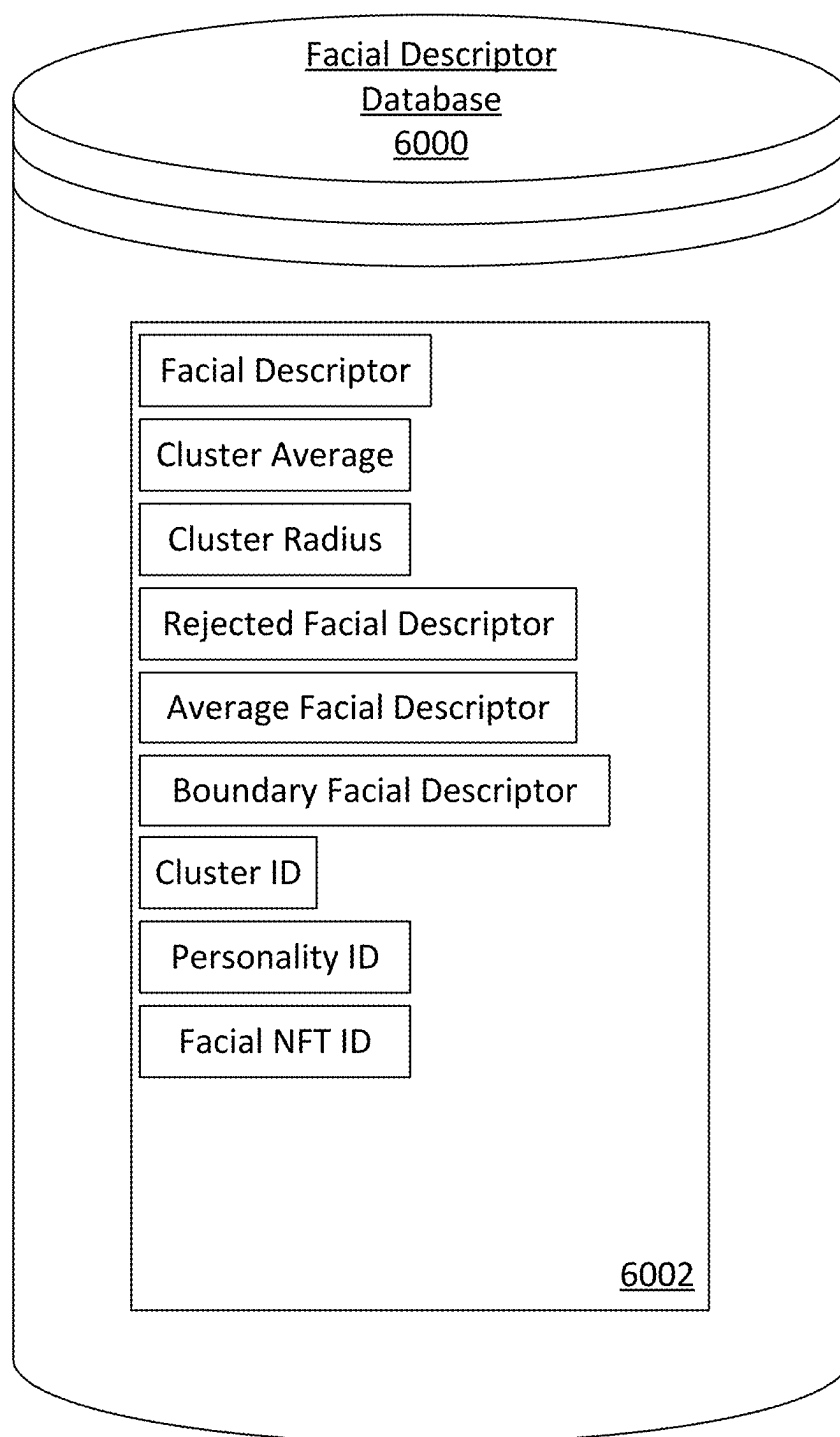


FIG. 60

**MULTIMEDIA CONTENT MANAGEMENT
AND PACKAGING DISTRIBUTED LEDGER
SYSTEM AND METHOD OF OPERATION
THEREOF**

RELATED APPLICATIONS

[0001] This application is a national phase application of and claims priority under 35 U.S.C. § 371 of PCT Patent Application Serial No. PCT/US23/64803 (Attorney Docket No. 6590.00006) filed on Mar. 29, 2023 and titled Multimedia Content Management and Packaging Distributed Ledger System and Method of Operation Thereof, which in turn claims priority under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/269,726 (Attorney Docket No. 6590.00002) filed on Mar. 22, 2022 and titled MULTIMEDIA CONTENT MANAGEMENT AND PACKAGING DISTRIBUTED LEDGER SYSTEM AND METHOD OF OPERATION THEREOF. The contents of these applications are incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates generally to a multimedia content management and packaging system, media guides, and more particularly, among other things, to a multimedia content management and packaging systems and media guides with distributed ledger (e.g., a blockchain system) capability.

BACKGROUND ART

[0003] Modern consumer and industrial electronics, especially devices with an image and video display capability, such as televisions, projectors, smart phones, and combination devices, are providing increasing levels of functionality to support modern life which require the display of managing multimedia information. The expansion of different display types coupled with larger display format sizes and resolutions require ever larger amounts of information to be stored on digital media to capture images and video recordings. Research and development in the existing technologies can take a myriad of different directions.

[0004] As users become more empowered with the growth of media and multi-media display options, new and old paradigms begin to take advantage of this new device space. There are many technological solutions to take advantage of this new multimedia content display opportunity. One existing approach is to display multimedia content on consumer, industrial, and mobile electronics, such as digital televisions, smart phones, tablet computers, digital projectors, monitors, gaming systems, or a combination devices, based on fixed schedules or according to manually selected schedules, such as on cable television systems.

[0005] Image and video display systems have been incorporated in televisions, smart phones, projectors, notebooks, and other portable products. Today, these systems aid users by providing viewing opportunities for available relevant information, such as images, graphics, text, or videos in a variety of conditions. The display of digital images provides invaluable relevant information for the user.

[0006] However, displaying multimedia content has become a paramount concern for the consumer. Mobile display systems, fixed display system, and other modern display systems must access ever increasing amounts of multimedia content with limited physical storage and screen

spaces. Larger multimedia content can consume more communication bandwidth during transmission, reducing the utility of remote display systems. The sheer volume of available multimedia content decreases the benefit of using the tools.

[0007] Thus, a need still remains for better multimedia content management and packaging systems to capture, package, and manage multimedia content. In view of the ever-increasing commercial competitive pressures, along with growing consumer expectations and the diminishing opportunities for meaningful product differentiation in the marketplace, it is increasingly critical that answers be found to these problems. Additionally, the need to reduce costs, improve efficiencies and performance, and meet competitive pressures adds an even greater urgency to the critical necessity for finding answers to these problems.

[0008] Solutions to these problems have been long sought but prior developments have not taught or suggested any solutions and, thus, solutions to these problems have long eluded those skilled in the art.

SUMMARY

[0009] The present invention embodiments provide a method of operation of a multimedia content management and packaging system, including: receiving a content object; detecting a media reference to the content object from an external media feed; calculating a usage score for the content object based on the media reference; and displaying the content object having the usage score greater than a usage score threshold.

[0010] The present invention embodiments provide a multimedia content management and packaging system, including: an ingest unit for receiving a content object; a content data storage unit, coupled to the ingest unit, for storing the content object; a usage unit, coupled to the content data storage unit, for updating the content object based on the detection of a media reference; a scoring and aggregation unit, coupled to the content data storage unit, for updating the usage score of the content object in the content data storage unit; and a display module, coupled to the scoring and aggregation unit, for displaying the content object having the usage score above a usage score threshold.

[0011] Certain embodiments of the invention have other steps or elements in addition to or in place of those mentioned above or are directed to other concepts such as a media guides for a plurality of programs, and a blockchain-based or other distributed-ledger media player. The steps or element will become apparent to those skilled in the art from a reading of the following appended claims and detailed description when taken with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a block diagram of a multimedia content management and packaging system in an embodiment of the present invention.

[0013] FIG. 2 is an example of a media guide.

[0014] FIG. 3 is an example of an ingest unit.

[0015] FIG. 4 is an example of a usage unit.

[0016] FIG. 5 is an example of a scoring and aggregation unit.

[0017] FIG. 6 is an example of a content data storage unit.

[0018] FIG. 7 is an example of a process flow for the multimedia content management and packaging system.

[0019] FIG. 8 is an example of a functional block diagram of the multimedia content management and packaging system.

[0020] FIG. 9 is a flow chart of a method of operation of the multimedia content management and packaging system in a further embodiment of the present invention.

[0021] FIG. 10 is a schematic diagram of a conventional multichannel streaming session between a streaming media server and a client media receiver.

[0022] FIGS. 11 to 16 are schematic diagrams of multiprogram streaming sessions.

[0023] FIGS. 17 to 20 show media guide examples.

[0024] FIG. 21 shows a display unit example.

[0025] FIGS. 22 to 33 show media guide examples.

[0026] FIG. 34 is a schematic diagram of a multiprogram streaming session.

[0027] FIGS. 35 and 36 show media guide examples.

[0028] FIG. 37 is a flow chart of a method for displaying a plurality of programs.

[0029] FIG. 38 is a flow chart of a method for a multimedia blockchain system.

[0030] FIG. 39 is a flow chart of a method of operating a media guide system.

[0031] FIG. 40 shows a media guide example.

[0032] FIG. 41 shows an NFT marketplace presentation.

[0033] FIG. 42 shows a multimedia distributed ledger system.

[0034] FIG. 43 shows a media guide example.

[0035] FIG. 44 shows a multimedia distributed ledger system.

[0036] FIG. 45 is a flowchart of a method for determining a relative usage score.

[0037] FIG. 46 is a flowchart of a method for determining an occurrence of a pre-identified content object in a digital video.

[0038] FIGS. 47, 48, and 49 are flowcharts of methods for generating NFTs.

[0039] FIG. 50 is a schematic diagram of a descriptor module.

[0040] FIG. 51 is a schematic diagram of an image engine.

[0041] FIG. 52 is a flow chart of a method for generating an NFT.

[0042] FIGS. 53.1, 53.2, and 53.3 show image examples.

[0043] FIGS. 54, 55, 56, and 57 show media guide examples.

[0044] FIGS. 58A and 58B are a flowchart of a method for determining facial descriptor cluster values.

[0045] FIGS. 59A and 59B are a flowchart of a method for facial recognition.

[0046] FIG. 60 is a schematic diagram of a facial descriptor database.

DETAILED DESCRIPTION

[0047] The following embodiments are described in sufficient detail to enable those skilled in the art to make and use the invention. It is to be understood that other embodiments would be evident based on the present disclosure, and that system, process, or mechanical changes may be made without departing from the scope of the present invention.

[0048] In the following description, numerous specific details are given to provide a thorough understanding of the invention. However, it will be apparent that the invention

may be practiced without these specific details. In order to avoid obscuring the present invention, some well-known circuits, system configurations, and process steps are not disclosed in detail.

[0049] The drawings showing embodiments of the system are semi-diagrammatic and not to scale and, particularly, some of the dimensions are for the clarity of presentation and are shown exaggerated in the drawing FIGs. Similarly, although the views in the drawings for ease of description generally show similar orientations, this depiction in the FIGs. is arbitrary for the most part. Generally, the invention can be operated in any orientation.

[0050] The same numbers are used in all the drawing FIGs. to relate to the same elements. The embodiments have been numbered first embodiment, second embodiment, etc. as a matter of descriptive convenience and are not intended to have any other significance or provide limitations for the present invention.

[0051] The term “image” is defined as a pictorial representation of an object. An image can include a two-dimensional image, three-dimensional image, video frame, a calculated file representation, an image from a camera, a video frame, or a combination thereof. For example, the image can be a machine-readable digital file, a physical photograph, a digital photograph, a motion picture frame, a video frame, an x-ray image, a scanned image, or a combination thereof. The image can be formed by pixels arranged in a rectangular array. The image can include an x-axis along the direction of the rows and a y-axis along the direction of the columns.

[0052] The term “content” is defined as a media object. Content can include video, images, audio, text, graphics, a social feed, RSS data, news, other digital information, or a combination thereof. The term “multimedia content” is defined as a media object that can include multiple types of media. For example, the multimedia content can include video and audio, video with graphics, graphics with audio, text with audio, a social feed, RSS data, news, other digital information, or a similar combination.

[0053] The horizontal direction is the direction parallel to the x-axis of an image. The vertical direction is the direction parallel to the y-axis of an image. The diagonal direction is the direction non-parallel to the x-axis and non-parallel to the y-axis.

[0054] The term “module” referred to herein can include software, hardware, or a combination thereof. For example, the software can be machine code, firmware, embedded code, and application software. Also for example, the hardware can be circuitry, processor, a graphical processing unit, digital signal processor, calculator, integrated circuit, integrated circuit cores, or a combination thereof.

[0055] The term “digital asset” may refer to a blockchain asset such as a cryptocurrency or other blockchain currency, non-fungible tokens, and other unique or limited-edition digital assets that may be transferred and associated with a wallet. “Wallet” refers to a digital wallet or digital wallet application interface. Wallets may include hardware and software implementations or combinations thereof (e.g., a Metamask web browser plug-in as an interface between a web3 application and a hardware wallet). A unique digital asset may have a unique ID to represent a particular digital asset. In some embodiments, a unique digital asset may also be unique in the sense that a unique ID represents a (unique) class of digital assets (e.g., a particular set of functional NFTs) and may offer semi-fungibility.

[0056] “Video content analysis” (VCA) includes the capability of automatically analyzing video to detect and determine temporal and spatial events via one or more algorithms. Example VCA techniques include object detection, face recognition, and alphanumeric recognition of digital images and/or videos. “Object detection” detects, in digital images and/or videos, instances of semantic objects of a certain class (e.g., humans, buildings, or cars).

[0057] Referring now to FIG. 1, therein is shown a block diagram of a multimedia content management and packaging system 100 in an embodiment of the present invention. The multimedia content management and packaging system 100 can ingest a set of content objects 102, monitor usage and references to content objects 102 in the real world, and dynamically update a usage score 106 for the content objects 102 in real time. Ingesting a set of content objects 102 is receiving and registering the content objects 102 so they can be recognized. Each of the content objects 102 can represent a portion of a multimedia item such as a song, a video, text, a document, a commercial, graphics (e.g., sport scores and time), a three-dimensional video, an audio track, a social feed, Real Simple Syndication (RSS) data, news, other digital information, or a combination thereof.

[0058] Alternatively or additionally, content objects 102 can represent a particular phrase (e.g., “My way or the highway.”; “Forget about it!”), particular animal or person (e.g., actor, streamer, journalist, politician, podcaster, reporter, host, and/or other media-exposed individuals), particular place (e.g., Austin, Texas downtown skyline; Zugspitze, the Royal Concertgebouw (e.g., the exterior (or some portion thereof) and/or interior (e.g., individual halls) of the building), or a particular thing.

[0059] In some embodiments, a content object 102 comprises or represents facial feature set and/or other data (e.g., audio spectrogram) that is typically unique to an individual and for detection via similar data being generated as a media reference 112 (e.g., facial and/or voice data matching a facial and/or voice data within a set of content objects 102, thus identifying a reference for the usage score 106). In some embodiments, content objects 102 may be hierarchically tagged such that, for example, an entire video program is a content object along with a plurality of identified content objects (e.g., objects identified by VCA) within said video program.

[0060] The multimedia content management and packaging system 100 can include an ingest unit 108, a content data storage unit 120, a usage unit 110, a scoring and aggregation unit 122, and a display unit 124. The usage unit 110 can receive a media reference 112 from external media feeds 128, such as a social media feed module 114, a usage feed module 116, and an environmental feed module 118.

[0061] The media reference 112 is an indicator that the media item associated with one of the content objects 102 has been used, referred to, played, or cited, either directly or indirectly in an internet web context (e.g., said actions occurring via a browser or other application of a network-connected device). For example, the media reference 112 can indicate that a movie or video associated with one of the content objects 102 has been played on Hulu® or YouTube®. In another example, the media reference 112 can indicate that a review of a television program associated with one of the content objects 102 has been published on Twitter™. In yet another example, the media reference 112

can indicate that a song associated with one of the content objects 102 has been played on the radio in a particular geographical area.

[0062] The ingest unit 108 is a module to enter, identify, and register the content objects 102. The ingest unit 108 can receive the content objects 102 and store a reference to each of the content objects 102 in the content data storage unit 120.

[0063] The content objects 102 are multimedia content elements. For example, the content objects 102 can be videos, audio recordings, web pages, movies, images, a social feed, RSS data, news, other digital information, electronic data from databases, or a combination thereof. Additionally or alternatively, content objects 102 may comprise or represent (e.g., pointers; hashed parameters) identification information that is sufficiently unique to an individual (e.g., facial and/or facial-feature recognition data; voice identification data). Each of the content objects 102 can have associated information that describes and defines the content.

[0064] Each of the content objects 102 can have the usage score 106. The usage score 106 is associated with the degree of usage or external references to the content objects 102. The usage score 106 is described in greater detail below.

[0065] The content data storage unit 120 is a module for storing information about the content objects 102. The content data storage unit 120 is described in further detail below. The content data storage unit 120 can include an entry for each of the content objects 102. Information associated with each of the content objects 102 can be stored in the content data storage unit 120.

[0066] The content data storage unit 120 can be coupled to the ingest unit 108, the scoring and aggregation unit 122, and the usage unit 110 in a variety of ways. For example, the content data storage unit 120 can be coupled to the ingest unit 108, the scoring and aggregation unit 122, and the usage unit 110 using a web sockets link, networking link, web real time communications, network sockets, or other communication technique.

[0067] The usage unit 110 is a module for associating the one of the content objects 102 with information about the external use of one of the content objects 102. The usage unit 110 can determine the usage or reference to one of the content objects 102 and update the information about one of the content objects 102. The usage unit 110 can receive and process information from the external media feeds 128. The external media feeds 128 are sources of media information. The external media feeds 128 can be a variety of sources including social media feeds, usage feeds, and environmental feeds.

[0068] The scoring and aggregation unit 122 is a module for updating and maintaining the current status of the usage score 106. The scoring and aggregation unit 122 can create and modify the usage score 106 for one of the content objects 102 using a variety of techniques. For example, the scoring and aggregation unit 122 can update the usage score 106 based on usage, time, duration, quality, location, last use, type of usage, or a combination thereof. The scoring and aggregation unit 122 can update the usage score 106 based on information about an aggregation of the content object 102. The scoring and aggregation unit 122 is described in greater detail below.

[0069] It has been discovered that updating the usage score 106 of each of the content objects 102 based on

external usage or reference provides an efficient way to identify the content objects **102** that are trending or important. The usage score **106** can be used to rank and rate the content objects **102**.

[0070] The multimedia content management and packaging system **100** can be implemented using hardware, software, or a combination thereof. For example, the ingest unit **108**, the usage unit **110**, and the scoring and aggregation unit **122** can be implemented with custom circuitry, a digital signal processor, microprocessor, or a combination thereof. The content data storage unit **120** can be implemented with magnetic media, electronic media, cloud media, optical media, magnetoresistive media, or a combination thereof.

[0071] In another example, the ingest unit **108**, the content data storage unit **120**, the usage unit **110** and the scoring and aggregation unit **122** can be implemented in software, in programmable hardware such as a field programmable gate array, or a combination thereof. The multimedia content management and packaging system **100** is a particular machine having hardware for calculating information regarding media content received from a variety of sources.

[0072] It has been discovered that scoring the usage of the content objects **102** improves the technical field of usage monitoring by measuring the real-world usage of the content objects **102** to return actual information on live and/or streaming use. Measuring the actual usage of the content objects **102** that have been registered within the content data storage unit **120** provides a more accurate method of determining the interest in one of the content objects **102** over time.

[0073] Referring now to FIG. 2, therein is shown an example of a media guide **202**. The media guide **202** is a representation of the usage score **106** associated with each of the content objects **102**.

[0074] The media guide **202** is a dynamic data structure that is updated as the usage score **106** of the content objects **102** is updated by the scoring and aggregation unit **122** of FIG. 1. The media guide **202** can have a variety of configurations. For example, the media guide **202** can be configured with channels **204** along the y-axis and the content objects **102** along the x-axis. The content objects **102** can be arranged from the highest value of the usage score **106** to the lowest.

[0075] The channels **204** represent a category of the content objects **102**. For example, the channels **204** can represent different categories such as movies, videos, sports, news, financial reports, or a combination thereof. The channels **204** can also represent different media sources such as broadcast television channels, cable television channels, satellite television channels, or a combination thereof. Yet further, the channels **204** can represent different Internet sources of media such as Hulu®, YouTube®, Crackle®, or other internet media sources. The channels **204** can be arranged according to user preference.

[0076] In another example, the channels **204** can be assigned a channel rating **206**. The media guide **202** can be configured to arrange the highest value for the channel rating **206** to the top of the media guide **202**.

[0077] Each of the content objects **102** in the media guide **202** can be associated with an activity meter **208** (e.g., a content object graphic) to show the current value of the usage score **106**. The activity meter **208** is an abstraction of the level of the usage score **106**. One example of the media guide **202** is a rectangular array comprising the activity

meter **208** for each of the content objects **102**, with the content objects **102** with the highest value of the usage score **106** sorted from left to right. The left most entry in one of the channels **204** of the media guide **202** has the highest value of the usage score **106**.

[0078] The media guide **202** is dynamically updated based on the information from the usage unit **110** of FIG. 1. As the usage score **106** of one of the content objects **102** changes, the activity meter **208** can be updated and the location of the activity meter **208** is updated based on the relative values of the usage score **106** of the other ones of the content objects **102**.

[0079] In an illustrative example, the media guide **202** is constantly updating based on the incoming information from the usage unit **110** and the scoring and aggregation unit **122**. As the usage score **106** of the content objects **102** changes, the configuration of the media guide **202** updates in real-time.

[0080] The activity meter **208** can be represented in a variety of ways. For example, the activity meter **208** can be represented as different graph types **210** such as a horizontal bar chart, a vertical bar chart, a pie chart, a line chart, a vector distribution, a grid, or a combination thereof.

[0081] The activity meter **208** can also be displayed using badging to form an intelligent graphical element, such as badges **212**. The badges **212** can be displayed in conjunction with representations of the channels **204**. The badges **212** can be a dynamic measurement of the popularity of each of the channels **204**. The badges **212** can vary over time.

[0082] The media guide **202** can represent an underlying multi-dimensional array of the usage score **106** for each of the content objects **102**. For example, the media guide **202** can be a representation based on big data analysis of the usage score **106** for the content objects **102**. In another example, the media guide **202** can include a time-varying representation that can show the change in the usage score **106** for each of the content objects **102** over a period of time or a contextual relevance to each other, such as other dimensions beyond time. The other dimensions can include location, subject matter, media type, source, format, size, quality, preferences, external factors, or a combination thereof. The period of time can be hourly, daily, weekly, monthly, or any useful time period or dimension. In yet another example, the media guide **202** can represent the usage score **106** for each of the content object **102** in the content data storage unit **120** of FIG. 1.

[0083] Referring now to FIG. 3, therein is shown an example of an ingest unit, such as the ingest unit **108** of FIG. 1. The ingest unit **108** can receive the content objects **102** and create entries in the content data storage unit **120**. The ingest unit **108** can include a receive content module **302**, a tagging module **304**, a hash module **306**, and a packaging module **307**. The receive content module **302** is a module to allow the entry of the content objects **102**. The receive content module **302** can receive the content objects **102** and allow the entry of additional information associated with each of the content objects **102**.

[0084] The receive content module **302** can receive one of the content objects **102** as entered by a user or operator or from a scheduling application. For example, the user can provide a set or list of the content objects **102** that can be read by the receive content module **302**. In another example, the user can input the set of the content objects **102** indi-

vidually. The list of the content objects **102** can be metadata describing the content objects **102** or the content objects **102** themselves.

[0085] Metadata from the content can be extracted in the packaging module **307** as described below. The packaging module **307** can transcode various versions of the content and then package the content into object with tagging indicating the next processing steps. In a further example, the receive content module **302** can automatically receive the content objects **102** by processing information directly from broadcast, cable, and satellite sources including digital and analog sources. The receive content module **302** can be coupled to a variety of the external media feeds **128** of FIG. **1**, such as analog television tuners, digital television tuners, analog radio, digital radio, satellite television, satellite audio, internet television sources, streaming media sources, digital media feed, a social feed, RSS data, news, other digital information, network connected devices, mobile phones, cameras, tablet computers, external feeds, cloud sources, private networks, public networks, or a combination thereof. Each of the external media feeds **128** can be sourced from a media device **318** that can provide a content stream having a combination of the content objects, voice, other audio, graphics, commercials, a social feed, RSS data, news, other digital information, or a combination thereof.

[0086] The media device **318** is a device for receiving media broadcasts, digital media streams, external feeds, or a combination thereof. For example, the media device **318** can be a television tuner, a radio receiver, a satellite receiver, a network interface, or a combination thereof. The media device **318** can receive broadcast or internet media and provide the external media feeds **128** to the receive content module **302**.

[0087] It has been discovered that automatically receiving the content objects **102** from the external media feeds improves the technology (e.g., big data analysis) for identifying said content objects **102**. By receiving local and remote media broadcasts and parsing the external media feeds to extract the content objects **102**, the receive content module **302** registers the content objects **102** more efficiently.

[0088] It has been discovered that automatically receiving the content objects **102** from the network media feeds improves the technology (e.g., big data analysis) for identifying said content objects **102**. By receiving local and remote internet media feeds and parsing the feeds to extract the content objects **102**, the receive content module **302** registers the content objects **102** to provide a near real-time feedback system.

[0089] The receive content module **302** can receive and parse the content objects **102** from the external media feeds **128** in a variety of ways. For example, the receive content module **302** can parse a continuous television stream from one of the external media feeds **128** based on available television schedule information, such as a programming guide. The receive content module **302** can parse one of the external media feeds **128**, such as a radio broadcast from a radio receiver, based on the detection of pauses between songs and the volume intensity profile of songs. In another example, the receive content module **302** can parse the content objects **102** from a digital radio broadcast from the internet based on the data in the related packaged content.

[0090] In another example, receive content module **302** can detect individuals in a video feed or audio feed. For

example, a content object **102** may include or be associated with a non-fungible token (NFT) or other unique digital asset (e.g., an audio NFT; facial-related NFTs) of a digital wallet (e.g., a hardware wallet and/or a Metamask browser extension or plug-in (i.e., a software wallet)).

[0091] The receive content module **302** can detect the occurrence of the content objects **102** by matching portions of the incoming content stream to pre-defined hash values identifying known songs, images, or other media or individuals. The receive content module **302** can also detect the content objects **102** using pre-defined signals within the content stream to mark the beginning and end of the content objects **102**.

[0092] Receiving the content objects **102** can include receiving information about the content objects **102**. For example, the receive content module **302** can receive the content objects **102** and associated data such as name, date, size, resolution, language, version, aspect ratio, or a combination thereof.

[0093] The receive content module **302** can receive one or more of the content objects **102** at a time. After receiving the content objects **102**, the receive content module **302** can pass the control flow to the tagging module **304**.

[0094] The packaging module **303** can transcode the content objects **102** into an array of formats based on predefined formulas. The packaging module **303** can be used to represent the content objects **102** in different forms for different usages. Once the packaging module **303** completes, then the control flow can pass to the tagging module **304**.

[0095] The tagging module **304** is a module for identifying and assigning additional information to the content objects **102**. The tagging module **304** can tag each of the content objects **102** and create a content record **310** in the content data storage unit **120** having a content identifier **314**, a content name **316**, and an ingestion datetime **308**. Tagging is the process of associating one of the content objects **102** with an identification and other information.

[0096] The content records **310** are data structure for representing the content objects **102**. The content identifier **314** is a unique identifier assigned to identify one of the content objects **102**. The content name **316** is a text string used to identify one of the content objects **102**. The ingestion datetime **308** is a timestamp indicating when one of the content objects **102** was received.

[0097] The tagging module **304** can detect the reception of one of the content objects **102** where a record already exists for that one of the content objects **102**. For example, if an identical copy of one of the content objects **102** is detected in the tagging module **304**, then the reference to that one of the content objects **102** can point to the existing entry for one of the content objects **102**. However, the tagging module **304** can detect and maintain separate records for different versions of one of the content objects **102**, such as the same video, but different language versions of the same video, lower resolution or quality of one of the content objects **102**, different aspect ratios, different editions, or a combination thereof.

[0098] The tagging module **304** can also attach workflow information to the content objects **102**. For example, the tagging module **304** can indicate that one of the content objects **102** can require additional processing, such as closed captioning processing, video refinement, text capture, format changing, size changing, audio formatting, or a combination thereof.

[0099] The tagging module 304 can tag one or more of the content objects 102 at a time. Once the tagging module 304 has created the content record 310 in the content data storage unit 120, the control flow can pass to the hash module 306.

[0100] The hash module 306 is a module for calculating a unique, deterministic, numerical value, a hash value 312, to identify the data of one of the content objects 102. Deterministic is defined as meaning that for a given input value, the same value for the hash value 312 must be generated. The hash value 312 is a representation the content objects 102. The hash value 312 is separate from the content identifier 314, which is not necessarily deterministic. The hash value 312 can be used to determine if two of the content objects 102 represent an identical piece of media. The hash value 312 can be calculated over the entirety of one of the content objects 102 or over smaller portions of the content objects 102.

[0101] Although the hash value 312 is described as a single value, multiple hash values can be associated with one of the content objects 102. For example, the content objects 102 can have individual hash values for different portions of the content objects 102. In another example, the content objects 102 can use different types of hash calculations for redundancy and efficiency.

[0102] The packaging module 303 can transcode the content into an array of predefined formulas. Once the packaging module 307 completes, then the control flow can pass to the tagging module 304.

[0103] It has been discovered that calculating the hash value 312 for one of the content objects 102 can increase the performance by enabling the detection of duplications of the content objects 102 by comparing the hash value 312 of one of the content objects 102 to the hash value 312 of others of the content objects 102. Comparing the numerical values is less compute intensive than comparing the entirety of the content objects 102.

[0104] It has been discovered that calculating the hash value 312 for each of the content objects 102 improves the technology of content object identification by assigning each of the content objects a unique instance of the hash value 312. The hash value 312 enables the detection of duplications of the content objects 102 by comparing the hash value 312 of one of the content objects 102 to the hash value 312 of others of the content objects 102. Comparing the numerical values is less compute intensive than comparing the entirety of the content objects 102.

[0105] Referring now to FIG. 4, therein is shown an example of a usage unit, such as the usage unit 110 of FIG. 1. The usage unit 110 is for detecting and registering the usage of the content objects 102 of FIG. 1 in the multimedia content management and packaging system 100 of FIG. 1. The usage unit 110 can include the social media feed module 114, the usage feed module 116, and the environmental feed module 118.

[0106] The social media feed module 114 can detect the usage and reference to one of the content objects 102 over social media channels, such as a social media feed 410. The social media channels can include Facebook®, Twitter®, LinkedIn®, YouTube®, 3rd party application programming interfaces (APIs), or other similar social media channels.

[0107] The social media feed module 114 can monitor usage and references to the content objects 102 in a variety of ways. For example, public uses of the content objects 102 can be detected by monitoring one or more accounts on a

social media site, by receiving a data feed describing the usage of the content objects 102 on the social media site, detecting a social media reference to stored copies of the content objects 102 available elsewhere on the Internet, processing data consolidation feeds summarizing references to the content objects 102 on the social media site, or a combination thereof. In another example, the public uses of the content objects 102 and their equivalents can be detected using an API of a social media site, aggregating and comparing the content objects 102 with active and archived usage data, or detecting the content objects 102 on a public network, cloud, or other data source.

[0108] The social media feed module 114 can detect usage and reference to the content objects 102 in a variety of ways. For example, the references to the content objects 102 can be exact by comparing a hash value from the social media reference to the hash value 312 of FIG. 3 of one of the content objects 102 in the content data storage unit 120. In another example, the reference to the content objects 102 can be approximate, such as when only the name or another identifier of one of the content objects 102 is detected on a social media site. In yet another example, the reference to the content objects 102 can be inferred when an indirect reference such as a partial name, abbreviation, time or date reference, slang reference, or other reference is used in a social media context. The social media feed module 114 can compare the indirect reference to a list or database correlating the indirect reference with the content objects 102.

[0109] The usage feed module 116 is a module for detecting the direct usage and reference to the media of the content objects 102. This can include websites such as video websites, movie websites, streaming media sites, video aggregators, APIs, or similar media sources. The usage feed module 116 can monitor a variety of usage feeds including a usage feed 412 such as YouTube®, Hulu®, CNN®, Crackle®, Vimeo®, Vevo®, API feeds, or other similar websites for viewing online media.

[0110] The usage feed module 116 can monitor usage and references to the content objects 102 in a variety of ways. For example, uses of the content objects 102 can be detected by directly monitoring one or more accounts on a media usage site, by using an API interface to a website, by receiving a data feed describing the usage of the content objects 102 on the media usage site, detecting a reference to stored copies of the content objects 102 available elsewhere on the Internet, processing data consolidation feeds summarizing access and references to the content objects 102 on the media usage site, or a combination thereof.

[0111] The usage feed module 116 can detect usage and reference to the content objects 102 in a variety of ways. For example, the references to the content objects 102 can be exact by comparing the hash value 312 from the media usage reference to the hash value 312 of one of the content objects 102 in the content data storage unit 120. In another example, the reference to the content objects 102 can be approximate, such as when a data feed includes only the name or another identifier of one of the content objects 102. In yet another example, the reference to the content objects 102 can be inferred when an indirect reference such as a partial name, abbreviation, time or date reference, slang reference, or other reference is used in a data feed. The usage feed module 116 can compare the indirect reference to a list or database correlating the indirect reference with the content objects 102.

[0112] In yet another example, references to the content objects 102 can be determined based on an indirect reference or inference may be made between the content objects 102 or classes of the content objects 102 based on the usage patterns in real time and historically. Predictive analysis of gaps in a media stream can be made based on historical statistical data stored in a data archive. An average value for existing references to one of the content objects 102 can be established and used to interpolate the number of references during a gap or missing portion of the media feed.

[0113] The environmental feed module 118 can monitor the external usage and references to the content objects 102 in a variety of ways. Use of the content objects 102 can be detected by directly monitoring an environmental feed 414 such as television and radio signals using a television tuner, radio receiver, or other receiving device for receiving broadcast, cable, Internet, RSS, API feeds, or satellite content. The received signals can be analyzed to detect the content objects 102 using facial recognition, voice recognition, closed captioning, hash value comparison, video analysis, audio parsing, or other similar techniques.

[0114] The use of the content objects 102 can also be detected using an API from commercial providers for television, radio, Internet, or satellite content to integrate and cross-correlate usage of the content objects 102. The usage of the content objects 102 can also be analyzed using the result from the packaging module 303 of FIG. 3 correlated with data extracted from the content objects 102 such as voice recognition, closed captioning, hash value comparison, video analysis, audio parsing, text feeds, or other similar techniques.

[0115] The environmental feed module 118 can identify the content objects 102 in the incoming feed using a variety of techniques. For example, the environmental feed module 118 can parse songs or other media by detecting the pause between media items, parsing based on timing, parsing by on express signaling, parsing based on audio volume variation, parsing based on video signal variation, or a combination thereof. The content objects 102 can also be parsed using packaging delimiters, 3rd party API data, feed data, or a combination thereof.

[0116] The environmental feed module 118 can detect the content objects 102 having multiple potential durations and parsing durations by calculating different values for the hash value 312 for different portions of the potential example of the content objects 102. For example, the environmental feed module 118 can calculate the hash value 312 for different potential parsing configurations and compare each of the different values of the hash value 312 to known content objects 102.

[0117] For example, radio signals in a localized market can be monitored to detect the usage of songs, audio books, lectures, or other content by monitoring the radio broadcasts using a radio receiver device. Video content and usage can be monitored by receiving broadcast television and detecting programming content. In another example, the media reference 112 of FIG. 1 to one of the content objects 102 can be performed by receiving a data feed describing the usage of the content objects 102 on the external media, such as by receiving a programming guide, or other descriptive content and scheduling feed. In another example, the data from APIs and external data feeds can identify the usage of the content objects 102.

[0118] The environmental feed module 118 can detect usage and reference to the content objects 102 in a variety of ways. For example, the references to the content objects 102 can be exact by comparing a hash value from the media usage reference to the hash value 312 of a portion of one of the content objects 102 in the content data storage unit 120. In another example, the reference to the content objects 102 can be approximate, such as when a data feed includes only the name or another identifier of one of the content objects 102. In yet another example, the reference to the content objects 102 can be inferred when an indirect reference such as a partial name, abbreviation, time or date reference, slang reference, or other reference is used. The environmental feed module 118 can compare the indirect reference to a list or database correlating the indirect reference with the content objects 102.

[0119] In an illustrative example, the environmental feed module 118 can perform speech to text analysis of a television program received from a television receiver to determine the title of a television program at a given time and geographical location. The environmental feed module 118 can compare the associated metadata or text extracted from the audio portion of the television program to known title or introductory sequences to detect the use of the content objects 102. Further, the environmental feed module 118 can use appropriate metadata to determine titles or other identifiers for the content object 102.

[0120] The usage unit 110 can receive the information about references to the content objects 102 from the social media feed module 114, the usage feed module 116, and the environmental feed module 118 and update a usage count 402 for the content objects 102. The usage count 402 can provide an indicator of the importance and popularity of the content objects 102 based on how often they are viewed, referred to, downloaded, used, or cited.

[0121] The usage count 402 can be a single element or it can be represented by a multi-valued array or other data structure. For example, the usage count 402 can represent specific channel information, feed information, date and time information, location information, language information, user demographics, user profiles, market information, or a combination thereof. The usage count 402 can be one of the contributing factors in calculating the usage score 106 of FIG. 1 for the content objects 102.

[0122] In an illustrative example, the usage score 106 can be updated based on a usage location 406 of the media reference 112 being within a location threshold 408 of a target location 404. The usage location 406 is the location associated with one of the media reference 112. The target location 404 is a location used to calculate the media guide 202 of FIG. 2. The location threshold 408 is a value representing a distance away from a location that can be considered the same location.

[0123] The content data storage unit 120 can be coupled to a business intelligence module 416. The business intelligence module 416 is a module for processing archived information from the content data storage unit 120. The business intelligence module 416 can archive data older than a business intelligence threshold and consolidate the information in an archive for processing. The archived data can be used to detect long term trends, correlations between different time periods, specific usage models, or a combination thereof. The business intelligence module 416 can receive external feeds including business support services

(BSS) feeds and business intelligence (BI) feeds providing additional information related to the content objects 102, the markets, the feeds, other usages, or a combination thereof.

[0124] It has been discovered that determining the usage count 402 for the content objects 102 increases system performance by identifying the content objects 102 that are important based on use. This allows the early detection of spiking events like viral videos and other content objects 102 by measuring the level of public interest in the content objects 102 based on how they are used and accessed across a wide range of real-time data sources representing real-world usage.

[0125] It has been discovered that updating the usage count 402 based on detecting usage of the content objects 102 in broadcast (e.g., streamed) television or radio increases system performance and functionality by providing geographically sensitive usage information.

[0126] It has been discovered that calculating multiple values of the hash value 312 to detect one of the content objects 102 having different durations or representing different portions of the content objects 102 improves the technology of media parsing. Because the parsing may be inexact, calculating different values of the hash value 312 for different portions or durations of the content objects 102 improves the likelihood of finding a matching value of the hash value 312.

[0127] Referring now to FIG. 5, therein is shown an example of a scoring and aggregation unit, such as the scoring and aggregation unit 122 of FIG. 1. The scoring and aggregation unit 122 is for assigned and updating the usage score 106 for each of the content objects 102. The usage score 106 can rank the importance or popularity of the content objects 102 or a group of the content objects 102.

[0128] The usage score 106 is a value indicating the relative importance or popularity of the content objects 102. The content objects 102 with higher values of the usage score 106 are considered more important than those with lower values.

[0129] Although the usage score 106 is described as a value, it is understood that the usage score 106 can be a multivalued data object, such as an array, list, or structure. For example, the usage score 106 can include different values for different categories, different content types, different languages, different location, or different times.

[0130] The usage score 106 can be dynamically updated based on usage, time, type, location, quality, length, language, or other similar factors. For example, the usage score 106 can vary over time by gradually reducing in value over time to implement a time decay effect. The usage score 106 can have different values in different locations, such as for the content objects 102 representing a foreign movie in an international market, a regional television program, a commercial, a video, a regional video, a song in a foreign language, or other similar situations.

[0131] In another example, the usage score 106 can be updated based on information based on the aggregation of a set of the content objects 102. The display of multiple copies of a commercial having one of the content objects 102 can be aggregated over several different television channels to be used to update the usage score 106 because of the increased buzz over the entire advertising campaign as measured by the size of the aggregation of the content objects 102. Similarly, the aggregation of a television and radio campaign linked to a group of the content objects 102

can increase the usage score 106 based on the correlation between the usage of the content objects 102.

[0132] The usage score 106 can be updated using a scoring modifier 510. The scoring modifier 510 is a value used to modify the usage score 106. For example, the scoring modifier 510 can be used to multiply, divide, add, offset, or subtract from the usage score 106. In another example, the usage score 106 can be updated based on the quality score 614. A reference to the usage of one of the content objects 102 having a high-quality value for the quality score 614 can increase the usage score 106 more than if the quality score 614 was low to indicate a good usage of one of the content objects 102.

[0133] The scoring and aggregation unit 122 can include a base score module 502, a decay module 504, a timing module 506, a location module 508, and a content module 509. The scoring and aggregation unit 122 can integrate the results of each of the modules to update the usage score 106.

[0134] The base score module 502 is a module for initially assigning the usage score 106 for the content objects 102. The base score module 502 can create the usage score 106 entry for each of the content objects 102 and update it based on the initial parameters associated with the content objects 102.

[0135] The base score module 502 can calculate the usage score 106 in a variety of ways. For example, the base score module 502 can calculate the usage score 106 based on the length of the content objects 102. A very short video clip associated with one of the content objects 102 can result in a low value for the usage score 106, while a longer length can indicate that a higher values for the usage score 106 is appropriate. Similarly, the usage score 106 can be modified based on language, location, length, quality, type, or other similar parameters.

[0136] The base score module 502 can assign and update the usage score 106 in when the content objects 102 are first input into the multimedia content management and packaging system 100 of FIG. 1. After the base score module 502 has updated the usage score 106 for the content objects 102, the control flow can pass to the decay module 504.

[0137] The decay module 504 is a module for modifying the usage score 106 for the content objects 102. For example, the decay module 504 can be based on a time decay model that lowers the usage score 106 for one of the content objects 102 over time to represent a gradual loss on importance or interest over time.

[0138] In an illustrative example, the usage score 106 of a popular movie or video can be high when it is first released. The number of social media references and usage references can increase the usage score 106 to indicate the importance or the viral nature for one of the content objects 102. However, over time the interest of the movie or video wanes and the usage score 106 can be automatically reduced using a decay model.

[0139] The decay module 504 can update the usage score 106 in a variety of ways. For example, the decay module 504 can update the usage score 106 by calculating the time between current time and the time of the peak of the usage score 106 for one of the content objects 102 and multiply by a decay factor 512 indicating the rate of decay over time. The decay factor 512 can be a linear value, an exponential factor, an equation, a segmented value, or a combination thereof. The decay factor 512 can be the scoring modifier 510.

[0140] It has been discovered that using the decay module 504 to modify the usage score 106 of the content objects 102 increases the accuracy and usability of the usage score 106. The relative importance of the content objects 102 varies over time in the real world. The decay module 504 implements a variety of decay model to update the usage score 106 to reflect the time-based decline in importance over time, allowing the usage score 106 to more accurately reflect the importance and true popularity of the content objects 102.

[0141] The decay module 504 can determine the decay model used to calculate the decay factor 512 for the content objects 102 in a variety of ways. The decay factor 512 can be based on a content type 514, the content size, language, location, the usage count 402 of FIG. 4, date of last usage, usage frequency, or a combination thereof. For example, the decay module 504 can select the decay factor 512 for each of the content objects 102 by looking up the content objects 102 in a pre-defined decay factor lookup table showing the a preferred value for the decay factor 512 for each of the content objects 102. In another example, the decay factor 512 can be determined by applying a weighted selection criteria based on a list of parameters such as the usage count 402, the content type 514, and the date of last usage.

[0142] The decay module 504 can update the usage score 106 in a variety of ways. For example, the decay module 504 can update the usage score 106 on a time-scheduled basis, on a demand driven basis, or based on availability of computational resources either synchronously or asynchronously. After the decay module 504 has updated the usage score 106 for the content objects 102, the control flow can pass to the timing module 506.

[0143] The timing module 506 is a module for updating the usage score 106 based on time-sensitive factors. The usage score 106 may vary over time because of timing of other events, social factors, financial factors, or a combination thereof. For example, the timing module 506 can update the usage score 106 of the content objects 102 based on real-time events, the day of the week, the season, the day of the month, time of year, the proximity to pay or bonus periods, the proximity to holidays, the weather, the location, or a combination thereof.

[0144] In an illustrative example, the usage score 106 of the content objects 102 representing movies or videos that have been released within the last week in the United States can be increased on the following Friday and Saturday. Similarly, the usage score 106 of the content objects 102 representing television summer specials in the Northern Hemisphere can be increased during the months of June, July, and August. Alternatively the usage score 106 of the content objects 102 representing television summer specials in the Southern Hemisphere can be decreased during the months of June, July, and August.

[0145] It has been discovered the updating the usage score 106 of the content objects 102 due to time-dependent factors increases the accuracy of the measurement of importance or popularity. By accommodating real-world time and seasonality factors, the usage score 106 more accurately represents the importance and value of each of the content objects 102.

[0146] The timing module 506 can calculate the scoring modifier 510 for each of the time-based events. For example, the timing module 506 can perform a table lookup to determine the scoring modifier 510 for a weekend modifier.

In another example, the timing module 506 can apply a pre-defined value for the scoring modifier 510 for summer months.

[0147] The timing module 506 can update the usage score 106 in a variety of ways. For example, the decay module 504 can update the usage score 106 on a time-scheduled basis, on a demand driven basis, or based on availability of computational resources. After the decay module 504 has updated the usage score 106 for the content objects 102, the control flow can pass to the location module 508.

[0148] The location module 508 is a module for updating the usage score 106 based on location. The usage score 106 may vary by location because of the popularity of local events, local newsworthiness of the content objects 102, language, location as related to the content objects 102, availability, or a combination thereof. For example, the location module 508 can update the usage score 106 of the content objects 102 based on the location associated with the content objects 102, the location of the display unit 124 of FIG. 1, location of a user, importance of a location, or a combination thereof.

[0149] The location module 508 can update the usage score 106 in a variety of ways. For example, the usage score 106 can be increased when the location associated with the content objects 102 matches the location of the user. If a French language video associated with one of the content objects 102 is displayed in France, then the usage score 106 can be higher to show more relative interest in that one of the content objects 102.

[0150] After the location module 508 has updated the location data, the control flow can pass to the content module 509. The content module 509 can update the usage score 106 in a number of ways by providing a content provider quality index score. The content module 509 can evaluate the content objects 102 in a variety of ways including the content size, content type, content resolution, content format, or other content-specific criteria.

[0151] It has been discovered that updating the usage score 106 based on a location associated with one of the content objects 102 improves the technology for media usage detection by increasing the accuracy of the measurement of importance or popularity. By accommodating real-world location factors, the usage score 106 more accurately represents the importance and value of each of the content objects 102.

[0152] Referring now to FIG. 6, therein is shown an example of a content data storage unit, such as the content data storage unit 120 of FIG. 1. The content data storage unit 120 can store information about the multimedia content management and packaging system 100 of FIG. 1. The content data storage unit 120 can store a variety of live information about the content objects 102 of FIG. 1.

[0153] For example, the content data storage unit 120 can include a media identifier 602, a length 604, the content type 514 of FIG. 5, a version 606, and the content name 316 of FIG. 3. The media identifier 602 is a number used to uniquely identify the media associated with the content objects 102. The length 604 is a value to indicate the duration of the content objects 102. The content type 514 is a value indicating the type of the content objects 102. The content type 514 can be an enumerated value to represent types such as movies, television programs, videos, audio tracks, songs, slideshows, images, multimedia presentations, a social feed, RSS data, news, other digital information, or

other similar media types. The version **606** is a value used to differentiate between multiple versions of the same media item associated with one of the content objects **102**. The version **606** can be an enumerated value representing different versions such as the broadcast television version, the director's cut, the original theatric release version, or similar version variations.

[0154] The content data storage unit **120** can include an aspect ratio **608**, a creation location **610**, an ingestion location **612**, a quality score **614**, a creation datetime **616**, and the ingestion datetime **308**. The aspect ratio **608** is a value indicating the ratio between horizontal and vertical dimensions of the media. The creation location **610** is the location where the content objects **102** were created. The ingestion location **612** is a value indicating the location where the content objects **102** were input into the system. The quality score **614** is a value measuring of the quality of one of the content objects **102** as compared to another version of the content, such as a benchmark version. The quality score **614** can indicate the quality of the content objects **102** as a numerical value, an enumerated value, a non-numerical rating, or a combination thereof.

[0155] The creation datetime **616** is a value representing the date and time when the content objects **102** were created. The ingestion datetime **308** is a value representing the date and time when the content objects **102** was entered into the system.

[0156] The content data storage unit **120** can include data about the media item directly associated with the content objects **102** including a content blob **620**, a content size **622**, and a content hash **624**. The content blob **620** is a database entity that can encode the media item associated with the content objects **102**. The content blob **620** can also be a pointer providing a reference to the media items. For example, the content blob **620** can be a serialized digital data structure representing an entire movie, video, television episode, song, or other media item. The content blob **620** can be a database blob data type, a cloud data element, a pointer, or other similar data structure. The content size **622** is a value representing the size of data of the content objects **102**. The content hash **624** is a deterministic value representing an identification of the content objects **102**.

[0157] The content data storage unit **120** can include relationship information between the content objects **102**. For example, the content data storage unit **120** can include the content identifier **314** of FIG. 3, a parent identifier **626**, and a child identifier **628**. The parent identifier **626** and the child identifier **628** are values indicating a hierarchical relationship between one of the content objects **102** having the content identifier **314**. For example, one of the content objects **102**, such as a standard definition movie, can have a parent that is a high-definition version of the movie, and a child that is a shortened version of the movie. The children can represent derivative editions of a parent.

[0158] The content data storage unit **120** can be coupled to a data warehouse archive **630**. The data warehouse archive **630** is a data storage element for archiving and consolidating older information from the content data storage unit **120**. The real-time data and operating data can be transferred from the content data storage unit **120**. Aggregated information and summaries can be stored in the data warehouse archive **630** for off-line processing and business intelligence analysis. The data warehouse archive **630** can be used by business intelligence module **416** of FIG. 4 to process

archived data from the content data storage unit **120** and external business feeds **418** such as BSS and BI feeds.

[0159] The data warehouse archive **630** can archive older copies of the data in the content data storage unit **120**. Content older than a content threshold can be automatically moved from the content data storage unit **120** to the data warehouse archive **630**. In addition, the information from the content data storage unit **120** can be analyzed and consolidated with the results stored in the data warehouse archive **630**.

[0160] It has been discovered that structuring the content data storage unit **120** to represent the hierarchical structure using the content identifier **314**, the parent identifier **626**, and the child identifier **628** improves the technology used to identify and group the content objects **102** that are related to one another. Operating on a family of the content objects **102** can increase the efficiency of processing by performing operations on an entire related family at one time.

[0161] Referring now to FIG. 7, therein is shown an example of a process flow for the multimedia content management and packaging system **100**. The process flow can include a get content module **702**, a detection module **704**, a scoring module **706**, and a display module **708**.

[0162] The multimedia content management and packaging system **100** can receive new media and form the content objects **102** in the get content module **702**. The get content module **702** can receive content from a variety of sources and create the content objects **102** using the ingest unit **108** of FIG. 1, the receive content module **302** of FIG. 3, the packaging module **303** of FIG. 3, the tagging module **304** of FIG. 3, and the hash module **306** of FIG. 3.

[0163] The get content module **702** can receive media items **714** and create the initial records for the content objects **102**. The get content module **702** can be implemented using the ingest unit **108** and other units of the multimedia content management and packaging system **100**.

[0164] The media items **714** are multimedia elements that can be embodied as one of the content objects **102**. The media items **714** can be portions of a media stream, such as an individual image object, a chair, soda can, person, or other element, that are part of a television broadcast. For example, media items can be pictures, sounds, words, actions, scenery, water, fire, persons, media clips, time delimited radio programs, talk shows, or similar items.

[0165] The get content module **702** can receive the content objects **102** manually by an operator entering a media item individually or as a bulk upload. Each of the media items can be used to create one or more of the content objects **102**.

[0166] The get content module **702** can receive the content objects **102** automatically by receiving an external media feed, such as from an internet data feed, an API feed, a television receiver, a radio receiver, a satellite receiver, or a combination thereof, and parsing the media feed to identify the media items **714**.

[0167] In an illustrative example, the get content module **702** can use the ingest unit **108** to automatically detect the media items **714**, such as a video, in the media feed from a television receiver using the timing information provided by a programming guide. In another example, the media items **714** can be detected along with metadata in a commercial field or internet feed or API-based feed. In yet another example, the get content module **702** can automatically detect a portion of song from a radio receiver by parsing gaps in the volume of the media feed from the radio receiver.

[0168] The get content module 702 can identify the media items 714 either by direct entry of identifying information, comparison of the media items to a database of known items, using an external service to identify, or a combination thereof. The media items 714 of the content objects 102 can be modified to assist with the identification process. For example, the media items 714 can be time-cropped to remove unnecessary content, volume adjusted, or a combination thereof.

[0169] The get content module 702 can receive the media items and form the content objects 102 in the content data storage unit 120 of FIG. 1. After the get content module 702 has completed, the control flow can pass to the detection module 704.

[0170] The detection module 704 can detect the media reference 112 of FIG. 1 to one of the content objects 102 using the usage unit 110 of FIG. 1, the social media feed module 114 of FIG. 1, the usage feed module 116 of FIG. 1, and the environmental feed module 118 of FIG. 1. When the media reference 112 is detected, the detection module 704 can increment the usage count 402 for the content objects 102.

[0171] The detection module 704 can use the social media feed module 114, the usage feed module 116, and the environmental feed module 118 to detect the media references in the real-world by monitoring network traffic, commercial media services, broadcast media channels, or a combination thereof. The detection module 704 can receive and scan the media feeds for keywords, matching hash values, specific text values, media clips, or other indicators to detect that the media reference 112 occurred in the media feed.

[0172] The degree of world-wide coverage for detecting the media reference 112 is intended to be as high and broad as possible. However, complete coverage of the entire Internet, all commercial media services, and all broadcast media around the world would require substantial resources, both technical and economic. It is understood that detecting the media reference 112 for a portion of the Internet or broadcast media would provide important information to enable the characterization of the usage of the content objects 102. Although coverage may increase over time, even a partial set of the media references can provide a valuable benefit.

[0173] The detection module 704 can detect the media reference 112 and update the information associated with the content objects 102 in a variety of ways. For example, the detection module 704 can update the usage count 402 of one of the content objects 102 when the media reference 112 is detected. In another example, the detection module 704 can associate the location of the media reference 112 to the content objects 102 to determine hot locations for one of the content objects 102.

[0174] The detection module 704 can search for the media reference 112 on a continuous basis or at scheduled intervals. The detection module 704 can operate on the live media feeds or on buffered copies of the live media feeds.

[0175] The detection module 704 is for updating the usage count 402 for the content objects 102 based on the detection of the media reference 112. After completion, the control flow can pass to the scoring module 706.

[0176] The scoring module 706 can update the usage score 106 of the content objects 102 to provide a relative ranking of the content objects 102. The scoring module 706 can be

implemented using the scoring and aggregation unit 122, the base score module 502 of FIG. 5, the decay module 504 of FIG. 5, the timing module 506 of FIG. 5, and the location module 508 of FIG. 5.

[0177] The content objects 102 can be organized based on the channels 204 of FIG. 2. The channels 204 are categories used to partition the content objects 102. For example, the content objects 102 can be partitioned by categories such as content type, location, quality, media type, or a combination thereof.

[0178] Although the usage score 106 can be described as a single value, it is understood that the usage score 106 can be a complex value having multiple aspects. For example, the usage score 106 can be a multidimensional matrix or array with different values of the usage score 106 for different categories or criteria.

[0179] The scoring module 706 can update the usage score 106 based on the usage count 402 of the content objects 102, where the usage count 402 is updated when the detection module 704 detects the media reference 112. For example, if one of the content objects 102 is popular and played or referenced a large number of times on social media website, then the number of the media reference 112 will be large and the usage count 402 will be incremented accordingly. The usage count 402 can then be used by the scoring module 706 to update the usage score 106.

[0180] In another example, the scoring module 706 can update the usage score 106 based on the location associated with the media reference 112. This can allow the measurement of the usage score 106 for the content objects 102 associated with a particular location.

[0181] The scoring module 706 can continuously update the usage score 106 of the content objects 102 to provide a real-time ranking of the content objects 102. The real-time ranking is based on the comparison between the usage score 106 of all of the content objects 102. The scoring module 706 can also compare current event information from the environmental feed module 118 of FIG. 1 against historical information from the data warehouse archive 630 of FIG. 6 to modify the usage score 106.

[0182] When the scoring module 706 has finished updating the usage score 106, the control flow can pass back to the detection module 704 if there are additional updates required. If no other updates are required, then the control flow can pass to the display module 708.

[0183] The scoring module 706 can include a user preferences structure 716 to support personalization for the content objects 102 having narrow areas of interest for each user. The scoring module 706 can update the usage score 106 based on the user preferences structure 716 to influence how the information should be prioritized and displayed. The user preference structure 716 can include user-based information such as device preference, content type preferences, timing preferences, transition preferences, or other content related user preferences.

[0184] The display module 708 is a module for displaying the content objects 102 based on different criteria. For example, the display module 708 can present one of the content objects 102 on the display unit 124 of FIG. 1. In another example, the display module 708 can display a portion of the media guide 202 of FIG. 2 on the display unit 124.

[0185] The selected one of the content objects 102 to be displayed on the display unit 124 can be determined in a

variety of ways. For example, the display module **708** can select one of the content objects **102** based on the usage score **106** and the creation location **610** of FIG. **6** within a distance threshold **712**. In another example, the display module **708** can select one of the content objects **102** based on the usage score **106** being above a usage score threshold **710**.

[0186] The usage score threshold **710** is a minimum value for the usage score **106**. For example, the usage score threshold **710** can be a value indicating a minimum level of interest. In another example, the usage score threshold **710** can be a value indicating a viral phenomenon level of interest. The usage score threshold **710** can be a multi-valued data structure having a variety of associated information. For example, the usage score threshold **710** can include a geographical location component, a language component, a genre component, a media type component, a topic component, or a combination thereof. Each of the components can be applied to the usage score **106** separately to identify the properties of the content objects **102**.

[0187] It has been discovered that comparing the usage score **106** to the usage score threshold **710** improves the technology for detecting the popularity of one of the content objects **102**. Changing the values of the usage score threshold **710** compensates for noise levels and identifies the content objects **102** having more significance.

[0188] Referring now to FIG. **8**, therein is shown an example of a functional block diagram of the multimedia content management and packaging system **100**. The multimedia content management and packaging system **100** can include a first device **801**, a second device **841** and a communication path **830**.

[0189] The first device **801** can communicate with the second device **841** over the communication path **830**. The first device **801** can send information in a first device transmission **832** over the communication path **830** to the second device **841**. The second device **841** can send information in a second device transmission **834** over the communication path **830** to the first device **801**.

[0190] For illustrative purposes, the multimedia content management and packaging system **100** is shown with the first device **801** as a client device, although it is understood that the multimedia content management and packaging system **100** can have the first device **801** as a different type of device. For example, the first device can be a server. In some examples, devices **801** and **841** may be nodes communicating over an application-defined overlay network as the communication path **830**.

[0191] Also for illustrative purposes, the multimedia content management and packaging system **100** is shown with the second device **841** as a server, although it is understood that the multimedia content management and packaging system **100** can have the second device **841** as a different type of device. For example, the second device **841** can be a client device.

[0192] For brevity of description in this embodiment of the present invention, the first device **801** will be described as a client device, such as a video camera, smart phone, or a combination thereof. The present invention is not limited to this selection for the type of devices. The selection is an example of the present invention.

[0193] The first device **801** can include a first control unit **808**. The first control unit **808** can include a first control interface **814**. The first control unit **808** can execute a first

software **812** to provide the intelligence of the multimedia content management and packaging system **100**.

[0194] The first control unit **808** can be implemented in a number of different manners. For example, the first control unit **808** can be a processor, an embedded processor, a microprocessor, a graphical processing unit, a hardware control logic, a hardware finite state machine (FSM), a digital signal processor (DSP), or a combination thereof.

[0195] The first control interface **814** can be used for communication between the first control unit **808** and other functional units in the first device **801**. The first control interface **814** can also be used for communication that is external to the first device **801**.

[0196] The first control interface **814** can receive information from the other functional units or from external sources, or can transmit information to the other functional units or to external destinations. The external sources and the external destinations refer to sources and destinations external to the first device **801**.

[0197] The first control interface **814** can be implemented in different ways and can include different implementations depending on which functional units or external units are being interfaced with the first control interface **814**. For example, the first control interface **814** can be implemented with electrical circuitry, microelectromechanical systems (MEMS), optical circuitry, wireless circuitry, wireline circuitry, or a combination thereof.

[0198] The first device **801** can include a first storage unit **804**. The first storage unit **804** can store the first software **812**. The first storage unit **804** can also store the relevant information, such as images, syntax information, videos, profiles, display preferences, sensor data, or any combination thereof.

[0199] The first storage unit **804** can be a volatile memory, a nonvolatile memory, an internal memory, an external memory, or a combination thereof. For example, the first storage unit **804** can be a nonvolatile storage such as non-volatile random-access memory (NVRAM), Flash memory, disk storage, or a volatile storage such as static random-access memory (SRAM).

[0200] The first storage unit **804** can include a first storage interface **818**. The first storage interface **818** can be used for communication between the first storage unit **804** and other functional units in the first device **801**. The first storage interface **818** can also be used for communication that is external to the first device **801**.

[0201] The first storage interface **818** can receive information from the other functional units or from external sources, or can transmit information to the other functional units or to external destinations. The external sources and the external destinations refer to sources and destinations external to the first device **801**.

[0202] The first storage interface **818** can include different implementations depending on which functional units or external units are being interfaced with the first storage unit **804**. The first storage interface **818** can be implemented with technologies and techniques similar to the implementation of the first control interface **814**.

[0203] The first device **801** can include a first media feed unit **806**. The first media feed unit **806** can receive the external media feeds for forming the content objects **102** of FIG. **1** and detecting the media items **714** of FIG. **7**. The first media feed unit **806** can include a television receiver, a radio receiver, a satellite receive, or a combination thereof.

[0204] The first media feed unit **806** can include a first media feed interface **816**. The first media feed interface **816** can be used for communication between the first media feed unit **806** and other functional units in the first device **801**.

[0205] The first media feed interface **816** can receive information from the other functional units or from external sources, or can transmit information to the other functional units or to external destinations. The external sources and the external destinations refer to sources and destinations external to the first device **801**.

[0206] The first media feed interface **816** can include different implementations depending on which functional units or external units are being interfaced with the first media feed unit **806**. The first media feed interface **816** can be implemented with technologies and techniques similar to the implementation of the first control interface **814**.

[0207] The first device **801** can include a first communication unit **810**. The first communication unit **810** can be for enabling external communication to and from the first device **801**. For example, the first communication unit **810** can permit the first device **801** to communicate with the second device **841**, an attachment, such as a peripheral device or a computer desktop, and the communication path **830**.

[0208] The first communication unit **810** can also function as a communication hub allowing the first device **801** to function as part of the communication path **830** and not limited to be an end point or terminal unit to the communication path **830**. The first communication unit **810** can include active and passive components, such as microelectronics or an antenna, for interaction with the communication path **830**.

[0209] The first communication unit **810** can include a first communication interface **820**. The first communication interface **820** can be used for communication between the first communication unit **810** and other functional units in the first device **801**. The first communication interface **820** can receive information from the other functional units or can transmit information to the other functional units.

[0210] The first communication interface **820** can include different implementations depending on which functional units are being interfaced with the first communication unit **810**. The first communication interface **820** can be implemented with technologies and techniques similar to the implementation of the first control interface **814**.

[0211] The first device **801** can include a first user interface **802**. The first user interface **802** allows a user (not shown) to interface and interact with the first device **801**. The first user interface **802** can include a first user input (not shown). The first user input can include touch screen, gestures, motion detection, buttons, sliders, knobs, virtual buttons, voice recognition controls, or any combination thereof.

[0212] The first user interface **802** can include the first display interface **803**. The first display interface **803** can allow the user to interact with the first user interface **802**. The first display interface **803** can include a display, a video screen, a speaker, or any combination thereof.

[0213] The first control unit **808** can operate with the first user interface **802** to display image information generated by the multimedia content management and packaging system **100** on the first display interface **803**. The first control unit **808** can also execute the first software **812** for the other functions of the multimedia content management and pack-

aging system **100**, including receiving image information from the first storage unit **804** for display on the first display interface **803**. The first control unit **808** can further execute the first software **812** for interaction with the communication path **830** via the first communication unit **810**.

[0214] For illustrative purposes, the first device **801** can be partitioned having the first user interface **802**, the first storage unit **804**, the first control unit **808**, and the first communication unit **810**, although it is understood that the first device **801** can have a different partition. For example, the first software **812** can be partitioned differently such that some or all of its function can be in the first control unit **808** and the first communication unit **810**. Also, the first device **801** can include other functional units not shown in FIG. **8** for clarity.

[0215] The multimedia content management and packaging system **100** can include the second device **841**. The second device **841** can be optimized for implementing the present invention in a multiple device embodiment with the first device **801**. The second device **841** can provide the additional or higher performance processing power compared to the first device **801**.

[0216] The second device **841** can include a second control unit **848**. The second control unit **848** can include a second control interface **854**. The second control unit **848** can execute a second software **852** to provide the intelligence of the multimedia content management and packaging system **100**.

[0217] The second control unit **848** can be implemented in a number of different manners. For example, the second control unit **848** can be a processor, an embedded processor, a microprocessor, a graphical processing unit, a hardware control logic, a hardware finite state machine (FSM), a digital signal processor (DSP), or a combination thereof.

[0218] The second control interface **854** can be used for communication between the second control unit **848** and other functional units in the second device **841**. The second control interface **854** can also be used for communication that is external to the second device **841**.

[0219] The second control interface **854** can receive information from the other functional units or from external sources, or can transmit information to the other functional units or to external destinations. The external sources and the external destinations refer to sources and destinations external to the second device **841**.

[0220] The second control interface **854** can be implemented in different ways and can include different implementations depending on which functional units or external units are being interfaced with the second control interface **854**. For example, the second control interface **854** can be implemented with electrical circuitry, microelectromechanical systems (MEMS), optical circuitry, wireless circuitry, wireline circuitry, or a combination thereof.

[0221] The second device **841** can include a second storage unit **844**. The second storage unit **844** can store the second software **852**. The second storage unit **844** can also store the relevant information, such as images, syntax information, video, profiles, display preferences, sensor data, or any combination thereof.

[0222] The second storage unit **844** can be a volatile memory, a nonvolatile memory, an internal memory, an external memory, or a combination thereof. For example, the second storage unit **844** can be a nonvolatile storage such as non-volatile random access memory (NVRAM), Flash

memory, disk storage, or a volatile storage such as static random access memory (SRAM).

[0223] The second storage unit **844** can include a second storage interface **858**. The second storage interface **858** can be used for communication between the second storage unit **844** and other functional units in the second device **841**. The second storage interface **858** can also be used for communication that is external to the second device **841**.

[0224] The second storage interface **858** can receive information from the other functional units or from external sources, or can transmit information to the other functional units or to external destinations. The external sources and the external destinations refer to sources and destinations external to the second device **841**.

[0225] The second storage interface **858** can include different implementations depending on which functional units or external units are being interfaced with the second storage unit **844**. The second storage interface **858** can be implemented with technologies and techniques similar to the implementation of the second control interface **854**.

[0226] The second device **841** can include a second media feed unit **846**. The second media feed unit **846** can receive the external media feeds for forming the content objects **102** and detecting the media reference **112** of FIG. 1. The second media feed unit **846** can include a television receiver, a radio receiver, a satellite receiver, or a combination thereof.

[0227] The second media feed unit **846** can include a second media feed interface **856**. The second media feed interface **856** can be used for communication between the second media feed unit **846** and other functional units in the second device **841**.

[0228] The second media feed interface **856** can receive information from the other functional units or from external sources, or can transmit information to the other functional units or to external destinations. The external sources and the external destinations refer to sources and destinations external to the second device **841**.

[0229] The second media feed interface **856** can include different implementations depending on which functional units or external units are being interfaced with the second media feed unit **846**. The second media feed interface **856** can be implemented with technologies and techniques similar to the implementation of the first control interface **814**.

[0230] The second device **841** can include a second communication unit **850**. The second communication unit **850** can enable external communication to and from the second device **841**. For example, the second communication unit **850** can permit the second device **841** to communicate with the first device **801**, an attachment, such as a peripheral device or a computer desktop, and the communication path **830**.

[0231] The second communication unit **850** can also function as a communication hub allowing the second device **841** to function as part of the communication path **830** and not limited to be an end point or terminal unit to the communication path **830**. The second communication unit **850** can include active and passive components, such as microelectronics or an antenna, for interaction with the communication path **830**.

[0232] The second communication unit **850** can include a second communication interface **860**. The second communication interface **860** can be used for communication between the second communication unit **850** and other functional units in the second device **841**. The second

communication interface **860** can receive information from the other functional units or can transmit information to the other functional units.

[0233] The second communication interface **860** can include different implementations depending on which functional units are being interfaced with the second communication unit **850**. The second communication interface **860** can be implemented with technologies and techniques similar to the implementation of the second control interface **854**.

[0234] The second device **841** can include a second user interface **842**. The second user interface **842** allows a user (not shown) to interface and interact with the second device **841**. The second user interface **842** can include a second user input (not shown). The second user input can include touch screen, gestures, motion detection, buttons, sliders, knobs, virtual buttons, voice recognition controls, or any combination thereof.

[0235] The second user interface **842** can include a second display interface **843**. The second display interface **843** can allow the user to interact with the second user interface **842**. The second display interface **843** can include a display, a video screen, a speaker, or any combination thereof.

[0236] The second control unit **848** can operate with the second user interface **842** to display information generated by the multimedia content management and packaging system **100** on the second display interface **843**. The second control unit **848** can also execute the second software **852** for the other functions of the multimedia content management and packaging system **100**, including receiving display information from the second storage unit **844** for display on the second display interface **843**. The second control unit **848** can further execute the second software **852** for interaction with the communication path **830** via the second communication unit **850**.

[0237] For illustrative purposes, the second device **841** can be partitioned having the second user interface **842**, the second storage unit **844**, the second control unit **848**, and the second communication unit **850**, although it is understood that the second device **841** can have a different partition. For example, the second software **852** can be partitioned differently such that some or all of its function can be in the second control unit **848** and the second communication unit **850**. Also, the second device **841** can include other functional units not shown in FIG. 8 for clarity.

[0238] The first communication unit **810** can couple with the communication path **830** to send information to the second device **841** in the first device transmission **832**. The second device **841** can receive information in the second communication unit **850** from the first device transmission **832** of the communication path **830**.

[0239] The second communication unit **850** can couple with the communication path **830** to send image information to the first device **801** in the second device transmission **834**. The first device **801** can receive image information in the first communication unit **810** from the second device transmission **834** of the communication path **830**. The multimedia content management and packaging system **100** can be executed by the first control unit **808**, the second control unit **848**, or a combination thereof.

[0240] The functional units in the first device **801** can work individually and independently of the other functional units. For illustrative purposes, the multimedia content management and packaging system **100** is described by opera-

tion of the first device **801**. It is understood that the first device **801** can operate any of the modules and functions of the multimedia content management and packaging system **100**. For example, the first device **801** can be described to operate the first control unit **808**.

[0241] The functional units in the second device **841** can work individually and independently of the other functional units. For illustrative purposes, the multimedia content management and packaging system **100** can be described by operation of the second device **841**. It is understood that the second device **841** can operate any of the modules and functions of the multimedia content management and packaging system **100**. For example, the second device **841** is described to operate the second control unit **848**.

[0242] For illustrative purposes, the multimedia content management and packaging system **100** is described by operation of the first device **801** and the second device **841**. It is understood that the first device **801** and the second device **841** can operate any of the modules and functions of the multimedia content management and packaging system **100**. For example, the first device **801** is described to operate the first control unit **808**, although it is understood that the second device **841** can also operate the first control unit **808**.

[0243] The multimedia content management and packaging system **100** can be partitioned between the first software **812** and the second software **852** in a variety of ways. The modules can be implemented in the first software **812**, the second software **852**, or a combination thereof.

[0244] For example, the first software **812** of the first device **801** can implement portions of the multimedia content management and packaging system **100**. For example, the first software **812** can execute on the first control unit **808** to implement the ingest unit **108** of FIG. **1** and the usage unit **110** of FIG. **1**.

[0245] The second software **852** of the second device **841** can implement portions of the multimedia content management and packaging system **100**. For example, the second software **852** can execute on the second control unit **848** to implement the scoring and aggregation unit **122** of FIG. **1** and the display module **708** of FIG. **7**.

[0246] The multimedia content management and packaging system **100** describes the module functions or order as an example. Each of the modules can operate individually and independently of the other modules. Furthermore, data generated in one module can be used by any other module without being directly coupled to each other.

[0247] The modules can be implemented in a variety of ways. For example, the modules can be implemented using hardware accelerators (not shown) within the first control unit **808** or the second control unit **848**, or can be implemented in hardware accelerators (not shown) in the first device **801** or the second device **841** outside of the first control unit **808** or the second control unit **848**.

[0248] The physical transformation of the external media feeds for forming the content objects **102** to displaying the content objects **102** on the pixel elements of the display unit results in physical changes to the pixel elements in the physical world, such as the change of electrical state the pixel imaging elements. The physical transformation is based on the operation of the multimedia content management and packaging system **100**. As the changes in the physical world occurs, such as the display of the content objects **102** on the display unit **124** of FIG. **1**, the operation of the multimedia content management and packaging sys-

tem **100** creates additional information that are converted back into changes in the pixel elements of the display unit **124** for continued operation of the multimedia content management and packaging system **100**.

[0249] Referring now to FIG. **9**, therein is shown a flow chart of a method **900** of operation of the multimedia content management and packaging system in a further embodiment of the present invention. The method **900** includes: receiving a content object in a block **902**; detecting a media reference to the content object from an external media feed in a block **904**; calculating a usage score for the content object based on the media reference in a block **906**; and displaying the content object having the usage score greater than a usage score threshold in a block **908**.

[0250] FIG. **10** schematically illustrates a conventional multichannel streaming system **20** that provides a prioritized multichannel streaming session. The multichannel streaming system **20** includes a streaming media server **22**, a client media player or receiver **26**, and a display device **28**. During operation of multichannel streaming system **20**, bidirectional communication between streaming media server **22** and client media receiver **26** occurs through a communications network **24**, while client media receiver **26** outputs video (and possibly audio) signals over a wired or wireless connection to display device **28**.

[0251] During a multichannel streaming session, streaming media server **22** transmits a prioritized streaming channel bundle **30** through communications network **24** to client media receiver **26**. Streaming channel bundle **30** can contain Over-The-Top (OTT) linear television (TV) programming.

[0252] The streaming media server **22** can include one or more content sources **32**, which feeds one or more encoder modules **34** under the command one or more control modules **36**, and transmits the encoded content to client media server **22** over communications network **24**.

[0253] When engaged in a multichannel streaming session, client media receiver **26** outputs visual signals for presentation on display device **28**. Display device **28** can be integrated into client media receiver **26** as a unitary system or electronic device. This may be the case when receiver **26** assumes the form of a mobile phone, tablet, laptop computer, or similar electronic device having a dedicated display screen. Alternatively, display device **28** can assume the form of an independent device, such as a freestanding monitor or television set, which is connected to client media receiver **26** (e.g., a gaming console, DVR, STB, or similar peripheral device) via a wired or wireless connection.

[0254] Client media receiver **26** may contain a processor **40** configured to selectively execute software instructions, in conjunction with associated memory **42** and conventional Input/Output (I/O) features **44**. I/O features **44** can include a network interface, an interface to mass storage, an interface to display device **28**, and/or various types of user input interfaces. Client media receiver **26** may execute a software program or application **46** directing the various hardware features of client media receiver **26** to perform functions. Application **46** suitably interfaces with processor **40**, memory **42**, and I/O features **44** via any conventional operating system **48** to provide functionalities.

[0255] As schematically shown in FIG. **10**, application **46** suitably includes control logic **50** adapted to process user input, obtain prioritized streaming channel bundle **30** from one or more content sources **32**, decode received streams, and supply corresponding output signals to display device

28. The streaming channels contained in prioritized streaming channel bundle **30** are decoded utilizing known techniques. In implementations, each channel stream contained in bundle **30** may be simultaneously decoded by a separate decoding modules. The decoding module or modules may be implemented using specialized hardware or software executing on processor **40**. Decoded programming can be provided to a presentation module **54**, which then generates output signals to display device **28**. In some embodiments, presentation module **54** may combine decoded programming from multiple streaming channels to create a blended or composite image; e.g., as schematically indicated in FIG. **10**, one or more PIP images **56** may be superimposed (i.e., overlaid) over a main or primary image generated on a screen of display device **28**.

[0256] In operation, control logic **50** of client media receiver **26** obtains programming in response to end user inputs received at I/O features **44** of receiver **26**. Control logic **50** may establish a control connection with remote streaming media server **22** via communications network **24** enabling the transmission of commands from control logic **50** to control module **36**. Streaming media server **22** may operate by responding to commands received from a client media receiver **26** via network **24**, as indicated in FIG. **10** by arrow **58**. Such commands may include information utilized to initiate a multichannel streaming session with streaming media server **22** possibly including data supporting mutual authentication of server **22** and receiver **26**.

[0257] FIG. **11** shows a multimedia content management and packaging system **1100**, which includes nodes **1102** and **1104**. Each of the nodes **1102**, **1104** may include a node application **1114** which is operably coupled with I/O **1106**, processor **1108**, memory **1110** (which may include content data storage unit **120**), OS **1112**, and display **124**. Node **1102** may be a commercial server, but one or both of nodes **1102** and **1104** can instead be user devices for viewing content.

[0258] In contrast to a single commercial server providing a plurality of multichannel streams in a one-to-many configuration with user devices, as shown in FIG. **10**, one embodiment implementation includes both (1) decoding by decoder(s) **1122**, at least a sub-plurality of programs from a multi-program transport stream (MPTS) **1120** for local (node) viewing and (2) re-transmitting stream **1120** over a partial mesh network topology (as streams **1120a**, **1120b**, and **1120n**) at the viewer/user application layer, although other configurations and arrangements of software, hardware, and network topologies can be implemented in other embodiments.

[0259] Node application **1114** may be operable to receive (e.g. via streaming media module **1116**) and transmit a plurality of MPTSs **1120a**, **1120b**, and **1120n** as part of an application layer overlay network **1126**. The overlay network **1126** may include, among other techniques, an application layer streaming video data according to one or more small-world topologies within a self-healing and self-organizing network (e.g., a small world wide-area peer-to-peer network).

[0260] Such a network enables nodes **1102** and **1104** to enter and leave the overlay network **1126** with minimum-to-no disruption to the presentation of programs of the MPTS streams (e.g., multichannel video streams) by other nodes that remain a part of the network. More generally stated, one or more of nodes **1102** and **1104** may be part of a peer-to-peer (P2P) overlay network, typically imple-

mented at the application layer, or a hybrid network thereof (e.g., a commercial server commutatively coupled to a P2P overlay network, with said server not being coupled as a node in the sense that it does not re-transmit and/or receive further programs that the network **1126** is sharing).

[0261] Embodiment configurations may include node-specific content delivery, feedback, and advertisement, among other things. For example, each stream **1120a**, **1120b**, and **1120n** may include the same channels, however, it is envisioned that a particular node may “mute” particular channels such that a node is locked from decoding certain channels received from a multichannel-stream-transmitting node. Such muted content may be restricted via direct user feedback such as age restrictions or calculated based on user preferences and other user feedback.

[0262] Decoder(s) **1122** may selectively decode at least a main program (e.g., an audio and/or audio-visual presentation) and at least one dynamic channel **1128**. As described in the context of FIG. **2** and elsewhere, a media guide can dynamically display other program content according to, among other possibilities, a usage score that can take into account stream-related data (e.g., detected content objects of a video stream) and interface said data with large data aggregation and analysis techniques. User “likes”, “follows”, and “shares” along with a determined number of node viewers and/or their social activity within a node application overlay network may influence usage score metrics that are updated throughout said network. Dynamic channel **1128** may “auto-surf” in the sense that channel **1128** changes programs and/or program presentations based on, in some examples, a network-updated usage score that can indicate the relative popularity of a program.

[0263] Streaming Media Tx Modules **1118**, **1118a**, **1118b**, and **1118n** respectively stream to node **1102** and further nodes (not shown). That is, node application **1114** can dynamically implement one or more a streaming media TX modules, as needed as part of a possible self-organizing and self-healing network **1126**. For example, the number of streaming media Tx modules **1118a**, **1118b**, and **1118n** may increase or reduce such that the number of MPTSs transmitted by node **1102** is correspondingly increased or reduced or said module(s) may reroute an MPTS to a respective receiving node within the network **1126**.

[0264] Nodes **1102** and **1104** may further receive and (re) transmit social media feed **114**, environmental feed **1180**, and business intelligence feed **1416** over the same network. Encoder(s) **1124** may encode content locally captured or stored on node **1102** or **1104** to provide video programming for MPTSs **1120** and/or **1120a**, **1120b**, **1120n**.

[0265] Nodes **1102** and **1104** may further include one or more of the ingest unit **110**, usage unit **110**, and scoring and aggregation unit **122**. In some embodiments, as shown in FIG. **12**, a usage score may be transmitted over network **1126** as metadata of MPTSs **1120** in metadata stream **1140**. Metadata stream **1140** may be a separate stream from MPTS **1120** and/or combined (or a part of) program data (e.g., multimedia data) of the MPTS **1120**.

[0266] To explain further, as seen in applications **1214** and **1224** of system **1200** of FIG. **12**, a usage score may be shared over network **1126** via metadata streams **1140**, **1140a**, **1140b**, and **1140n** as separate, discrete streams or as a combined stream with respective MPTSs **1120a**, **1120b**, and **1120n**. In some embodiments, for example, Watch-to-Earn

and Tag-to-Earn embodiments, usage score data may be locally generated and/or combined with received usage score data.

[0267] For example, as seen in system 1300 of FIG. 13, nodes 1302 and 1304 respectively include node applications 1314 and 1324, which both include facial detection unit 1306 as a sub-module of detection module 704. Wallet 1308 may be a digital wallet (e.g., a blockchain wallet) and/or a wallet module that interacts with an external hardware wallet (not shown). In some embodiments, wallet 1308 may include a non-fungible token (NFT) 1310 or other unique digital asset associated with a particular content object such as media-exposed person. In some embodiments, a digital currency (e.g., Ethereum ETH, Fantom FTM, or Polygon MATIC) is earned for each time facial detection unit 1306 detects a face associated with said NFT that is owned by or otherwise associated with wallet 1308 (e.g., wallet 1308 being the “owner” address or staking address of an NFT that is staked on a blockchain platform in order to earn digital currency from a streaming media platform).

[0268] In some embodiments, usage score 106 is used to determine the payout to wallet 1308. Said usage score 106 may be based on a single node’s usage score calculation. Usage score 106 may be based on facial detection determination on a per stream basis and/or a per node basis.

[0269] For example, usage score 106 may be determined based on a program stream of MPTS 1120, independently or regardless of how many nodes are playing said program stream on display 124. In some embodiments, a usage score 106 may represent an accumulated number of facial detections by each node that is displaying a program stream of MPTS 1120. That is, such a representation of usage score 106 reflects how many times (and/or other value such as facial detection duration) a particular face has been cumulatively detected by each node of network 1126 on a decoded program stream of MPTS 1120 that is being presented on a respective display unit 124.

[0270] In some embodiments, usage score 106 may be a composite score based on the program stream itself (e.g., a base usage score), which is modified based on the number of nodes that are presenting said program stream on display unit 124 (e.g., a modified usage score). In some embodiments, said base usage score may be the modifier and said modified usage score may be the base score (e.g., determining the base score based on the number of nodes that are playing relevant content on a respective display unit 124).

[0271] System 1400 of FIG. 14 includes nodes 1402 and 1404 with, respectively, node application 1414 with match-to-earn module 1408, tag-to-earn module, streaming media Tx modules 1118a and 1118, and wallets 1308a and 1308b. Module 1408 monetizes a user’s viewing of display unit 124. A user may earn, among other things, a blockchain currency (e.g., ETH, VET, FTM, MATIC) that is sent to the viewer’s wallet 1308a or 1308b. A viewer may earn said currency in a multitude of ways, including the duration of watching, the number of ads watched, social media activity relevant to content objects, or a combination of thereof, among other possibilities.

[0272] Module 1406 monetizes a user’s participating in data characterization for, among other reasons, categorizing content objects. For example, a user may interact with a (graphic) outlined area for tagging and/or confirming tags of individuals presenting, participating, or otherwise shown on a program. Tags may include real names and/or usernames

of social media platforms such as Twitter, Facebook, LinkedIn, and the like. Module 1406 may provide a drop-down menu while a user types a full names. Additionally or alternatively, the drop-down menu may provide a list of social media usernames or handles.

[0273] Advert module 1410 may provide and/or monitor advertising for billing or other accounting purposes. Advert module 1410 may be conditioned on a user engaging with the watch-to-earn module 1408 so that monetization, from both the user-watcher and advertiser perspectives, is tied together. In some embodiments, such a system enables for an efficiently use of advertising budgets and may not require any use of cookies or the like since monetization (and calculations related thereto) can occur on a per-node basis.

[0274] In some embodiments, advert module 1410 presents advertisements based on a detected plurality of content objects of a program that is being viewed. Such embodiments can present relevant advertising (with respect to said program) without the use of data, but rather utilizing, for example, video content analysis such as facial detection and other object detecting techniques.

[0275] System 1500 of FIG. 15 includes nodes 1502 and 1504 with streaming module 1518 of node application 1514. In some embodiments, streaming module 1518 streams MPTS 1520 (and MPTS 1520a, 1520b, and 1520n), inclusive of metadata such as a usage score(s) for content objects that are detected or otherwise identified in one or more programs of MPTSs 1520, 1520a, 1520b, and 1520n, among other examples

[0276] Nodes 1502 and 1504 also include overlay network module 1506 for establishing and maintaining, for example, peer-to-peer overlay network connections with other nodes. Example topologies that module 1506 may utilize include partial and full-mesh arrangements. Partial-mesh topologies may be established according to a small-world paradigm, among other techniques.

[0277] System 1600 of FIG. 16 includes nodes 1602 and 1604 with dynamic media guide module 1606 of application 1614. In some embodiments, said module 1606 dynamically displays one or more programs as explained in further detail below and with reference to FIG. 2. In some embodiments, module 1606 provides an array of programs, arranged by channel and relative usage, either directly measured by a node or derived from a social media feed, among other examples.

[0278] It is understood that any of the nodes 1102, 1104, 1202, 1204, 1302, 1304, 1402, 1404, 1502, 1504, 1602, and/or 1604 may include any of the modules and functions that are shown for one node, but not for another. For example, overlay network module may also be included in the other nodes besides 1602 and 1604 of FIG. 16. The particular groupings of modules, units, and functionalities for each node of FIGS. 11 to 16 are example groupings and not mutually exclusive ones.

[0279] Node applications 1114, 1214, 1314, 1414, 1514, 1614 may include any application that can run on user equipment such as a browser. In some embodiments, one or modules of said applications may be plug-ins running on a web browser application. In some embodiments, said applications may be a plug-in running on a web browser application (e.g., application 4414). In some embodiments, node applications are omitted and one or more of the modules

may run on a user-viewer's user equipment as a plug-in and/or application that is communicatively coupled to a server.

[0280] FIG. 17 shows display unit 124, which shows media guide 1701 with a main program 1700, which shows a person 1702 to be identified. Outlined area 1704 may outline at least a facial region of person 1702 for, among other possibilities, allowing a user to identify person 1702 or confirm the identity of person 1702 via an interaction with interactive area 1706. Banner 1707 shows a user's token balance 1708, which may be a total balance in a user's wallet and/or a session balance from one or more sessions. In some embodiments, user may claim their earnings periodically before transferring to a user's application wallet or hardware wallet, among other examples.

[0281] Usage score 1710 may show one or more scores or graphical indicators thereof for the main program 1700 and/or other programs being displayed on display unit 124. Channel rank 1712 may show a ranking of program 1700 within a channel that program 1700 is grouped in. Said grouping may be pre-packaged from a streaming device and/or thematically grouped by a node application, with or without user input or other assistance.

[0282] Overall rank 1714 may be a ranking of program 1700 that is relative to all programs being streamed by a node network or subnetwork thereof. Wallet UI 1716 may interact with a node application wallet or an external wallet to perform functions like connecting operably coupling with to the node application to receive, send, stake, and perform other wallet-related functions. In some embodiments, a user may be rewarded in a blockchain currency for "tagging" or identifying people such as media personalities and entering an identifying name, such as a real name or username of a node application or of another platform (e.g., Twitter®).

[0283] In some embodiments, a user may be rewarded with a blockchain currency for watching programs displayed on display unit 124 (e.g., watch-to-earn).

[0284] The graphic components of banner 1707 may be differently arranged across display unit 124. For example, wallet UI 1716 maybe detached from the presentation of main program 1700 such that it resides in a distinct area above, below, or to the side of program 1700. Banner 1707 may "disappear" after a decay duration and re-appear after a user interaction (e.g., a cursor being moved across or towards a lower portion of program 1700).

[0285] FIG. 18 shows media guide 1801 with main program 1800 and interactive area 1806 for a user to provide an objective determination as to whether person 1702, with outline 1704, is correctly identified as "Jon Bath". Thumbs up icon 1802 indicates an affirmation and thumbs down icon 1804 indicates that person 1702 has been misidentified (e.g., a false positive). In response to a user selecting icon 1802 or 1804, said user's token balance 1708 may increase from a tag-to-earn reward. In some embodiments, identified facial descriptors (of a video stream) that triggered a false positive identification may be deemed, in response to a user-viewer's feedback, as a "rejected facial descriptor" and associated (e.g., via a cluster ID) with a data cluster representing confirmed, rejected and/or identified facial descriptors of a media personality or other individual.

[0286] FIG. 19 shows media guide 1901 with a main program 1900 that shows people 1902a, 1902b, and 1902c with respective outlined areas 1904a, 1904b, and 1904c. Graphic IDs 1914a, 1914b, and 1914c respectively include

interactive areas 1906a, 1906b, and 1906c; photo IDs 1908a, 1908b, and 1908c; identified name 1910a, 1910b, and 1910c; and username 1912a, 1912b, and 1912c.

[0287] Similar to FIG. 18, interactive areas 1906a, 1906b, and 1906c provide icons for a user to affirm or disaffirm the individual being of the identity shown in a respective graphic ID 1914a, 1914b, and 1914c. In some embodiments, interactive areas may include the entire graphic ID so that, for example, clickable links can be included with one or more of photo IDs 1908a, 1908b, and 1908c; identified name 1910a, 1910b, and 1910c; and username 1912a, 1912b, and 1912c. In some embodiments, a user may directly enter a name or username or correct a name or username displayed within a graphic ID.

[0288] FIG. 20 shows interactive areas 2006a, 2006b, 2006c as an interactive graphic ID and further includes plus icon 2002a, 2002b, and 2002c, and bell icon 2004a, 2004b, and 2004c. The plus icons 2002a, 2002b, and 2002c may respectively allow for user interaction to "follow" or other related functionality the identified person. Additionally or alternatively, a user may interact with one or more of bell icons 2004a, 2004b, and 2004c to receive alerts, both within and "outside" of main program (e.g., as shown in FIG. 21) when the associated identified person is being shown on a program.

[0289] In some embodiments, said alerts may be time stamped and/or trigger the relevant program to be buffered by the node application in, for example, memory accessible by the node application. In some embodiments, the alert may be displayed for a predetermined amount of time, particularly as it relates to a buffering limit, which may be expressed in time, a memory threshold, among other possibilities. In some examples, the buffer may use "local" memory, but additionally or alternatively, buffering memory may be provided by other nodes connected to the node that displayed the notification (e.g., neighboring nodes).

[0290] Display 2124 of FIG. 21 shows desktop 2100 with overlaid notifications 2101, 2102, and 2104. A user may click on either notification 2102 or 2104 and, in response, launch or otherwise present a node application for playing a buffered program beginning at the timestamp associated with one of said notifications 2102 and 2104. Additionally or alternatively, notifications 2101, 2102, and 2104 may include audio notifications.

[0291] Media guide 2201 of FIG. 22 shows main program 2200 with dynamic channel 2202 and interactive area 2204 graphically overlaid said program 2200. Area 2204 may be a graphical presentation to provide subject and/or objective feedback from a user-viewer concerning, for example, main program 2200 or dynamic channel 2202. For example, area 2204 may allow a user to "like" various main program 2200 content for modifying a base usage score. A modified usage score is then used to determine, out of a plurality of streaming programs, which program has the highest usage score and thus is presented via dynamic channel 2202. In some embodiments, the usage score is periodically updated during program presentations, thus allowing for dynamic channel 2202 to "auto-surf" according to, for example, a customized usage score taking account a user-viewer's preferences and a base usage score derived from content object detection and/or other metadata (e.g., social media shares, likes, re-tweets, and the like).

[0292] Media guide 2301 of FIG. 23 shows main program 2300 with interactive area 2304, which may provide subject

or objecting feedback. Objective feedback may include confirmation or correction of facial identification information, voice identification information, and/or other content object identification information.

[0293] Media guide **2401** of FIG. **24** shows main program **2400** with interactive area **2404**, which may provide subject or objecting feedback. Objective feedback may include confirmation or correction of facial identification information, voice identification information, and/or other content object identification information.

[0294] Media guide **2501** of FIG. **25** shows main program **2500** with dynamic channels **2502a**, **2502b**, and **2502c**. In some embodiments, dynamic channels **2502a-c** may be graphically overlayed on main program **2500**, but in alternative embodiments, dynamic channels **2502a-c** may reside outside of the presentation of main program **2500** (such as above, below, or besides main program **2500**).

[0295] The dynamic channels **2502a**, **2502b**, and **2502c** may move up or down depending on a respective usage score such that dynamic channel **2502a** has the top usage score and channel **2502c** has the lowest usage score of the three dynamic channels. Channel **2502c** may be replaced with another program (and thus no longer shown as a dynamic channel) or move up the dynamic channel rankings to exchange positions with channel **2502a** or **2502b**.

[0296] For example, as shown in FIGS. **26** and **27**, media guide **2601** presents main program **2600** and animation **2604**, which indicates that dynamic channel **2602c** is swapping places with dynamic channel **2602b**, leaving dynamic channel **2602a** in the top spot. In some embodiments, at least one of a usage score and a competitiveness score **2612** is used to rank dynamic channels **2602a**, **2602b**, and **2602c**. In some embodiments a competitiveness score **2612** modifies a usage score. In some embodiments, the competitiveness score **2612** is periodically updated with, for example, a point differential between players or teams such that game programs with smaller point differentials are ranked as more competitive than other game programs and ranked as such. In some embodiments, a competitiveness score **2612** is weighted in relation to how many rounds or how much time is remaining in a game program. For example, a node application may detect, from video data, scores and remaining (or current) game time for modifying a competitiveness score weight for game programs that are near the end of a game and thus possibly achieving a higher overall rank **2614** than on a point-differential point basis alone.

[0297] As shown in FIG. **28**, media guide **2801** shows main program **2800** and a frame animation **2804** for highlighting dynamic channel **2602b**. In some embodiments, animation **2804** may indicate a competitiveness score while the presented order reflects a usage score ordering. In some embodiments, animation **2804** indicates which of the dynamic channels **2602a-c** has the relative highest score, which may be represented by meters **2908**.

[0298] As shown in FIGS. **29** to **33**, media guide **2901** shows an array **2900** of programs **2906** with accompanying meters **2908**. In some embodiments, array **2900** shows a decoded sub-plurality and/or all of the programs of an MPTS to display a plurality of, for example, video streams.

[0299] Programs **2906** may be organized, along the x-axis, in channels such as channels **2910** and **2912**. Programs **2906**, within a given channel, may be ranked according to, for example, a respective program usage score along pro-

gram ranking axis **2904**. Channels may be ranked according to, for example, a respective channel usage score along channel ranking axis **2902**.

[0300] As shown in FIG. **30**, animation **3000** shows channel **2910** moving up along the channel ranking axis **2902**. As shown in FIG. **31**, channel **2910** is now ranked higher than channel **2912**. As shown in FIG. **32**, program **2906a** is moving left, along program ranking axis **2904**, to indicate a higher usage score than program **2906b**. Animation **3202** may highlight this change in program ranking. As shown in FIG. **33**, program **2906a** is now ranked higher than program **2906b** along axis **2904**.

[0301] FIG. **34** shows a multimedia content management and packaging system **3400**, which includes nodes **3402** and **3404** with node application **3414**. Node application includes streaming Tx/Rx module **3406**, buffer-to-earn module **3408**, conditional unlock module **3410**, and share-to-earn module **3422**. Detection module **704** may include sports score and time detector **3412**, multimedia NFT detector module **3413**, song detector module **3416** (e.g., a Shazam® application or the like), and face detector module **3418**. Competitiveness score module **3420** may modify, replace, or supplement a usage score in determining, for example, a video program ranking.

[0302] Multimedia NFT detector module **3413** (e.g., an NFT-based application) may detect an audio or visual content object within a program of MPTS **3424**, particularly if said detected audio or visual content object is associated with an NFT or other unique digital asset that is associated with a wallet. In some embodiments, audio NFTs may represent an individual song or music group, with detections being detecting a particular song or song segment being played as main program content and/or background music of a main program. Visual NFTs may include personality NFTs that are linked to, for example, data clusters of facial detection parameters (e.g., grouped datasets of facial descriptors) for detecting individuals shown in a video program of MPTS **3424**. In some embodiments, by tagging shown individuals or confirming detection of shown individuals, facial descriptor datasets and/or cluster values may be expanded and/or refined via user-viewer feedback and triggering rewards or other earnings for the tagging/confirming user-viewer.

[0303] In some embodiments, module **704** may apply one or more video content analysis algorithms to a plurality of video programs of MPTS **3424** being transmitted over network **1126** for detecting content objects. In some embodiments, buffer-to-earn module **3408** allows nodes to buffer programs on behalf of neighboring nodes and earn a blockchain currency. In some embodiments, conditional unlock module **3410** may unlock one or more features of dynamic media guide module **1606** depending on, for example, a condition of a digital wallet. In some embodiments, share-to-earn module **3422** tracks a user's shares of content objects and/or programs of the MPTS **3424** and rewards a user in a blockchain currency for said sharing.

[0304] As an example of sports score and time detector **3412** in use, media guide **3501** of FIG. **35** shows sports video program **3500** with informational graphic **3502**. In some embodiment, sports score and time detector **3412** can detect information, through image and/or video analysis, such game period information **3504**, game time information **3506**, team information **3508**, and/or game score information **3510**. In some embodiments, detector **3412** detects

game time information, score information, and team information from a graphic of program **3500**.

[0305] In some embodiments, a competitiveness score **3512** can be calculated from said information. For example, an embodiment competitiveness score **3512** may use, as a base score, a point differential from information **3510**, such as $17-13=4$ as the base score. In some embodiments, a base score may be modified by at least one of game period information **3504** and game time information **3506**. In one example, a weight between 0 and 1 is provided in relation to how much time is remaining in a game, with, for example, the weight approaching 0 as a game nears the end of regulation (e.g., countdown to 0, last inning, nearing over-age time).

[0306] In some embodiments, the competitiveness score **3512** is determined by applying a “game time remaining” weight to the point-differential base score. Various sports programs can be ranked strictly by competitiveness score or in combination with usage score **1710**. In some embodiments, an overall rank **1714** may be calculated by the usage score being modified by an inverse of competitiveness score **3512** (e.g., a multiplicative inverse): a usage score of “10” being multiplied by “1/0.1” or “10”, which may be the inverse of a “0.1” competitiveness score **3512**.

[0307] In some embodiments, a reduction in a competitiveness score **3512** represents an increase in competitiveness. Said ranking may be displayed by showing the ranked sports program video in a ranked order as described above. Although the data for competitiveness score **3512** has been described as being extracted via analysis of a video signal, team, score, and game time information can also be obtained via metadata.

[0308] Media guide **3601** shows wallet interface **1716a** which shows audio NFT **3602**, facial recognition NFT **3604**, functional NFT **3606**, and claim button **3608**. Interface **1716a** may further shows blockchain currency data associated with the wallet address, including a wallet balance, a staked balance, NFTs, rewards received via NFTs, and session balances from watch-to-earn, tag-to-earn, and share-to-earn. NFTs **3602**, **3604**, and **3606** may be owned by a wallet and/or the NFTs **3602**, **3604**, and **3606** may be staked via a staking contract for earning blockchain currency.

[0309] Audio NFT **3602** may be an NFT that earns, for a wallet, a blockchain currency (e.g., EAT) when associated audio content is detected in a program. For example, audio NFT **3602** may be associated with a particular song or ensemble such that NFT **3602** earns EAT in response to detecting said song or a song from said ensemble in a program.

[0310] Facial recognition NFT **3604** earns, for an associated wallet address, a blockchain currency when an associated face is detected in a video program of, for example, a MPTS. NFT **3604** may represent facial recognition data of a media personality or the like to allow viewers, among others, to earn currency based on said media personality appearing in an MPTS.

[0311] Functional NFT **3606** may represent a particular media guide functionality. For example, NFT **3606** may allow for advertisement free displaying of programs of an MPTS and/or displaying one or more active channels. Other functional NFTs **3606** may include a rewards multiplier, decrease of claim fees, NFT rewards multiplier, transmit new programming to the network (e.g., an additional program to the MPTS), among other possibilities.

[0312] Conditional unlock module **3410** of FIG. **34** may utilize one or more of the wallet data points shown in FIG. **36** to unlock one or more functionalities of a media guide. Claim button **3608** may be a UI element for a user to transfer session balances to a wallet. In some embodiments, claiming may occur without a user manually claiming.

[0313] FIG. **37** shows method **3700** for displaying a plurality of programs. At step **3702**, a node receives a plurality of digital programs. In some embodiments, the plurality may be a MPTS (e.g., a plurality of video programs) received from another node via an application layer defined overlay network.

[0314] Step **3704** includes determining a content object occurrence in at least one digital program, thereby identifying a detected content object. In some embodiments, step **3704** may utilize facial recognition techniques to detect a particular individual (e.g., an individual that a user-view has “followed” and/or asked for notifications for when said individual appears in a video program as shown in, for example, FIG. **20**) and said detection may be verified, for example, via further subsequent detections in determining a content object occurrence.

[0315] Step **3706** includes in response to identifying the detected content object, buffering the at least one digital program, thereby providing a buffered digital program. The buffered digital program may be of a MPTS and would generally be “missed” unless the user-viewer was watching the program at a particular instance in time (e.g., a live peercast or other livecast or live streamed content).

[0316] Step **3708** includes providing, by the node and during the receiving step **3702**, a content object detection notification to a user of the node. One example of said notification can be found in FIG. **21** with notifications **2102** and **2104**. In some embodiments, step **3708** provides a timestamped notification associated with a time of the buffered program. In some embodiments, said time is approximately (or is) the time that the content object detection step **3704** occurs.

[0317] Step **3710a** includes determining if a buffer limit has been reached. Step **3710** includes discontinuing buffering the at least one digital program according to a buffer limit condition. The buffer limit condition may be, for example, temporal (e.g., a time limit) or a particular amount of buffered data (e.g., 1 GB). In some examples, neighboring nodes may “buffer-to-earn” to extend a buffer limit of a buffering node. In some examples, a user may be given an option to pay, in a blockchain currency, to access the surplus buffered program material provided by neighboring nodes after the local, user node reached a buffer limit.

[0318] In response to a user interaction with the content object detection notification that occurs before the buffer limit condition is met, step **3712** includes displaying the buffered digital program.

[0319] FIG. **38** shows method **3800** for a multimedia blockchain system. Step **3802** includes identifying a content object in at least one of a digital audio program, a digital video program, and a digital multimedia program, the content object being identified by analyzing at least one of content object voice data, content object image data, content object facial parameter data, content object digital video data, and content object digital audio data, thereby providing an identified content object. In some embodiments, step **3802** may utilize video content analysis (e.g., facial recognition techniques) for identifying a content object.

[0320] In some embodiments, content object data comprises or represents data clusters of one or more descriptors of content objects, which may be updated and further populated from multimedia program data. For example, said data clusters may populate content data storage unit 120 for identifying one or more facial features. As a consequence of a user-viewer tagging to earn, for example, datasets and/or data clusters may be expanded upon or otherwise updated based on a user providing input that an identified content object is correct or by manually tagging a content object by typing a name or entering an other identifier.

[0321] Step 3804 includes generating, in response to identifying the content object, a content object digital asset that is operable and unique within at least the blockchain system and representative of the identified content object. In some embodiments, step 3804 may be done automatically. Alternatively, a user may manually input an identified content object (e.g., a picture, a song, a video clip) into an NFT generator (e.g., an image generation engine comprising one or more image processors) for creating the content object digital asset (e.g., a unique image generated by an imputed image of a detected content object).

[0322] In some embodiments, the content object digital asset includes an NFT with metadata. In some embodiments, the NFT metadata may comprise unique identification data, unique class identification data, and/or image data such as generative image data based on an image of the identified content object. In some embodiments, the NFT metadata includes identification information that ties the identified content object to the NFT (e.g., a content object ID). In some embodiments, metadata includes data fields for controlling functionalities such as allowing ad-free viewing and/or earning currency based on one or more detections of the identified content object in a streamed video.

[0323] Optional step 3806 includes generating a digital visual representation of the identified content object for displaying the digital asset in a blockchain wallet interface. In some embodiments, the generating step 3806 includes receiving the content object itself (e.g., a picture of a media personality) and/or descriptor data of the content object (e.g., facial recognition data) as a basis for generating, via algorithms, image engines, or other process, the visual representation. In some embodiments, an image engine generates unique digital visual representations for each NFT based on an image input (e.g., a digital image depicting an identified content object).

[0324] Step 3808 includes detecting the identified content object in the at least one digital audio program, digital video program, and digital multimedia program and determining an accumulated detection value for the identified content object that represents a number of detections during a pre-determined time period. In some embodiments, the digital asset represents a facial recognition by video content analysis and step 3808 includes detecting, by facial recognition, the identified content object in the plurality of digital video programs, thereby identifying human individuals as the identified detected content objects.

[0325] Step 3810 includes determining a usage score based on the determined accumulated detection value, thereby providing a determined usage score. Step 3812 includes crediting, to a digital wallet associated with the digital asset, a blockchain currency amount that is based on at least the determined usage score. In some embodiments, the content object digital asset is a non-fungible token.

[0326] In some embodiments, the crediting step 3812 includes transferring, via a blockchain transaction, a blockchain currency (e.g., a cryptocurrency) to the digital wallet. In some embodiments, the crediting step 3812 may include a re-basing of the currency, reflections, and/or other techniques that increase the numerical number (quantity) of a cryptocurrency balance of a digital wallet. In some embodiments, step 3812 accumulates, via a smart contract, a claimable amount that only the digital wallet can claim. In response to a wallet owner claiming an accumulated balance (e.g., a user viewer interaction with claim button 3608 of FIG. 36), said balance is then credited to the digital wallet address.

[0327] FIG. 39 shows method 3900 of operation of a media guide system. Step 3902 includes operably coupling the system to a digital wallet of at least one blockchain. Operable coupling examples include a wallet or user-viewer providing a wallet address to the media guide system (e.g., an application running on a user equipment and/or a plugin of said application) and/or performing a signature via a digital wallet.

[0328] Step 3904 includes determining, by the system, if the digital wallet meets an unlocking condition. In some embodiments, step 3904 may be performed by conditional unlock module 3410. Unlocking conditions may be the presence of a type or number of NFTs, a type or balance of a blockchain currency of a wallet, a combination of NFT and currency requirements, a staked or unstaked condition of an NFT or currency, the amount of staked NFT and/or currency among other examples.

[0329] Step 3906 includes unlocking, by the system and at least partly based on the determining step affirming that the digital wallet meets the unlocking condition, at least one video program functionality of the system. Unlocked functionalities may be basic playback functionality, access to one or more programs, advertisement free playback by a player, allowing previously blocked programs of a MPTS to be decoded on a node associated with the wallet, display of active channel(s), and/or the use of any functionality such that a viewer/user cannot view any program of an MPTS or other audio or video stream without at least a minimum balance of a relevant blockchain currency, among other examples.

[0330] For example, the ability to encode and/or transmit program data as a “first” or “server” source of program data to share with the network as an “added” program to the MPTS (and not merely forward along a received MPTS) may require one or more functional NFTs and/or minimum balances on an associated wallet.

[0331] In some embodiments, an unlocking condition may be establishing an operable coupling with a wallet and media guide system (e.g., successfully performing step 3902). For example, watch-to-earn and tag-to-earn applications may require a connected wallet for modules 1406 and 1408 to be operable by a user-viewer. In some embodiments, earning rates are adjusted depending on the one or more states of a media guide player and/or a user device that is running the media guide player application. For example, muting a main presenting program (of an MPTS) may lower the watch-to-earn rate. Allowing access to a user device camera may raise a watch-to-earn rate for verifying that a user is actually viewing and/or listening to a program.

[0332] In some embodiments, earning rates for watch-to-earn and tag-to-earn may vary by program. For example,

advertising may pay higher rates than main programing and even variation among programs, when in a program a user starts viewing, and/or under what conditions (e.g., number of node viewers; competitiveness score).

[0333] In some embodiments, allotments for watch-to-earn tokens or other blockchain currency may be provided on a program and/or time basis (hourly, every half hour). That is, a fixed pool of a blockchain currency is allotted to be shared amongst the viewer-users that are watching (watch-to-earn), tagging (tag-to-earn), and/or buffering (buffer-to-earn).

[0334] In some embodiments, watch-to-earn rates may be contingent upon one or more settings of a node application (e.g., node application 3414) and/or a node (e.g., node 3402). For example, muting a program may reduce a watch-to-earn rate. In some embodiments, “muting” may be detected with “muting” (tag-to-earn), and/or lowering a program’s volume below a threshold.

[0335] In some embodiments, allowing access to a camera or microphone of a node for verification by the node application regarding a user-viewer being actively engaged with program content. In some embodiments, a camera may be a broadband imaging camera and/or a narrowband (e.g., infrared) camera for monitoring if a user-viewer is near a display. In some embodiments, a microphone may be detecting if multiple programs are being played at once and/or a minimum playback volume is being met for determining a watch-to-earn rate for a node.

[0336] FIG. 40 shows media guide 4001, which shows main program 4000 with an overlaid graphic ID 4006, which shows NFT image 4008, plus icon 4002, bell icon 4004, and NFT icon 4010. NFT image 4008 is an image associated with a detected content object of main program 4000. In FIG. 40, the detected content object is person 1902a, “Jon Bath”. In some embodiments, NFT image 4008 may represent the image data stored on a distributed ledger according to a token standard (e.g., ERC-721 or ERC-1155). Additionally or alternatively, NFT image and/or video data may be stored “off-chain” in servers and/or nodes. In some embodiments, NFT metadata that is stored “on-chain” includes hash value data for image, audio, and/or multimedia data that is stored in off-chain storage.

[0337] However NFT image 4008 is stored, guide 4001 presents a user-viewer the NFT image 4008 associated with detected person 1902a. As shown in FIG. 41, marketplace presentation 4100 shows NFT image 4008 with an NFT graphic 4008a, shown as a dodecagon that surrounds an eye feature 1902i. In some embodiments, NFT graphic 4008a may be based on VCA techniques including edge detection and facial recognition and generate parameter data (e.g., detected edges and/or facial features) that is visually highlighted in a content object image by an imagine engine utilizing, for example, image processing techniques such as filters and applying other image transform functions to the content object image for generating an NFT image. In this example, NFT graphic 4008a may represent an eye or facial detection data and may be generated based on at least said data.

[0338] In some embodiments, NFT image 4008 may generally show a previously identified personality with one or more NFT graphics 4008a. In some embodiments, the generated image of NFT image 4008 is unique due to the combination of the underlying content object image data and the NFT graphic(s) 4008a.

[0339] Returning to FIG. 40, NFT icon 4010 may include an active link to, for example, an NFT marketplace (e.g., marketplace presentation 4100) and/or other NFT presentations such as a data presentation (e.g., data presentation 4300 of FIG. 43). Presentations 4100 and 4300 may be presented within a media guide or as an external presentation (e.g., as a separate tab within a web browser (e.g., Firefox, Chrome, Brave, Edge)).

[0340] Returning to FIG. 41, interactive area 4106 may display identified name 1910a and a purchase icon 4102, which would initiate a purchase or bid, for example, for the NFT represented by NFT image 4008.

[0341] FIG. 42 shows method 4200 for a multimedia distributed ledger system. At step 4202, a content object occurrence is determined in a program. In some embodiments, a content object occurrence is determined by one or more content object detections. In some embodiments, an “occurrence” may be determined based on a plurality of content objection detections within a time subperiod. In some embodiments, a recent detection list is populated, refreshed, and checked to determine if a threshold number of detections for a particular content object is reflected in the recent detection list and thereby determining the content object occurrence. In some embodiments, each further detection added to the recent detection list that is above a threshold number may be determined as an occurrence.

[0342] One inventive feature is providing accurate and timely content object occurrence notifications to a user-viewer across, for example, multiple video program streams based on a plurality of content object detections for each respective stream.

[0343] At step 4204, method 4200 determines if the content object is associated with a content object digital asset. If not, method 4200 may offer tag-to-earn to a user-viewer at step 4206. In some embodiments, in response to a user-viewer tagging a content object (e.g., a media personality; famous animal; song title and performer), an NFT may be generated that is associated with the tagged content object. In some embodiments, a user-viewer is provided with one or more social media accounts via, for example, social media API 4424, for tagging an individual.

[0344] At step 4208, a visual representation of the content object digital asset is displayed to a user-viewer. Displaying may include graphic ID overlays onto a video stream. In some embodiments, a further displaying step 4210 may include displaying, in response to a user-viewer interaction with the visual representation of step 4208, the content object digital asset in a digital asset marketplace (e.g., OpenSea, Unify, Rarible). Step 4212 may include offering, via the digital asset marketplace, at least partial ownership of the content object digital asset. Ownership may be wholly transferred into a single “owner” wallet or shared among multiple wallets. Shared ownership may include, for example, proportionate shares in accumulated cryptocurrency generated by and in relation to a number of determined content object detections over a given time period.

[0345] Depending on step 4214, method 4200 may, at step 4216, associate, on a distributed digital ledger, a digital wallet with the purchased content object digital asset. In some embodiments, a smart contract or interface thereof facilitates said association. At step 4218, method 4200 may end.

[0346] FIG. 43 shows media guide 4301 with NFT presentation 4300, which may include data card graphic 4302,

which shows identified name **1910a**, NFT Type, NFT Total Rewards, NFT Pending Rewards, Wallet Owner Address, Average Daily Occurrences, Average Daily Detected Time. Data card graphic **4302** may further provide NFT icon **4304** for an NFT marketplace URL or the like.

[0347] FIG. **44** shows multimedia distributed ledger system **4401**, which may include NFT smart contract **4402**, content object NFT database **4404**, browser application **4406**, wallet **4408**, image engine **4422**, and social media API **4424**. Browser application **4406** may include application **4414** and/or one or more modules thereof (e.g., modules implemented as a plug-in). In some embodiments, application **4414** may receive a video stream and/or other media stream from a content delivery network (CDN), which may be a server, in addition or alternatively to receiving a MPTS from an overlay network. In some embodiments, application **4414** may be a node application.

[0348] Contract **4402** may have fields for or otherwise process data structures **4410** such as a smart contract address, a token ID, a token URI, which is generally a reference to associated NFT data (e.g., image or video data associated with the token ID), a token name, a token owner (e.g., a wallet address), functional metadata **4412** such as token type (e.g., ad-blocker, content object, personality, transmission allowance) and content object ID, and media metadata (e.g., an image) that may be stored on-chain.

[0349] In some embodiments, an NFT is uniquely identified by the combination of a contract address and token ID. In some embodiments, functional metadata **4412** may enable, disable, or otherwise modify one or more functionalities of application **4414**, including functionalities related to the playback of video content and/or monetization thereof.

[0350] Contract **4420** may include functions such as token burn (e.g., “destroy” an NFT by sending it to a wallet address that is generally inaccessible (e.g., 0x . . . 0000 and 0x . . . dead)), token mint (e.g., create an NFT), token transfer between wallets, credit pending tokens to an NFT owner’s wallet, get the token ID, get the token type, display the media metadata, and/or toggle an ad-free mode setting.

[0351] Content object NFT database **4404** may have data structures **4410** (e.g., database field formats) or otherwise process and/or store data related to detections of associated content objects in a program presentation such as an detection channel, detection time, last detection time, ID information of a detected content object (e.g., a content object ID associated with one or more VCA descriptors), a content object NFT ID, recently detected content object IDs, pending NFT credit balance, and/or a content object image (e.g., an image of a content object (e.g., a facial image) or a reference thereto (e.g., a URL to a content object image)).

[0352] In some embodiments, total detections may reflect a historical accumulated number of detections. In some embodiments, periodic total detections reflect a number of detections over an hourly, daily, or other time period (e.g., 1 to 24 hours) that is refreshed after said time period. In some embodiments, fungible digital assets and/or NFTs may be earned (e.g., increase a credit balance) based on the periodic total detections of a content object that is associated with an NFT.

[0353] In some embodiments, content object IDs of recent detections may be periodically updated in real or near-real time in database **4404** to reflect a raw number of content object detections. In some embodiments, recently detected

content object IDs is a list with a fixed number of entries (e.g., 3 to 15 content object IDs). In some embodiments, an occurrence is at least partially determined if a content object ID is entered multiple times (e.g., a content ID appearing two or more times in a fixed data structure (e.g., a fixed array or buffer) may result in a determined occurrence).

[0354] FIG. **45** shows method **4500** for determining a relative usage score. Step **3810a** includes determining a total of accumulated detections of at least a subset of identified content objects. In some embodiments, the subset may be restricted to identified content objects that are associated with an NFT. In some embodiments, the subset may be restricted by requiring a minimum number of detections of a content object. In some embodiments, the pre-determined time period is 24 hours.

[0355] Step **3810b** includes determining the usage score for the identified content object based on a ratio between the cumulative detections of the identified content object and, for example, the sum of cumulative detections of the at least subset of identified content objects. For example, a given identified content object may have been detected 10 times over a time period and the at least subset has a cumulative detection sum of 100 over the same time period. In such cases, in some embodiments, an associated wallet and/or NFT is credited, directly to a wallet or as a pending credit to be claimed, one-tenth of the daily rewards pool, typically comprising a pool of one or more digital assets (e.g., a pool of one or more cryptocurrencies, NFTs, and the like).

[0356] FIG. **46** shows method **4600** for determining an occurrence of a pre-identified content object in a digital video. Step **4602** includes sampling a video stream for content object descriptors. In some embodiments, sampling rates may be at least one frame per second. In some embodiments, content object descriptors may include facial descriptors. In some embodiments, content object descriptors may include a plurality of descriptors such as those related to facial detection and character recognition of on-screen text and/or closed captions data (e.g., subtitles). In some embodiments, a face detection model samples facial descriptors for each detected face. In some embodiments, the face detection model generates vector values as the facial descriptors.

[0357] Step **4604** includes determining a minimum Euclidean distance between the sampled content object descriptor and the closest content object descriptor of a plurality of content object descriptors. In some embodiments, the plurality of content object descriptors may be clustered facial descriptors of previously identified individuals. In some embodiments, step **4604** includes calculating a squared Euclidean distance between the sampled content object descriptor and the closest descriptor of a plurality of descriptors.

[0358] Step **4606** includes applying at least one threshold to the determined minimum Euclidean distance of step **4604**. Step **4606** may include a maximum threshold such that distance values determined by step **4604** that are below (or, in some embodiments, below or equal to) the maximum threshold value are processed as a positive identification of a pre-identified content object. If a determined distance value is above (or, in some embodiments, above or equal to) the maximum threshold value, method **4600** may end **4622**, at least with respect to further steps related to those particular content object descriptor samples. In some embodiment, step **4602** is continuously occurring on a video stream

independently of any end step **4622** that is reached in example embodiments, thereby continuously providing, in real-time or near real-time, content object descriptor samples of a video.

[0359] Step **4610** includes adding, to a content object detection list, a content object ID that is associated with the closest content object descriptor. In some embodiments, a content object ID is associated with a cluster of detected features for identifying a particular content object. Step **4612** includes determining if a content object ID has been detected at least a minimum number of times to meet an occurrence threshold. If not, method **4600** may end **4622**.

[0360] In some embodiments, if a content object ID occurs at least two (or more) times on the list, an occurrence of the content object ID has been determined, and step **4618** includes providing a content object occurrence notification to a user viewer. In some embodiments, a recent detection list of content object IDs with a fixed maximum number of entries (e.g., 10 to 20 content object IDs) is updated in real time or near-real time. In some embodiments, a content object ID must appear at least three times on a recent detection list before a content object occurrence (vs. a mere detection) has been determined.

[0361] In some embodiments, the recent detection list is updated based on a content object sampling rate (e.g., newly detected content object IDs are provided every second or a period thereof (e.g., 2 to 5 seconds)). In some embodiments, a recent detection list provides a list of content IDs in an order related to their relative detection times (e.g., a list of 10 content object IDs are the last 10 content objects that were detected).

[0362] FIG. **47** shows method **4700** for generating content object NFTs. Step **4702** includes identifying, in a video stream and utilizing video content analysis (VCA), a content object. Step **4704** includes determining if an NFT associated with the identified content object has already been created or “minted”. If so, method **4700** may return to step **4702**. In some embodiments, step **4702** is continuously being performed regardless of the determination made by step **4704** (e.g., steps **4702** and **4704** may be performed in parallel). If not, step **4706** may check if a content object has met a particular condition such as being identified a minimum number of times, identify confirmation by one or more user-viewers, minimum number or data size of content object descriptors, among other examples. If the condition is not met, method **4700** may return to step **4702**.

[0363] At step **3804a**, an image engine (e.g., image engine **4422**) processes at least one image of the identified content object, thereby producing a content object graphic. Step **3804b** includes utilizing a smart contract to associate the content object graphic with an NFT ID of a distributed ledger system. In some embodiments, the NFT ID is a smart contract address and a token ID. Step **4708** may include offering at least partial ownership of the newly created NFT in an NFT marketplace.

[0364] FIG. **48** shows method **4800** for generating an NFT. Step **4802** includes detecting, by video content analysis of a video stream, facial descriptors. Step **4804** includes associating, in a database, the facial descriptor cluster values with an identification value. In some embodiments, the identification value may represent a particular individual. Step **4806** includes comparing the detected facial descriptors with a facial image of a social media profile. In some embodiments, the profile may be of the particular individual

associated with the identification value. Step **4808** includes determining if the results of the comparison step of **4806** satisfies a similarity threshold such as minimum distance between a facial descriptor of the facial image and a facial descriptor cluster value. If not, method **4800** may reach an end step **4816** with respect to a particular facial image. In some embodiments, method **4800** may select further facial images for step **4806**.

[0365] If the similarity threshold is satisfied, step **4810** may process, in response to step **4808**, the facial image utilizing one or more transforms and/or style transfers (e.g., a neural style transfer), thereby producing a facial image graphic. An example method is provided in FIG. **51**. Step **4812** includes associating, via a smart contract and in response to the image engine processing step **4810**, the facial image graphic with an NFT ID of a distributed ledger system (e.g., a blockchain system), thereby providing an NFT. Optional step **4814** includes associating, in a database (e.g., an NFT database), the NFT ID with the identification value. Thus, in some embodiments, the NFT minted by step **4812** is associated with an individual that is represented by the identification value.

[0366] FIG. **49** shows method **4900** for generating an NFT. Step **4902** includes determining if the results of the comparison step of **4806** satisfy a similarity threshold such as minimum distance between a facial descriptor of the facial image and a facial descriptor cluster value. If not, method **4900** may select a further facial image for step **4806**. Step **4904** may determine if only a single face is shown in the facial image. If not, method **4900** may return to step **4806** for a further facial image. If so, step **4906** may scale and/or crop the facial image. These steps may provide relatively uniform facial images. Step **4908** may apply text recognition to determine if the facial image includes text. In some embodiments, method **4900** returns to step **4806** for comparing a further facial image with one or more facial descriptor cluster values.

[0367] Step **4910** includes processing, in response to not detecting text, the facial image, thereby producing a facial image graphic. Method **4900** may then advance to step **4812**.

[0368] FIG. **50** shows descriptor module **5000** including content object descriptor sampler module **5002**, similarity module **5004**, cluster module **5006**, occurrence module **5008**, and detection module **5010**. In some embodiments, sampler module **5002** may sample one or more descriptors at a given frame rate or sampling period (e.g., every second). In some embodiments, similarity module **5004** may calculate similarity scores between, for example, sampled content object descriptors and content object cluster values (e.g., cluster radius, cluster center/average (mean and/or median), a descriptor closest to a cluster average, a descriptor farthest away (e.g., a Euclidian distance) from the cluster average).

[0369] Content object descriptor cluster module **5006** may calculate and/or cause to store a plurality of content object cluster values. In some embodiments, a cluster value may be a descriptor of a cluster, such as descriptors that are closest and/or farthest away from a cluster center. In some embodiments, a cluster value may be characteristics of a cluster such as an average or center of a cluster and/or a cluster radius.

[0370] Content object occurrence module **5008** determines if a content object has been detected a sufficient number of times and/or meets other occurrence thresholds (e.g., multiple facial detection plus on-screen character

recognition of the media-exposed personality's name that confirms the facial detections). Content object detection module **5010** detects content objects within a program presentation. In some embodiments, module **5010** includes a face detection module that provides facial descriptors for sampler module **5002**.

[0371] FIG. **51** shows image engine **4422**, which includes style transfer module **5102**, facial feature parameter module **5104**, image segmentation module **5106**, and composite image module **5106**. In some embodiments, module **5102** transfers a style from a source image to an NFT image. In some embodiments, module **5102** applies neural style transfer techniques.

[0372] Facial feature parameter module **5104** obtains, for example, facial key points from image data. In some embodiments, the facial key points serve as a basis for image segmentation module **5106** to segment an image. In some embodiments, a different style transfer is applied for each image segment. Composite image module **5106** may receive a plurality of stylized images and generate a composite image of said stylized images. In some embodiments, the composite image is stylized differently in each image segment of the composite image.

[0373] FIG. **52** shows a method **5200** for generating an NFT. FIGS. **53.1**, **53.2**, and **53.3** show example images **5300**, **5308**, **5310**, **5312**, and **5314** that may be processed and/or generated by method **5200**. Step **5202** includes obtaining facial key points based on a selected facial image. Step **5204** includes segmenting, based on the facial key points, the selected facial image into a plurality of facial image segments (e.g., performing image segmentation). For example, selected image **5300** of FIG. **53.1** depicts face **5301** with key points **5303** outlining a face or head area and dividing face **5301** into a left facial section **5302**, a right facial section **5304**. In some embodiments, background section **5306** may be the area outside of outermost key points. Selected image **5300** has been segmented into three sections although fewer or more segments are possible. In some embodiments, image segments are distinct, possibly non-overlapping areas of an image.

[0374] Step **5206** includes generating, via image style transfer, a stylized image for each facial image segment. In some embodiments, the image style transfer step **5206** includes a neural style transfer using a respective source style image for each image segment. For example, FIG. **53.2** shows a respective style source image **5308**, **5310**, and **5312** for left facial section **5302**, right facial section **5304**, and background section **5306**.

[0375] Step **5208** includes generating a composite image of each stylized image, thereby producing a facial image graphic, with an example composite image **5314** shown in FIG. **53.3**. Step **5210** includes minting, via a smart contract and on a blockchain system, an NFT comprising at least one of a facial image graphic and a facial image graphic location. In some embodiments, NFT metadata may store the facial image graphic on-chain. Alternatively or additionally, NFT metadata may store an address (URL, URI) to where, off-chain, the facial image graphic is stored.

[0376] FIG. **54** shows media guide **5401** with main program **5400**, coming next notification **5402**, and address bar **5404**. Graphic IDs **1914a** and **1914b** respectively identify people **1902a** and **1902b**. Notification **5402**, in some embodiments, is shown based on a facial detection or occurrence by an upstream node and/or server, which may

be receiving a stream slightly ahead of an overlay network node. In some embodiments, an upstream server may transmit graphic IDs, "coming next", and other identifications to one or more nodes of an overlay network.

[0377] FIG. **55** shows main program **5400** advanced to showing only Gert Mann **1902c** with accompanying graphic ID **1914c**. For example, the main program **5400** of FIG. **54** may be a previous scene or camera that switches, in a few seconds or fewer, to showing Gert Mann **1902c**. Media guide **5401** is thus capable of providing, in real time, interactive icons and information related to currently displayed media personalities and soon-to-be-displayed media personalities.

[0378] Media Guide **5401** may interact with cursor **5502**. For example, frame **1904c** may be clickable by bringing cursor **5502** close to or within frame **1904c**, as shown by FIG. **56** with frame **1904c** graphically changing from a dashed lined to a solid line. In some embodiments, a frame may change colors or other graphic change to indicate to a user-viewer that said frame is interactable with mouse clicks or similar inputs. If cursor **5502** is outside this interactive area of frame **1904c**, as show in FIG. **55**, a mouse click will not trigger an interaction with frame **1904c**.

[0379] As shown in FIG. **57**, one interactive response is providing text entry box **5702** (e.g., a pop-up window). Box **5702** accepts a real name and/or a social media handle or username **5704** text for mapping a social media profile to facial detections (or confirmation of previous manual and/or algorithmic mappings).

[0380] FIGS. **58A** and **58B** show a method **5800** for determining facial descriptor cluster values. After step **4804**, step **5802** includes assigning each detected facial descriptor to a respective cluster. Step **5804** includes merging the clusters based on a square (Euclidean) distance threshold value. For example, clusters with a distance value that is equal and/or under a square distance of 0.44 are merged. Step **5806** determines if the merge is complete (e.g., all "mergeable" clusters have been merged). If so, step **5808** includes identifying which of the merged clusters includes the most facial descriptors. In some embodiments, step **5808** establishes the cluster and data derived therefrom for a personality identification value.

[0381] Step **5810** includes determining based on the identified cluster of step **5808**, at least one cluster value. Example cluster values may include an average of all facial descriptors (e.g., a cluster center), a facial descriptor closest to the average, a facial descriptor farthest from the average, and a distance of the farthest facial descriptor from the average (e.g., a cluster radius). Cluster values may be facial descriptors, distance values derived from facial descriptors, and/or average values derived from clustered facial descriptors.

[0382] Step **5812** includes associating, in the database, the determined cluster values with the personality identification value. The cluster values of step **5812** may server as a basis for determining facial identification and/or occurrences in a video stream. Optional step **5814** may include associating, in the database, rejected facial descriptors with the personality identification value. In some embodiments, a user-viewer provides feedback on a false identification of a personality. The facial descriptors that trigger the false identification may be utilized in further detection and/or occurrence processes (e.g., in method **5900**). Optional step **5816** may include associating, in the database, user-viewer confirmed

facial descriptors with at least one of the personality identification value and the merged cluster. For example, the user-viewer confirmed facial descriptors may be added to a dataset that is tied to a personality identification value. Optional step **5818** may include updating the merged cluster value(s) based on at least the user-viewer confirmed facial descriptors. For example, step **5818** may include re-calculating or otherwise updating a cluster center or average, a facial descriptor closest to the average, a facial descriptor farthest from the average, and/or a cluster radius at least partly based on the user-viewer confirmed facial descriptors for a particular personality.

[0383] FIGS. **59A** and **59B** show method **5900** for facial recognition. Step **5902** includes calculating a distance value based on a sampled facial descriptor and a plurality of cluster averages. In some embodiments, each cluster average may be representative of an individual. Step **5904** includes calculating a potential distance value based on at least the calculated distance value and a cluster radius value. In some embodiments, the cluster radius value is subtracted from the calculated distance value. Step **5906** determines if the potential distance value is below a current minimum distance, which may be initially set to a large value. If below, step **5908** includes calculating a distance value based on the detected facial descriptor and a closest rejected descriptor of the cluster. Step **5910** includes calculating a distance value based on the detected facial descriptor and a closest clustered descriptor, with respect to the detected facial descriptor, of the cluster. Step **5912** determines if the determined distance value of **5910** is less than the determined distance value of **5908**. If not, method **5900** may return to step **5908**.

[0384] If so, step **5914** determines if the determined distance value is less than the current minimum distance. If not, method **5900** may return to step **5908**. If so, method **5900** may progress to step **5916**, which designates the current cluster as the closest cluster. If there are further distance values to calculate or otherwise process at step **5917**, step **5924** may set the current minimum distance equal to the distance value that was calculated by step **5910** and method **5900** may return to step **5904**.

[0385] If not, method **5900** may progress to step **5918**, which includes providing, for the personality ID associated with the closest cluster, a facial detection indication. Said indication may include adding a personality ID to a recent detection list of a video program/stream. Said indication may include updating a daily counter, which accumulates the number of discrete detections over a given time period. Said indication may include providing a notice to a user-viewer of detecting a face, in a video stream, that is associated with a personality ID.

[0386] Optional step **5920** includes updating at least one closest cluster value based on at least the detected facial descriptor. In some embodiments, step **5920** may include re-calculating or otherwise updating a cluster center or average, a facial descriptor closest to the average, a facial descriptor farthest from the average, and/or a cluster radius. In some embodiments, step **5920** may include adding the detected facial descriptor to the dataset of a particular cluster (e.g., a clustered dataset). Additionally or alternatively, cluster values are updated based on user-viewer confirmations (e.g., step **5818**). Method **5900** may then end at step **5922**.

[0387] FIG. **60** shows facial descriptor database **6000**, which may have data structures **6002** (e.g., database field

formats) or otherwise process and/or store data related to facial descriptors and rejected facial descriptors. Facial descriptor data may include one or more of identified facial descriptors from a video stream. Facial descriptors may be pre-populated and/or updated with, for example, further sampled facial descriptors from subsequent video streams, with or without user-viewer input or feedback. Rejected facial descriptors may be expanded or otherwise updated via user feedback concerning one or more facial descriptors that triggered a false identification of a personality.

[0388] Data structures **6002** may further include a cluster average (and/or cluster center), a cluster radius, an average facial descriptor (e.g., a facial descriptor closest to the cluster center), a boundary facial descriptor (e.g., a facial descriptor furthest away from the cluster center), a cluster ID, a personality ID, and/or a facial NFT ID.

[0389] It has been discovered that the present invention thus has numerous aspects. The present invention valuably supports and services the historical trend of reducing costs, simplifying systems, and increasing performance. These and other valuable aspects of the present invention consequently further the state of the technology to at least the next level.

[0390] Thus, it has been discovered that the multimedia content management and packaging system of the present invention furnishes important and heretofore unknown and unavailable solutions, capabilities, and functional aspects for processing image content. The resulting processes and configurations are straightforward, cost-effective, uncomplicated, highly versatile and effective, can be surprisingly and unobviously implemented by adapting known technologies, and are thus readily suited for efficiently and economically manufacturing devices fully compatible with conventional manufacturing processes and technologies. The resulting processes and configurations are straightforward, cost-effective, uncomplicated, highly versatile, accurate, sensitive, and effective, and can be implemented by adapting known components for ready, efficient, and economical manufacturing, application, and utilization.

[0391] While the invention has been described in conjunction with a specific best mode, it is to be understood that many alternatives, modifications, and variations will be apparent to those skilled in the art in light of the above description. Accordingly, it is intended to embrace all such alternatives, modifications, and variations that fall within the scope of the included claims. All matters set forth herein or shown in the accompanying drawings are to be interpreted in an illustrative and non-limiting sense. For example, distance calculations between, for example, facial descriptor vector values may be Euclidian distances, Manhattan distances, Average Distances (e.g., a modified Euclidean distance calculation), weighted Euclidian distances, Chord distances, and/or non-Euclidian distances, among other examples.

[0392] As another example, embodiments may include Video-on-demand streams provided by a commercial server to a user device (e.g., computer, smart phone, streaming device, gaming console) that is running a watch-to-earn application (e.g., a media player application) that is operably coupled to a digital wallet. Embodiment streams include audio streams, video streams, and interactive streams (e.g., video game streaming or “cloud gaming”), among other media streaming examples.

[0393] As another example, embodiments may implement one or more modules (e.g., watch-to-earn module **1408**) as

a browser plug-in, a standalone application, an embedded module in a video player, or an embedded module in a web page containing a video player, among other possible software architectures. In some embodiments, said one or more modules may be communicatively coupled to a server.

1. A method of operation of a multimedia content management and packaging system comprising:

detecting, by video content analysis, content objects of at least one video program, thereby identifying detected content objects;

displaying the at least one video program;

detecting, from an external media feed, media references to at least a sub-plurality of the detected content objects;

calculating, while the at least one video program is being displayed, a usage score for the at least sub-plurality of detected content objects based on at least the media references, the calculating step comprising utilizing a time decay factor for at least partially determining the usage score, thereby providing a calculated usage score; and

displaying, while the at least one video program is being displayed, a usage score graphic according to the calculated usage score.

2. The method of claim 1 wherein the external media feed includes a social media feed for detecting the media references on a social media site.

3. The method of claim 1 wherein the external media feed includes a usage feed for detecting the media references.

4. The method of claim 1 wherein the external media feed includes an external media feed for detecting the media references on a broadcast or digital media system.

5. The method of claim 1 with the calculating the usage score step comprising updating the usage score based on a quality score.

6. The method of claim 1 with the calculating the usage score step comprising modifying the usage score according to a competitiveness score.

7. The method of claim 1 with the detected content objects step comprising, detecting, by facial recognition, content objects in the at least one video program, thereby identifying human individuals as the identified detected content objects.

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29. A method for displaying a plurality of programs, the method comprising:

receiving, by a node, a plurality of video programs;

transmitting, by the node and during the receiving step, the plurality of video programs on an application-layer defined overlay network;

displaying, by the node and during the receiving and transmitting steps, at least two distinct programs from the plurality of video programs, the displaying step comprising displaying a first video program;

detecting, by video content analysis, a content object in at least a second video program, thereby identifying a detected content object; and

the displaying step further comprises displaying, based on the detected content object, the second video program overlayed on or next to the first video program.

30. The method of claim 29 the detection step comprising detecting content objects in at least the second video program, thereby identifying detected content objects, and the displaying step comprising displaying, based on the detected content objects, the second video program overlayed on or next to the first video program.

31. The method of claim 29 further comprising receiving, by the node, node-specific subjective feedback for the detected content object of the first video program, the displaying step comprising displaying, based on the node-specific subjective feedback, the second video program overlayed on or next to the first video program.

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62. A method for providing at least near real-time notifications for a plurality of digital programs, the method comprising:

receiving, by a server, the plurality of digital programs; transmitting, by the server and during the receiving step, the plurality of digital programs to at least one node of an overlay application network;

detecting, by the server and during the receiving and transmitting steps, a content object in at least one digital program of the plurality of digital programs, thereby identifying a detected content object; and least partially in response to identifying the detected content object, transmitting, by the server and via the overlay application network, a content object detection notification to the at least one node.

63. The method of claim **62** further comprising providing, by the at least one node and in response to receiving the content object detection notification, a content object detection notification to a user of the at least one node.

64. The method of claim **62** with the plurality of digital programs comprising a plurality of video programs.

65. The method of claim **62** with the detecting the content object step comprising, detecting, by facial recognition, content objects in at least one video program, thereby identifying human individuals as the identified detected content objects.

66. The method of claim **62** with the detection step comprising determining, by the server, a minimum Euclidean distance between a content object descriptor and a closest content object descriptor of a plurality of content object descriptors.

67. The method of claim **62** further comprising determining, by the server, a content object occurrence, the content object occurrence step comprising determining if the detected content object detection has been previously detected, the transmitting the content object detection noti-

fication step comprising transmitting, by the server and in response to a determined content object occurrence, a content object occurrence notification to the at least one node.

68. The method of claim **67** further comprising including, by the server and in a content object detection list, a content object ID in response to the detecting the content object step and the determining the content object occurrence step comprises determining if the content object ID is included at least twice in the content object detection list.

69. The method of claim **68** with the including step comprising including in the content object detection list with a predetermined maximum list size, the content object ID in response to the detecting the content object step.

70. The method of claim **68** with the including step comprising continuously including, in real or near-real time, the content object ID in the content object detection list in response to further detections of the content object in the at least one digital program.

71. The method of claim **63** with the providing step comprising providing, by the at least one node and in response to receiving the content object detection notification, the content object detection notification before the detected content object is presented, by the at least one node, in the at least one digital program.

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